Chapter 4

Linear Programming: The Simplex Method

"Checking the results of a decision against its expectations shows executives what their strengths are, where they need to improve, and where they lack knowledge or information."

– Peter Drucker

PREVIEW

The purpose of this chapter is to help you gain an understanding of how the simplex method works. Understanding the underlying principles help to interpret and analyze solution of any LP problem.

LEARNING OBJECTIVES

After studying this chapter, you should be able to

- understand the meaning of the word 'simplex' and logic of using simplex method.
- convert an LP problem into its standard form by adding slack, surplus and/or artificial variables.
- set-up simplex tables and solve LP problems using the simplex algorithm.
- interpret the optimal solution of LP problems.
- recognize special cases such as degeneracy, multiple optimal solution, unbounded and infeasible solutions.

CHAPTER OUTLINE

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4.1 INTRODUCTION

Most real-life problems when formulated as an LP model have more than two variables and therefore need a more efficient method to suggest an optimal solution for such problems. In this chapter, we shall discuss a procedure called the *simplex method* for solving an LP model of such problems. This method was developed by G B Dantzig in 1947.

The *simplex*, an important term in mathematics, represents an object in an *n*-dimensional space, connecting $n + 1$ points. In one dimension, the simplex represents a line segment connecting two points; in two dimensions, it represents a triangle formed by joining three points; in three dimensions, it represents a four-sided pyramid.

The concept of simplex method is similar to the graphical method. In the graphical method, extreme points of the feasible solution space are examined in order to search for the optimal solution that lies at one of these points. For LP problems with several variables, we may not be able to graph the feasible region, but the optimal solution will still lie at an extreme point of the many-sided, multidimensional figure (called an *n*-dimensional polyhedron) that represents the feasible solution space. The simplex method examines these extreme points in a systematic manner, repeating the same set of steps of the algorithm until an optimal solution is found. It is for this reason that it is also called the *iterative method*.

Since the number of extreme points of the feasible solution space are finite, the method assures an improvement in the value of objective function as we move from one iteration (extreme point) to another and achieve the optimal solution in a finite number of steps. The method also indicates when an unbounded solution is reached.

4.2 STANDARD FORM OF AN LP PROBLEM

Simplex method examines corner points of the feasible region, using matrix row operations, until an optimal solution is found.

The use of the simplex method to solve an LP problem requires that the problem be converted into its standard form. The standard form of the LP problem should have the following characteristics:

(i) All the constraints should be expressed as equations by adding slack or surplus and/or artificial variables.

(ii) The right-hand side of each constraint should be made non-negative (if not). This is done by multiplying both sides of the resulting constraint by -1.

(iii) The objective function should be of the maximization type.

The standard form of the LP problem is expressed as:

$$\text{Optimize (Max or Min) } Z = c_1 x_1 + c_2 x_2 + \ldots + c_n x_n + 0s_1 + 0s_2 + \ldots + 0s_m$$

subject to the linear constraints

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n + s_1 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n + s_2 &= b_2 \\ \cdot \quad\quad \cdot \quad\quad\quad\quad \cdot & \\ \cdot \quad\quad \cdot \quad\quad\quad\quad \cdot & \\ \cdot \quad\quad \cdot \quad\quad\quad\quad \cdot & \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n + s_m &= b_m \end{aligned}$$

and $$x_1, x_2, \ldots, x_n, s_1, s_2, \ldots, s_m \geq 0$$

This standard form of the LP problem can also be expressed in the compact form as follows:

$$\text{Optimize (Max or Min) } Z = \sum_{j=1}^{n} c_j x_j + \sum_{i=1}^{m} 0s_i \qquad \text{(Objective function)}$$

subject to the linear constraints

$$\sum_{j=1}^{n} a_{ij}x_j + s_i = b_i; \quad i = 1, 2, \ldots, m \qquad \text{(Constraints)}$$

and $$x_j, s_i \geq 0, \quad \text{for all } i \text{ and } j \qquad \text{(Non-negativity conditions)}$$

In matrix notations the standard form is expressed as:

Optimize (Max or Min) $Z = \mathbf{cx} + \mathbf{0s}$

subject to the linear constraints

$$\mathbf{Ax} + \mathbf{s} = \mathbf{b}, \text{ and } \mathbf{x}, \mathbf{s} \geq \mathbf{0}$$

where, $\mathbf{c} = (c_1, c_2, \ldots, c_n)$ is the row vector, $\mathbf{x} = (x_1, x_2, \ldots, x_n)^T$, $\mathbf{b} = (b_1, b_2, \ldots, b_m)^T$ and $\mathbf{s} = (s_1, s_2, \ldots, s_m)$ are column vectors, and $\mathbf{A}$ is the $m \times n$ matrix of coefficients, a_{ij} of variables $x_1, x_2, \ldots, x_n$ in the constraints.

Remarks Any LP problem (maximization or minimization) may have

1. (a) no feasible solution, i.e. value of decision variables, x_j ($j = 1, 2, \ldots, n$) may not satisfy every constraint.
   (b) a unique optimum feasible solution.
   (c) more than one optimum feasible solution, i.e. alternative optimum feasible solutions.
   (d) a feasible solution for which the objective function is *unbounded* (value of the objective function can be made as large as possible in a maximization problem or as small as possible in a minimization problem).
2. Any minimization LP problem can be converted into an equivalent maximization problem by changing the sign of c_j's in the objective function. That is,

$$\text{Minimize } \sum_{j=1}^{n} c_j x_j = \text{Maximize } \sum_{j=1}^{n} (-c_j) x_j$$

3. Any constraint expressed by equality (=) sign may be replaced by two weak inequalities. For example, $a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n = b_1$ is equivalent to following two simultaneous constraints,

$$a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n \leq b_1 \quad \text{and} \quad a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n \geq b_1$$

Slack variable represents an unused quantity of resource; it is added to less-than or equal-to type constraints in order to get an equality constraint.

4. Three types of additional variables, namely (i) slack variables (s) (ii) surplus variables (– s), and (iii) artificial variables (A) are added in the given LP problem to convert it into the standard form for the following reasons:
   (a) These variables allow us to convert inequalities into equalities, thereby converting the given LP problem into its standard form. Such form will help in getting solution of the LP problem.
   (b) These variables help decision-makers to draw economic interpretation from the final solution.
   (c) These variables are used to get an initial feasible solution represented by the columns of the identity matrix.

The summary of the additional variables to be added in the given LP problem in order to convert it into a standard form is given in Table 4.1.

Table 4.1 Summary of Additional Variables Added in an LP Problem

Types of Constraint	*Extra Variable Needed*	*Coefficient of Additional Variables in the Objective Function*		*Presence of Additional Variables in the Initial Solution*
		Max Z	*Min Z*	
• Less than or equal to (≤)	A slack variable is added	0	0	Yes
• Greater than or equal to (≥)	A surplus variable is subtracted, and	0	0	No
	an artificial variable is added	$-M$	$+M$	Yes
• Equal to (=)	Only an artificial variable is added	$-M$	$+M$	Yes

Remark *A slack variable* represents an unused resource, either in the form of time on a machine, labour hours, money, warehouse space, etc. Since these variables don't yield any profit, therefore such variables are added to the objective function with zero coefficients.

A surplus variable represents the amount by which solution values exceed a resource. These variables are also called *negative slack variables*. Surplus variables, like slack variables carry a zero coefficient in the objective function.

Definitions

Basic solution Given a system of m simultaneous linear equations with n ($> m$) variables: $\mathbf{Ax} = \mathbf{B}$, where $\mathbf{A}$ is an $m \times n$ matrix and rank ($\mathbf{A}$) = m. Let $\mathbf{B}$ be any $m \times m$ non-singular submatrix of $\mathbf{A}$ obtained by reordering m linearly independent columns of $\mathbf{A}$. Then, a solution obtained by setting $n - m$ variables not associated with the columns of $\mathbf{B}$, equal to zero, and solving the resulting system is called a *basic solution* to the given system of equations.

The m variables (may be all non zero) are called *basic variables.* The $m \times m$ non-singular sub-matrix $\mathbf{B}$ is called a *basis matrix* and the columns of $\mathbf{B}$ as *basis vectors.*

If $\mathbf{B}$ is the basis matrix, then the basic solution to the system of equations will be $\mathbf{x_B} = \mathbf{B}^{-1}\mathbf{b}$.

Basic feasible solution A basic solution to the system of simultaneous equations, $\mathbf{Ax} = \mathbf{b}$ is called *basic feasible* if $\mathbf{x_B} \geq 0$.

Degenerate solution A basic solution to the system of simultaneous equations, $\mathbf{Ax} = \mathbf{b}$ is called *degenerate* if one or more of the basic variables assume zero value.

Cost vector Let $\mathbf{x_B}$ be a basic feasible solution to the LP problem.

Maximize $Z = \mathbf{cx}$

subject to the constraints

$$\mathbf{Ax} = \mathbf{b}, \quad \text{and} \quad \mathbf{x} \geq 0.$$

Then the vector $\mathbf{c_B} = (c_{B1}, c_{B2}, \ldots, c_{Bm})$, is called *cost vector* that represents the coefficient of basic variable, $\boldsymbol{x_B}$ in the basic feasible solution.

4.3 SIMPLEX ALGORITHM (MAXIMIZATION CASE)

The steps of the simplex algorithm for obtaining an optimal solution (if it exists) to a linear programming problem are as follows:

Step 1: Formulation of the mathematical model

(i) Formulate the LP model of the given problem.

(ii) If the objective function is of minimization, then convert it into equivalent maximization, by using the following relationship

$$\text{Minimize } Z = -\text{Maximize } Z^*, \text{ where } Z^* = -Z.$$

(iii) Check whether all the b_i ($i = 1, 2, \ldots, m$) values are positive. If any one of them is negative, then multiply the corresponding constraint by -1 in order to make $b_i > 0$. In doing so, remember to change a $\leq$ type constraint to a $\geq$ type constraint, and vice versa.

(iv) Express the given LP problem in the standard form by adding additional variables in constraints (as per requirement) and assign a *z*ero-cost coefficient to these variables in the objective function.

(v) Replace each *unrestricted* variable (if any) with the difference of the two non-negative variables.

Surplus variable represents the amount of resource usage above the minimum required and is added to greater-than or equal-to constraints in order to get equality constraint.

Step 2: Set-up the initial solution Write down the coefficients of all the variables in the LP problem in a tabular form, as shown in Table 4.2, in order to get an initial basic feasible solution [$\mathbf{x_B} = \mathbf{B^{-1}}\,\mathbf{b}$].

		$c_j \rightarrow$	c_1	c_2	...	c_n	0	0	...	0
Basic Variables Coefficient (c_B)	*Basic Variables* B	*Basic Variables Value* b ($= x_B$)	*Variables* x_1	x_2	...	x_n	s_1	s_2	...	s_m
c_{B1}	s_1	$x_{B1} = b_1$	a_{11}	a_{12}	...	a_{1n}	1	0	...	0
c_{B2}	s_2	$x_{B2} = b_2$	a_{21}	a_{22}	...	a_{2n}	0	1	...	0
.	.	.	.	.		.	.	.		.
.	.	.	.	.		.	.	.		.
.	.	.	.	.		.	.	.		.
c_{Bm}	s_m	$x_{Bm} = b_m$	a_{m1}	a_{m2}	...	a_{mn}	0	0	...	1
$Z = \Sigma\, c_{Bi} x_{Bi}$		$z_j = \Sigma\, c_{Bi} x_j$	0	0	...	0	0	0	...	0
		$c_j - z_j$	$c_1 - z_1$	$c_2 - z_2$	...	$c_n - z_n$	0	0	...	0

Table 4.2 Initial Simplex Table

After having set up the initial simplex table, locate the identity matrix (contains all *z*eros except positive elements 1's on the diagonal). The identity matrix so obtained is also called a *basis matrix* [because basic feasible solution is represented by $\mathbf{B} = \mathbf{I}$].

The columns of identity matrix represent the coefficients of slack variables that have been added to the constraints. Each column of the identity matrix also represents a basic variable.

Assign values of the constants (b_i's) to the column variables in the identity matrix [because $\mathbf{x_B} = \mathbf{B}^{-1}\,\mathbf{b} = \mathbf{I}\,\mathbf{b} = \mathbf{b}$].

Degeneracy arises when there is a tie in the minimum ratio value that determines the variable to enter into the next solution.

The variables corresponding to the columns of the identity matrix are called *basic variables* and the remaining ones are called *non-basic variables.* In general, if an LP model has n variables and m ($< n$) constraints, then m variables would be basic and $n - m$ variables non-basic. In certain cases one or more than one basic variables may also have *z*ero values. It one or more basic variables have zero value, then this situation is called *degeneracy* and will be discussed later.

The first row in Table 4.2 indicates the coefficients c_j of variables in the objective function. These values represent the cost (or profit) per unit associated with a variable in the objective function and these are used to determine the variable to be entered into the basis matrix **B**.

The column '$\mathbf{c_B}$' lists the coefficients of the current basic variables in the objective function. These values are used to calculate the value of Z when one unit of any variable is brought into the solution. Column '$\mathbf{x_B}$' represents the values of the basic variables in the current basic solution.

Basis is the set of variables present in the solution, have positive non-zero value; these variables are also called **basic variables.**

Numbers, a_{ij} in the columns under each variable are also called *substitution rates* (or *exchange coefficients*) because these represent the rate at which resource i ($i = 1, 2, \ldots, m$) is consumed by each unit of an activity j ($j = 1, 2, \ldots, n$).

The values z_j represent the amount by which the value of objective function Z would be decreased (*or increased*) *if one unit of the given variable is added to the new solution. Each of the values in the $c_j - z_j$ row represents the net amount of increase* (*or decrease*) *in the objective function that would occur when one unit of the variable represented by the column head is introduced into the solution.* That is:

$$c_j - z_j \text{ (net effect)} = c_j \text{ (incoming unit profit/cost)} - z_j \text{ (outgoing total profit/cost)}$$

where z_j = Coefficient of basic variables column × Exchange coefficient column j

Step 3: Test for optimality Calculate the $c_j - z_j$ value for all non-basic variables. To obtain the value of z_j multiply each element under *'Variables'* column (columns, $\mathbf{a_j}$ of the coefficient matrix) with the corresponding elements in $\mathbf{c_B}$-column. Examine values of $c_j - z_j$. The following three cases may arise:

(i) If all $c_j - z_j \le 0$, then the basic feasible solution is optimal.

(ii) If at least one column of the coefficients matrix (i.e. $\mathbf{a}_k$) for which $c_k - z_k > 0$ and all other elements are negative (i.e. $a_{ik} < 0$), then there exists an unbounded solution to the given problem.

(iii) If at least one $c_j - z_j > 0$, and each of these columns have at least one positive element (i.e. a_{ij}) for some row, then this indicates that an improvement in the value of objective function Z is possible.

Non-basic variables are those which are not in the 'basis' and have zero-value.

Step 4: Select the variable to enter the basis If Case (iii) of Step 3 holds, then select a variable that has the largest $c_j - z_j$ value to enter into the new solution. That is,

$$c_k - z_k = \text{Max}\ \{(c_j - z_j);\quad c_j - z_j > 0\}$$

The column to be entered is called the *key* or *pivot* column. Obviously, such a variable indicates the largest per unit improvement in the current solution.

Step 5: Test for feasibility (variable to leave the basis) After identifying the variable to become the basic variable, the variable to be removed from the existing set of basic variables is determined. For this, each number in $\mathbf{x_B}$-column (i.e. b_i values) is divided by the corresponding (but positive) number in the key column and a row is selected for which this ratio is non-negative and minimum. This ratio is called the *replacement* (*exchange*) *ratio.* That is,

$$\frac{x_{Br}}{a_{rj}} = \text{Min}\left\{\frac{x_{Bi}}{a_{rj}};\ a_{rj} > 0\right\}$$

This ratio restricts the number of units of the incoming variable that can be obtained from the exchange. *It may be noted that the division by a negative or by a zero element in key column is not permitted.*

The row selected in this manner is called the *key* or *pivot row* and it represents the variable which will leave the solution.

The element that lies at the intersection of the key row and key column of the simplex table is called *key* or *pivot* element.

Infeasibility is the situation in which no solution satisfies all constraints of an LP problem.

Step 6: Finding the new solution

(i) If the key element is 1, then the row remains the same in the new simplex table.

(ii) If the key element is other than 1, then divide each element in the key row (including the elements in $\mathbf{x_B}$-column) by the key element, to find the new values for that row.

(iii) The new values of the elements in the remaining rows of the new simplex table can be obtained by performing elementary row operations on all rows so that all elements except the key element in the key column are *z*ero. In other words, for each row other than the key row, we use the formula:

$$\text{Number in new row} = \begin{pmatrix}\text{Number in}\\ \text{old row}\end{pmatrix} \pm \left[\begin{pmatrix}\text{Number above or below}\\ \text{key element}\end{pmatrix} \times \begin{pmatrix}\text{Corresponding number in}\\ \text{the new row, that is row}\\ \text{replaced in Step 6 (ii)}\end{pmatrix}\right]$$

The new entries in $\mathbf{c_B}$ and $\mathbf{x_B}$ columns are updated in the new simplex table of the current solution.

Step 7: Repeat the procedure Go to Step 3 and repeat the procedure until all entries in the $c_j - z_j$ row are either negative or *z*ero.

$c_j - z_j$ row values represent net profit or loss resulting from introducing one unit of any variable into the 'basis' or solution mix.

Remark The flow chart of the simplex algorithm for both the maximization and the minimization LP problem is shown in Fig. 4.1.

Example 4.1 Use the simplex method to solve the following LP problem.

Maximize $Z = 3x_1 + 5x_2 + 4x_3$

subject to the constraints

(i) $2x_1 + 3x_2 \le 8$, (ii) $2x_2 + 5x_3 \le 10$, (iii) $3x_1 + 2x_2 + 4x_3 \le 15$

and $x_1, x_2, x_3 \ge 0$

[*Rewa MSc* (*Maths*)*, 1995; Meerut MSc* (*Stat.*)*, 1996; MSc* (*Maths*)*, 1998; Kurukshetra, MSc* (*Stat.*)*, 1998*]

Solution Step 1: Introducing non-negative slack variables s_1, s_2 and s_3 to convert the given LP problem into its standard form:

Maximize $Z = 3x_1 + 5x_2 + 4x_3 + 0s_1 + 0s_2 + 0s_3$

subject to the constraints

(i) $2x_1 + 3x_2 + s_1 = 8$, (ii) $2x_2 + 5x_3 + s_2 = 10$, (iii) $3x_1 + 2x_2 + 4x_3 + s_3 = 15$

and $x_1, x_2, x_3, s_1, s_2, s_3 \ge 0$

Step 2: Since all b_i (RHS values) > 0, (i = 1, 2, 3), therefore choose the initial basic feasible solution as:

$$x_1 = x_2 = x_3 = 0; s_1 = 8, s_2 = 10, s_3 = 15 \text{ and Max } Z = 0$$

This solution can also be read from the initial simplex Table 4.3 by equating row-wise values in the basis (**B**) column and solution values ($\mathbf{x_B}$) column.

Step 3: To see whether the current solution given in Table 4.3 is optimal or not, calculate

$$c_j - z_j = c_j - \mathbf{c_B}\,\mathbf{B^{-1}}\,\mathbf{a}_j = c_j - \mathbf{c_B}\,\mathbf{y}_j$$

for non-basic variables x_1, x_2 and x_3 as follows.

z_j = (Basic variable coefficients, $\mathbf{c_B}$) × (jth column of data matrix)

That is,

$z_1 = 0\,(2) + 0\,(0) + 0\,(3) = 0$ for x_1-column

$z_2 = 0\,(3) + 0\,(2) + 0\,(2) = 0$ for x_2-column

$z_3 = 0\,(0) + 0\,(5) + 0\,(4) = 0$ for x_3-column

These z_j values are now subtracted from c_j values in order to calculate the net profit likely to be gained by introducing (performing) a particular variable (activity) x_1, x_2 or x_3 into the new solution mix. This is done by introducing one unit of each variable x_1, x_2 and x_3 into the new solution mix.

$$c_1 - z_1 = 3 - 0 = 3, \qquad c_2 - z_2 = 5 - 0 = 5, \qquad c_3 - z_3 = 4 - 0 = 4$$

The z_j and $c_j - z_j$ rows are added into the Table 4.3.

The values of basic variables, s_1, s_2 and s_3 are given in the *solution values* ($\mathbf{x_B}$) column of Table 4.3. The remaining variables that are non-basic at the current solution have zero value. The value of objective function at the current solution is given by

$$Z = (\text{Basic variables coefficient, } \mathbf{c_B}) \times (\text{Basic variables value, } \mathbf{x_B})$$
$$= 0\,(8) + 0\,(10) + 0\,(15) = 0$$

Table 4.3
Initial Solution

		$c_j \rightarrow$	3	5	4	0	0	0	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\,(= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3	*Min Ratio* x_B/x_2
0	s_1	8	2	(3)	0	1	0	0	8/3 →
0	s_2	10	0	2	5	0	1	0	10/2
0	s_3	15	3	2	4	0	0	1	15/2
$Z = 0$		z_j	0	0	0	0	0	0	
		$c_j - z_j$	3	5 ↑	4	0	0	0	

Since all $c_j - z_j \geq 0$ ($j = 1, 2, 3$), the current solution is not optimal. Variable x_2 is chosen to enter into the basis because $c_2 - z_2 = 5$ is the largest positive number in the x_2-column, where all elements are positive. This means that for every unit of variable x_2, the objective function will increase in value by 5. The x_2-column is the key column.

Step 4: The variable that is to leave the basis is determined by dividing the values in the $\mathbf{x_B}$-column by the corresponding elements in the key column as shown in Table 4.3. Since the ratio, 8/3 is minimum in row 1, the basic variable s_1 is chosen to leave the solution (basis).

Step 5 (***Iteration 1***): Since the key element enclosed in the circle in Table 4.3 is not 1, divide all elements of the key row by 3 in order to obtain new values of the elements in this row. The new values of the elements in the remaining rows for the new Table 4.4 can be obtained by performing the following elementary row operations on all rows so that all elements except the key element 1 in the key column are zero.

$$R_1 \text{ (new)} \rightarrow R_1 \text{ (old)} \div 3 \text{ (}\textit{key element}\text{)}$$
$$\rightarrow (8/3,\ 2/3,\ 3/3,\ 0/3,\ 1/3,\ 0/3,\ 0/3) = (8/3,\ 2/3,\ 1,\ 0,\ 1/3,\ 0,\ 0)$$

R_2 (new) $\rightarrow R_2$ (old) $- 2R_1$ (new)
$10 - 2 \times 8/3 = 14/3$
$0 - 2 \times 2/3 = -4/3$
$2 - 2 \times 1 = 0$
$5 - 2 \times 0 = 5$
$0 - 2 \times 1/3 = -2/3$
$1 - 2 \times 0 = 1$
$0 - 2 \times 0 = 0$

R_3 (new) $\rightarrow R_3$ (old) $- 2R_1$ (new)
$15 - 2 \times 8/3 = 29/3$
$3 - 2 \times 2/3 = 5/3$
$2 - 2 \times 1 = 0$
$4 - 2 \times 0 = 4$
$0 - 2 \times 1/3 = -2/3$
$0 - 2 \times 0 = 0$
$1 - 2 \times 0 = 1$

	$c_j \rightarrow$		3	5	4	0	0	0	
Basics Variables Coefficient c_B	Basic Variables B	Basic Variables Value b $(= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3	Min Ratio x_B/x_3
5	x_2	8/3	2/3	1	0	1/3	0	0	–
0	s_2	14/3	– 4/3	0	(5)	– 2/3	1	0	(14/3)/5 →
0	s_3	29/3	5/3	0	4	– 2/3	0	1	(29/3)/4
$Z = 40/3$		z_j	10/3	5	0	5/3	0	0	
		$c_j - z_j$	– 1/3	0	4 ↑	– 5/3	0	0	

Table 4.4 Improved Solution

The improved basic feasible solution can be read from Table 4.4 as: $x_2 = 8/3$, $s_2 = 14/3$, $s_3 = 29/3$ and $x_1 = x_3 = s_1 = 0$. The improved value of the objective function becomes:

$$Z = (\text{Basic variable coefficients, } \mathbf{c_B}) \times (\text{Basic variable values, } \mathbf{x_B})$$
$$= 5\,(8/3) + 0\,(14/3) + 0\,(29/3) = 40/3$$

Once again, calculate values of $c_j - z_j$ to check whether the solution shown in Table 4.4 is optimal or not. Since $c_3 - z_3 > 0$, the current solution is not optimal.

Iteration 2: Repeat Steps 3 to 5. Table 4.5 is obtained by performing following row operations to enter variable x_3 into the basis and to drive out s_2 from the basis.

$$R_2 \text{ (new)} = R_2 \text{ (old)} \div 5 \text{ (}\textit{key element}\text{)} = (14/15, -4/15, 0, 1, -2/15, 1/5, 0)$$

R_3 (new) $\rightarrow$ R_3 (old) $- 4R_2$ (new)
$29/3 - 4 \times 14/15 = 89/15$
$5/3 - 4 \times -4/15 = 41/15$
$0 - 4 \times 0 = 0$
$4 - 4 \times 1 = 0$
$-2/3 - 4 \times -2/15 = -2/15$
$0 - 4 \times 1/5 = -4/5$
$1 - 4 \times 0 = 0$

Table 4.5 is completed by calculating the new z_j and $c_j - z_j$ values and the new value of objective function:

$z_1 = 5\,(2/3) + 4\,(-4/15) + 0\,(41/15) = 34/15$ $\quad$ $c_1 - z_1 = 3 - 34/15 = 11/15$ for x_1-column

$z_4 = 5\,(1/3) + 4\,(-2/15) + 0\,(2/15) = 17/15$ $\quad$ $c_4 - z_4 = 0 - 17/15 = -17/15$ for s_1-column

$z_5 = 5\,(0) + 4\,(1/5) + 0\,(-4/5) = 4/5$ $\quad$ $c_5 - z_5 = 0 - 4/5 = -4/5$ for s_2-column

The new objective function value is:

$$Z = (\text{Basic variables coefficient, } \mathbf{c_B}) \times (\text{Basic variables value, } \mathbf{x_B})$$
$$= 5\,(8/3) + 4\,(14/15) + 0\,(89/15) = 256/15$$

The improved basic feasible solution is shown in Table 4.5.

	$c_j \rightarrow$		3	5	4	0	0	0	
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value b $(= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3	Min Ratio x_B/x_1
5	x_2	8/3	2/3	1	0	1/3	0	0	(8/3)/(2/3) = 4
4	x_3	14/15	– 4/15	0	1	– 2/15	1/5	0	–
0	s_3	89/15	(41/15)	0	0	2/15	– 4/5	1	(89/15)/(41/15) = 2.17 →
$Z = 256/15$		z_j	34/15	5	4	17/15	4/5	0	
		$c_j - z_j$	11/15 ↑	0	0	– 17/15	– 4/5	0	

Table 4.5 Improved Solution

Iteration 3: In Table 4.5, since $c_1 - z_1$ is still a positive value, the current solution is not optimal. Thus, the variable x_1 enters the basis and s_3 leaves the basis. To get another improved solution as shown in Table 4.5 perform the following row operations in the same manner as discussed earlier.

R_3 (new) $\rightarrow R_3$ (old) $\times$ 15/41 (*key element*)
$\rightarrow$ (89/15 $\times$ 15/41, 41/15 $\times$ 15/41, 0 $\times$ 15/41, 0 $\times$ 15/41,
$-$ 2/15 $\times$ 15/41, $-$ 4/5 $\times$ 15/41, 1 $\times$ 15/41)
$\rightarrow$ (89/41, 1, 0, 0, $-$ 2/41, $-$ 12/41, 15/41)

R_1 (new) $\rightarrow R_1$ (old) $-$ (2/3) R_3 (new)				R_2 (new) $\rightarrow R_2$ (old) + (4/15) R_3 (new)			
8/3	$-$ 2/3 $\times$	89/3	= 50/41	14/15	+ 4/15 $\times$	89/41	= 62/41
2/3	$-$ 2/3 $\times$	1	= 0	$-$ 4/15	+ 4/15 $\times$	1	= 0
1	$-$ 2/3 $\times$	0	= 1	0	+ 4/15 $\times$	0	= 0
0	$-$ 2/3 $\times$	0	= 0	1	+ 4/15 $\times$	0	= 1
1/3	$-$ 2/3 $\times$	$-$ 2/41	= 15/41	$-$ 2/15	+ 4/15 $\times$	$-$ 2/41	= $-$ 6/41
0	$-$ 2/3 $\times$	$-$ 2/41	= 8/41	1/5	+ 4/15 $\times$	$-$ 12/41	= 5/41
0	$-$ 2/3 $\times$	15/41	= $-$ 10/41	0	+ 4/15 $\times$	15/41	= 4/41

Table 4.6 Optimal Solution

		$c_j \rightarrow$	3	5	4	0	0	0
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* b (= x_B)	x_1	x_2	x_3	s_1	s_2	s_3
5	x_2	50/41	0	1	0	15/41	8/41	$-$ 10/41
4	x_3	62/41	0	0	1	$-$ 6/41	5/41	4/41
3	x_1	89/41	1	0	0	$-$ 2/41	$-$ 12/41	15/41
Z = 765/41		z_j	3	5	4	45/41	24/41	11/41
		$c_j - z_j$	0	0	0	$-$ 45/41	$-$ 24/41	$-$ 11/41

In Table 4.6, all $c_j - z_j < 0$ for non-basic variables. Therefore, the optimal solution is reached with, x_1 = 89/41, x_2 = 50/41, x_3 = 62/41 and the optimal value of Z = 765/41.

Example 4.2 A company makes two kinds of leather belts, belt A and belt B. Belt A is a high quality belt and belt *B* is of lower quality. The respective profits are Rs 4 and Rs 3 per belt. The production of each of type *A* requires twice as much time as a belt of type B, and if all belts were of type B, the company could make 1,000 belts per day. The supply of leather is sufficient for only 800 belts per day (both A and B combined). Belt A requires a fancy buckle and only 400 of these are available per day. There are only 700 buckles a day available for belt B.

What should be the daily production of each type of belt? Formulate this problem as an LP model and solve it using the simplex method.

Solution Let x_1 and x_2 be the number of belts of type A and B, respectively, manufactured each day. Then the LP model would be as follows:

Maximize (total profit) $Z = 4x_1 + 3x_2$

subject to the constraints

(i) $2x_1 + x_2 \le 1{,}000$ (Time availability), (ii) $x_1 + x_2 \le 800$ (Supply of leather)

(iii) $\left.\begin{matrix} x_1 \le 400 \\ x_2 \le 700 \end{matrix}\right\}$ (Buckles availability)

and $x_1, x_2 \ge 0$

Standard form Introducing slack variables s_1, s_2, s_3 and s_4 to convert given LP model into its standard form as follows.

Maximize $Z = 4x_1 + 3x_2 + 0s_1 + 0s_2 + 0s_3 + 0s_4$

subject to the constraints

(i) $2x_1 + x_2 + s_1 = 1{,}000$, (ii) $x_1 + x_2 + s_2 = 800$

(iii) $x_1 + s_3 = 400$, (iv) $x_2 + s_4 = 700$

and $x_1, x_2, s_1, s_2, s_3, s_4 \geq 0$

Solution by simplex method An initial feasible solution is obtained by setting $x_1 = x_2 = 0$. Thus, the initial solution is: $s_1 = 1{,}000$, $s_2 = 800$, $s_3 = 400$, $s_4 = 700$ and Max $Z = 0$. This solution can also be read from the initial simplex Table 4.7.

	$c_j \rightarrow$		4	3	0	0	0	0	
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value b $(= x_B)$	x_1	x_2	s_1	s_2	s_3	s_4	Min Ratio x_B/x_1
0	s_1	1,000	2	1	1	0	0	0	1,000/2 = 500
0	s_2	800	1	1	0	1	0	0	800/1 = 800
0	s_3	400	(1)	0	0	0	1	0	400/1 = 400 →
0	s_4	700	0	1	0	0	0	1	not defined
$Z = 0$		z_j	0	0	0	0	0	0	
		$c_j - z_j$	4 ↑	3	0	0	0	0	

Table 4.7 Initial Solution

In Table 4.7, since $c_1 - z_1 = 4$ is the largest positive number, we apply the following row operations in order to get an improved basic feasible solution by entering variable x_1 into the basis and removing variable s_3 from the basis.

R_3 (new) $\rightarrow R_3$ (old) $\div$ 1 (*key element*) R_1 (new) $\rightarrow R_1$ (old) $- 2R_3$ (new)

R_2 (new) $\rightarrow R_2$ (old) $- R_3$ (new)

The new solution is shown in Table 4.8.

	$c_j \rightarrow$		4	3	0	0	0	0	
Basic Variables Coefficient c_B	Basis Variables B	Basic Variables Value b $(= x_B)$	x_1	x_2	s_1	s_2	s_3	s_4	Min Ratio x_B/x_2
0	s_1	200	0	(1)	1	0	− 2	0	200/1 = 200 →
0	s_2	400	0	1	0	1	− 1	0	400/1 = 400
4	x_1	400	1	0	0	0	1	0	–
0	s_4	700	0	1	0	0	0	1	700/1 = 700
Z = 1,600		z_j	4	0	0	0	4	0	
		$c_j - z_j$	0	3 ↑	0	0	− 4	0	

Table 4.8 An Improved Solution

The solution shown in Table 4.8 is not optimal because $c_2 - z_2 > 0$ in x_2-column. Thus, again applying the following row operations to get a new solution by entering variable x_2 into the basis and removing variable s_1 from the basis, we get

R_1 (new) $\rightarrow R_1$ (old) $\div$ 1 (*key element*) R_2 (new) $\rightarrow R_2$ (old) $- R_1$ (new)

R_4 (new) $\rightarrow R_4$ (old) $- R_1$ (new)

The improved solution is shown in Table 4.9.

Table 4.9
Improved Solution

		$c_j \rightarrow$	4	3	0	0	0	0	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b(= x_B)$	x_1	x_2	s_1	s_2	s_3	s_4	*Min Ratio* x_B/s_3
3	x_2	200	0	1	1	0	−2	0	—
0	s_2	200	0	0	−1	1	(1)	0	200/1 = 200 →
4	x_1	400	1	0	0	0	1	0	400/1 = 400
0	s_4	500	0	0	−1	0	2	1	500/2 = 250
Z = 2,200		z_j	4	3	3	0	−2	0	
		$c_j - z_j$	0	0	−3	0	2	0	
							↑		

The solution shown in Table 4.9 is not optimal because $c_5 - z_5 > 0$ in s_3-column. Thus, again applying the following row operations to get a new solution by entering variable s_3, into the basis and removing variable s_2 from the basis, we get

R_2 (new) $\rightarrow R_2$ (old) ÷ 1 (*key element*) $\qquad R_1$ (new) $\rightarrow R_1$ (old) + $2R_2$ (new)

R_3 (new) $\rightarrow R_3$ (old) − R_2 (new); $\qquad R_4$ (new) $\rightarrow R_4$ (old) − $2R_2$ (new)

The new improved solution is shown in Table 4.10.

Table 4.10
Optimal Solution

		$c_j \rightarrow$	4	3	0	0	0	0
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* b $(= x_B)$	x_1	x_2	s_1	s_2	s_3	s_4
3	x_2	600	0	1	− 1	2	0	0
0	s_3	200	0	0	− 1	1	1	0
4	x_1	200	1	0	1	− 1	0	0
0	s_4	100	0	0	1	− 2	0	1
Z = 2,600		z_j	4	3	1	2	0	0
		$c_j - z_j$	0	0	− 1	− 2	0	0

Since all $c_j - z_j < 0$ correspond to non-basic variables columns, the current basic feasible solution is also the optimal solution. Thus, the company must manufacture, x_1 = 200 belts of type A and x_2 = 600 belts of type B in order to obtain the maximum profit of Rs 2,600.

Example 4.3 A pharmaceutical company has 100 kg of A, 180 kg of B and 120 kg of C ingredients available per month. The company can use these materials to make three basic pharmaceutical products namely 5-10-5, 5-5-10 and 20-5-10, where the numbers in each case represent the percentage of weight of A, B and C, respectively, in each of the products. The cost of these raw materials is as follows:

Ingredient	*Cost per kg (Rs)*
A	80
B	20
C	50
Inert ingredients	20

The selling prices of these products are Rs 40.5, Rs 43 and 45 per kg , respectively. There is a capacity restriction of the company for product 5-10-5, because of which the company cannot produce more than 30 kg per month. Determine how much of each of the products the company should produce in order to maximize its monthly profit. [*Delhi Univ., MBA, 2004, AMIE, 2005*]

Solution Let the P_1, P_2 and P_3 be the three products to be manufactured. The data of the problem can then be summarized as follows:

Product	*Product Ingredients*			*Inert*
	A	*B*	*C*	
P_1	5%	10%	5%	80%
P_2	5%	5%	10%	80%
P_3	20%	5%	10%	65%
Cost per kg (Rs)	80	20	50	20

Cost of $P_1 = 5\% \times 80 + 10\% \times 20 + 5\% \times 50 + 80\% \times 20 = 4 + 2 + 2.50 + 16 =$ Rs 24.50 per kg

Cost of $P_2 = 5\% \times 80 + 5\% \times 20 + 10\% \times 50 + 80\% \times 20 = 4 + 1 + 5 + 16 =$ Rs 26 per kg

Cost of $P_3 = 20\% \times 80 + 5\% \times 20 + 10\% \times 50 + 65\% \times 20 = 16 + 1 + 5 + 13 =$ Rs 35 per kg

Let x_1, x_2 and x_3 be the quantity (in kg) of P_1, P_2 and P_3, respectively to be manufactured. The LP problem can then be formulated as

Maximize (net profit) Z = (Selling price – Cost price) × (Quantity of product)

$$= (40.50 - 24.50)\,x_1 + (43 - 26)\,x_2 + (45 - 35)\,x_3 = 16x_1 + 17x_2 + 10x_3$$

subject to the constraints

$$\frac{1}{20}x_1 + \frac{1}{20}x_2 + \frac{1}{5}x_3 \le 100 \quad \text{or} \quad x_1 + x_2 + 4x_3 \le 2{,}000$$

$$\frac{1}{10}x_1 + \frac{1}{20}x_2 + \frac{1}{20}x_3 \le 180 \quad \text{or} \quad 2x_1 + x_2 + x_3 \le 3{,}600$$

$$\frac{1}{20}x_1 + \frac{1}{10}x_2 + \frac{1}{10}x_3 \le 120 \quad \text{or} \quad x_1 + 2x_2 + 2x_3 \le 2{,}400$$

$$x_1 \le 30$$

and $x_1, x_2, x_3 \ge 0$.

Standard form Introducing slack variables s_1, s_2 and s_3 to convert the given LP model into its standard form as follows:

Maximize $Z = 16x_1 + 17x_2 + 10x_3 + 0s_1 + 0s_2 + 0s_3 + 0s_4$

subject to the constraints

(i) $x_1 + x_2 + 4x_3 + s_1 = 2{,}000$, (ii) $2x_1 + x_2 + x_3 + s_2 = 3{,}600$

(iii) $x_1 + 2x_2 + 2x_3 + s_3 = 2{,}400$, (iv) $x_1 + s_4 = 30$

and $x_1, x_2, x_3, s_1, s_2, s_3, s_4 \ge 0$

Solution by simplex method An initial basic feasible solution is obtained by setting $x_1 = x_2 = x_3 = 0$. Thus, the initial solution shown in Table 4.11 is: $s_1 = 2{,}000$, $s_2 = 3{,}600$, $s_3 = 2{,}400$, $s_4 = 30$ and Max $Z = 0$.

	$c_j \rightarrow$	*16*	*17*	*10*	*0*	*0*	*0*	*0*	
Basic Variables Coefficient c_B	*Basis Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3	s_4	*Min Ratio* x_B/x_2
0	s_1	2,000	1	1	4	1	0	0	0	2,000/1 = 2,000
0	s_2	3,600	2	1	1	0	1	0	0	3,600/1 = 3,600
0	s_3	2,400	1	(2)	2	0	0	1	0	2,400/2 = 1,200 →
0	s_4	30	1	0	0	0	0	0	1	—
$Z = 0$		z_j	0	0	0	0	0	0	0	
		$c_j - z_j$	16	17 ↑	10	0	0	0	0	

Table 4.11
Initial Solution

Since $c_2 - z_2 = 17$ in x_2-column is the largest positive value, we apply the following row operations in order to get a new improved solution by entering variable x_2 into the basis and removing variable s_3 from the basis.

R_3 (new) → R_3 (old) ÷ 2 (*key element*) R_1 (new) → R_1 (old) – R_3 (new)

R_2 (new) → R_2 (old) – R_3 (new)

The new solution is shown in Table 4.12.

Table 4.12 Improve Solution

	$c_j \rightarrow$	16	17	10	0	0	0	0	
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3	s_4	Min Ratio x_B/x_1
0	s_1	800	1/2	0	3	1	0	– 1/2	0	800/(1/2) = 1,600
0	s_2	2,400	3/2	0	0	0	1	– 1/2	0	2,400/(3/2) = 1,600
17	x_2	1,200	1/2	1	1	0	0	1/2	0	1,200/(1/2) = 2,400
0	s_4	30	(1)	0	0	0	0	0	1	30/1 = 30 →
Z = 20,400		z_j	17/2	17	17	0	0	17/2	0	
		$c_j - z_j$	15/2 ↑	0	– 7	0	0	– 17/2	0	

The solution shown in Table 4.12 is not optimal because $c_1 - z_1 > 0$ in x_1-column. Thus, applying the following row operations to get a new improved solution by entering variable x_1 into the basis and removing the variable s_4 from the basis, we get

R_4 (new) $\rightarrow R_4$ (old) ÷ 1 (*key element*); R_1 (new) $\rightarrow R_1$ (old) – (1/2) R_4 (new)

R_2 (new) $\rightarrow R_2$ (old) – (3/2) R_4 (new); R_3 (new) $\rightarrow R_3$ (old) – (1/2) R_4 (new)

The new solution is shown in Table 4.13.

Table 4.13 Optimal Solution

	$c_j \rightarrow$	16	17	10	0	0	0	0
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3	s_4
0	s_1	785	0	0	3	1	0	– 1/2	– 1/2
0	s_2	2,355	0	0	0	0	1	– 1/2	– 3/2
17	x_2	1,185	0	1	1	0	0	1/2	– 1/2
16	x_1	30	1	0	0	0	0	0	1
Z = 20,625		z_j	16	17	17	0	0	17/2	15/2
		$c_j - z_j$	0	0	– 7	0	0	– 17/2	– 15/2

Since all $c_j - z_j < 0$ corresponding to non-basic variables columns, the current solution is an optimal solution. Thus, the company must manufacture, x_1 = 30 kg of P_1, x_2 = 1,185 kg of P_2 and x_3 = 0 kg of P_3 in order to obtain the maximum net profit of Rs 20,625.

4.4 SIMPLEX ALGORITHM (MINIMIZATION CASE)

In certain cases, it is difficult to obtain an initial basic feasible solution of the given LP problem. Such cases arise

(i) when the constraints are of the ≤ type,

$$\sum_{j=1}^{n} a_{ij}\, x_j \le b_i, \quad x_j \ge 0$$

and value of few right-hand side constants is negative [i.e. $b_i < 0$]. After adding the non-negative slack variable s_i (i = 1, 2, . . ., m), the initial solution so obtained will be $s_i = -b_i$ for a particular resource, i. This solution is not feasible because it does not satisfy non-negativity conditions of slack variables (i.e. $s_i \ge 0$).

(ii) when the constraints are of the ≥ type,

$$\sum_{j=1}^{n} a_{ij}\, x_j \ge b_i, \quad x_j \ge 0$$

After adding surplus (negative slack) variable s_i, the initial solution so obtained will be $-s_i = b_i$ or $s_i = -b_i$

$$\sum_{j=1}^{n} a_{ij}\, x_j - s_i = b_i, \quad x_j \geq 0,\ s_i \geq 0$$

This solution is not feasible because it does not satisfy non-negativity conditions of surplus variables (i.e. $s_i \geq 0$). In such a case, artificial variables, A_i ($i = 1, 2, \ldots, m$) are added to get an initial basic feasible solution. The resulting system of equations then becomes:

$$\sum_{j=1}^{n} a_{ij}\, x_j - s_i + A_i = b_i$$

$$x_j, s_i, A_i \geq 0, \quad i = 1, 2, \ldots, m$$

These are m simultaneous equations with $(n + m + m)$ variables (n decision variables, m artificial variables and m surplus variables). An initial basic feasible solution of LP problem with such constraints can be obtained by equating $(n + 2m - m) = (n + m)$ variables equal to zero. Thus the new solution to the given LP problem is: $A_i = b_i$ ($i = 1, 2, \ldots, m$), which is not the solution to the original system of equations because the two systems of equations are not equivalent. Thus, to get back to the original problem, artificial variables must be removed from the optimal solution. There are two methods for removing artificial variables from the solution.

- *Two-Phase Method*
- *Big-M Method or Method of Penalties*

The simplex method, both for the minimization and the maximization LP problem, may be summarized through a flow chart shown in Fig. 4.1.

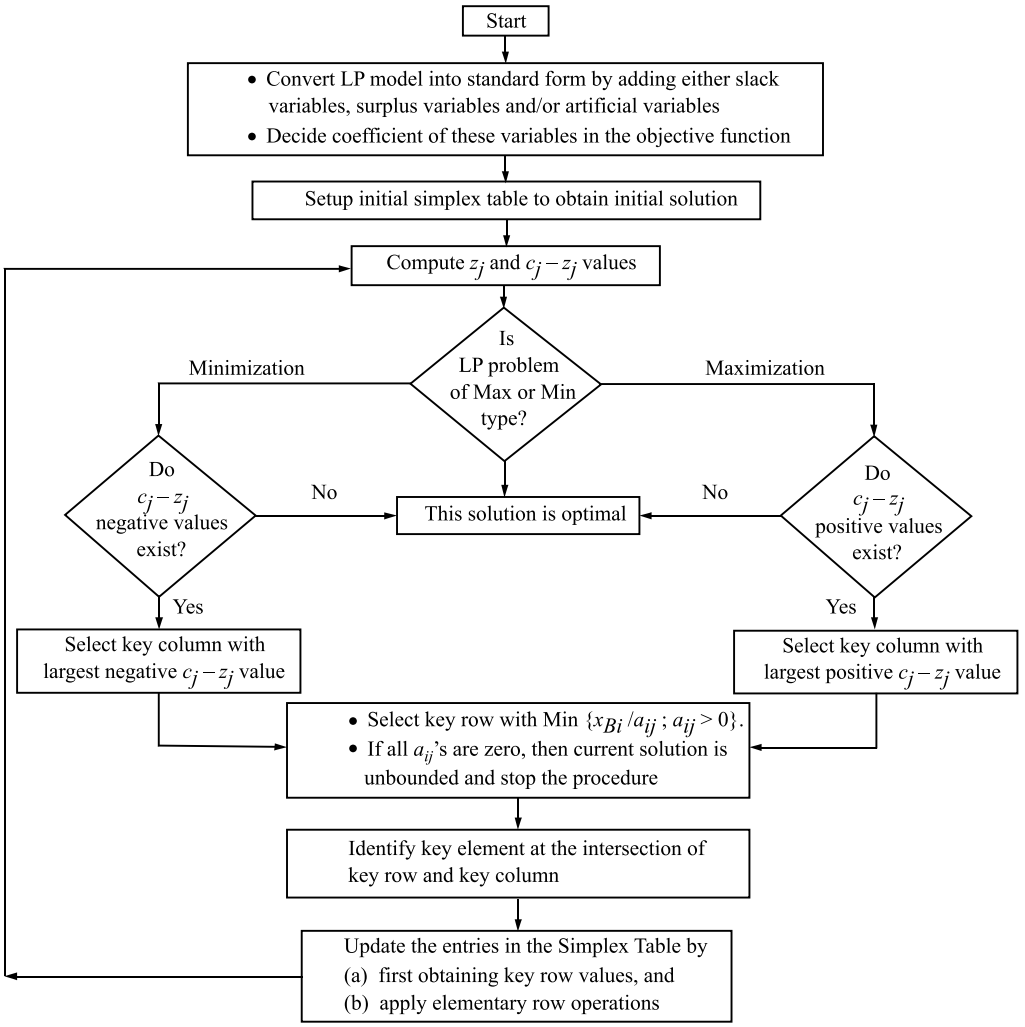


Fig. 4.1 Flow Chart of Simplex Algorithm

Remark Artificial variables have no meaning in a physical sense and are only used as a tool for generating an initial solution to an LP problem. Before the optimal solution is reached, all artificial variables must be dropped out from the solution mix. This is done by assigning appropriate coefficients to these variables in the objective function. These variables are added to those constraints with equality (=) and greater than or equal to (≥) sign.

4.4.1 Two-Phase Method

In the first phase of this method, the sum of the artificial variables is minimized subject to the given constraints in order to get a basic feasible solution of the LP problem. The second phase minimizes the original objective function starting with the basic feasible solution obtained at the end of the first phase. Since the solution of the LP problem is completed in two phases, this method is called the *two-phase method.*

Advantages of the method

1. No assumptions on the original system of constraints are made, i.e. the system may be redundant, inconsistent or not solvable in non-negative numbers.
2. It is easy to obtain an initial basic feasible solution for Phase I.
3. The basic feasible solution (if it exists) obtained at the end of phase I is used as initial solution for Phase II.

An artificial variable is added to the constraints to get an initial solution to an LP problem.

Steps of the Algorithm: *Phase I*

Step 1 (a): If all the constraints in the given LP problem are 'less than or equal to' (≤) type, then Phase II can be directly used to solve the problem. Otherwise, the necessary number of surplus and artificial variables are added to convert constraints into equality constraints.

(b) If the given LP problem is of minimization, then convert it to the maximization type by the usual method.

Step 2: Assign zero coefficient to each of the decision variables (x_j) and to the surplus variables; and assign – 1 coefficient to each of the artificial variables. This yields the following auxiliary LP problem.

$$\text{Maximize } Z^* = \sum_{i=1}^{m} (-1)\, A_i$$

subject to the constraints

$$\sum_{i=1}^{n} a_{ij}\, x_j + A_i = b_i, \qquad i = 1, 2, \ldots, m$$

and $$x_j, A_j \geq 0$$

Step 3: Apply the simplex algorithm to solve this auxiliary LP problem. The following three cases may arise at optimality.

(a) Max $Z^* = 0$ and at least one artificial variable is present in the basis with positive value. This means that no feasible solution exists for the original LP problem.

(b) Max $Z^* = 0$ and no artificial variable is present in the basis. This means that only decision variables (x_j's) are present in the basis and hence proceed to Phase II to obtain an optimal basic feasible solution on the original LP problem.

(c) Max $Z^* = 0$ and at least one artificial variable is present in the basis at *z*ero value. This means that a feasible solution to the auxiliary LP problem is also a feasible solution to the original LP problem. In order to arrive at the basic feasible solution, proceed directly to Phase II or else eliminate the artificial basic variable and then proceed to Phase II.

Remark Once an artificial variable has left the basis, it has served its purpose and can, therefore, be removed from the simplex table. An artificial variable is never considered for re-entry into the basis.

Phase II: Assign actual coefficients to the variables in the objective function and *z*ero coefficient to the artificial variables which appear at *z*ero value in the basis at the end of Phase I. The last simplex table of

Phase I can be used as the initial simplex table for Phase II. Then apply the usual simplex algorithm to the modified simplex table in order to get the optimal solution to the original problem. Artificial variables that do not appear in the basis may be removed.

Example 4.4 Use two-phase simplex method to solve the following LP problem:

Minimize $Z = x_1 + x_2$

subject to the constraints

(i) $2x_1 + x_2 \geq 4$, (ii) $x_1 + 7x_2 \geq 7$

and $x_1, x_2 \geq 0$

Solution Converting the given LP problem objective function into the maximization form and then adding surplus variables s_1 and s_2 and artificial variables A_1 and A_2 in the constraints, the problem becomes:

Maximize $Z^* = -x_1 - x_2$

subject to the constraints

(i) $2x_1 + x_2 - s_1 + A_1 = 4$, (ii) $x_1 + 7x_2 - s_2 + A_2 = 7$

and $x_1, x_2, s_1, s_2, A_1, A_2 \geq 0$

where $Z^* = -Z$

Phase I: This phase starts by considering the following auxiliary LP problem:

Maximize $Z^* = -A_1 - A_2$

subject to the constraints

(i) $2x_1 + x_2 - s_1 + A_1 = 4$, (ii) $x_1 + 7x_2 - s_2 + A_2 = 7$

and $x_1, x_2, s_1, s_2, A_1, A_2 \geq 0$

The initial solution is presented in Table 4.14.

	$c_j \rightarrow$	*0*	*0*	*0*	*0*	*– 1*	*– 1*
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b(= x_B)$	x_1	x_2	s_1	s_2	A_1	A_2
–1	A_1	4	2	1	– 1	0	1	0
– 1	A_2	7	1	(7)	0	– 1	0	1 →
$Z^* = -11$		z_j	– 3	– 8	1	1	– 1	– 1
		$c_j - z_j$	3	8 ↑	– 1	– 1	0	0

Table 4.14 Initial Solution

Artificial variables A_1 and A_2 are now removed, one after the other, maintaining the feasibility of the solution.

Iteration 1: Applying the following row operations to get an improved solution by entering variable x_2 in the basis and first removing variable A_2 from the basis. The improved solution is shown in Table 4.15. Note that the variable x_1 cannot be entered into the basis as this would lead to an infeasible solution.

R_2 (new) $\rightarrow R_2$ (old) $\div$ 7 (*key element*); R_1 (new) $\rightarrow R_1$ (old) $- R_2$ (new)

	$c_j \rightarrow$	*0*	*0*	*0*	*0*	*– 1*	*– 1*
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b(= x_B)$	x_1	x_2	s_1	s_2	A_1	A_2*
– 1	A_1	3	13/7	0	– 1	(1/7)	1	– 1/7 →
0	x_2	1	1/7	1	0	– 1/7	0	1/7
$Z^* = -3$		z_j	– 13/7	0	1	– 1/7	– 1	1/7
		$c_j - z_j$	13/7	0	– 1	1/7 ↑	0	– 8/7

Table 4.15 Improved Solution

* This column can permanently be removed at this stage.

Iteration 2: To remove A_1 from the solution shown in Table 4.15, we enter variable s_2 in the basis by applying the following row operations. The new solution is shown in Table 4.16. It may be noted that if instead of s_2, variable x_1 is chosen to be entered into the basis, it will lead to an infeasible solution

$$R_1 \text{ (new)} \to R_1 \text{ (old)} \times 7; \qquad R_2 \text{ (new)} \to R_2 \text{ (old)} + (1/7)\, R_1 \text{ (new)}$$

Table 4.16 Improved Solution

		$c_j \to$	0	0	0	0	−1	−1
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b(=x_B)$	x_1	x_2	s_1	s_2	A_1^*	A_2^*
0	s_2	21	13	0	−7	1	7	−1
0	x_2	4	2	1	−1	0	1	0
$Z^* = 0$		z_j	0	0	0	0	0	0
		$c_j - z_j$	0	0	0	0	−1	−1

* Remove columns A_1 and A_2 from Table 4.16.

Since all $c_j - z_j \le 0$ correspond to non-basic variables, the optimal solution: $x_1 = 0$, $x_2 = 4$, $s_1 = 0$, $s_2 = 21$, $A_1 = 0$, $A_2 = 0$ with $Z^* = 0$ is arrived at. However, this solution may or may not be the basic feasible solution to the original LP problem. Thus, we have to move to Phase II to get an optimal solution to our original LP problem.

Phase II: The modified simplex table obtained from Table 4.16 is represented in Table 4.17.

Table 4.17

		$c_j \to$	−1	−1	0	0	
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b(=x_B)$	x_1	x_2	s_1	s_2	Min Ratio x_B/x_1
0	s_2	21	(13)	0	−7	1	21/13 →
−1	x_2	4	2	1	−1	0	4/2
$Z^* = -4$		z_j	−2	−1	1	0	
		$c_j - z_j$	1 ↑	0	−1	0	

Iteration 1: Introducing variable x_1 into the basis and removing variable s_2 from the basis by applying the following row operations:

$$R_1 \text{ (new)} \to R_1 \text{ (old)} \div 13 \text{ (\textit{key element})}; \qquad R_2 \text{ (new)} \to R_2 \text{ (old)} - 2R_1 \text{ (new)}$$

The improved basic feasible solution so obtained is given in Table 4.18. Since in Table 4.18, $c_j - z_j \le 0$ for all non-basic variables, the current solution is optimal. Thus, the optimal basic feasible solution to the given LP problem is: $x_1 = 21/13$, $x_2 = 10/13$ and Max $Z^* = -31/13$ or Min $Z = 31/13$.

Table 4.18 Optimal Solution

		$c_j \to$	−1	−1	0	0
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b(=x_B)$	x_1	x_2	s_1	s_2
−1	x_1	21/13	1	0	−7/13	1/13
−1	x_2	10/13	0	1	1/13	−2/13
$Z^* = -31/13$		z_j	−1	−1	6/13	1/13
		$c_j - z_j$	0	0	−6/13	−1/13

Example 4.5 Solve the following LP problem by using the two-phase simplex method.

Minimize $Z = x_1 - 2x_2 - 3x_3$

subject to the constraints

(i) $-2x_1 + x_2 + 3x_3 = 2$, (ii) $2x_1 + 3x_2 + 4x_3 = 1$

and $x_1, x_2, x_3 \ge 0$. [*Meerut, MSc (Math), 2003*]

Solution After converting the objective function into the maximization form and by adding artificial variables A_1 and A_2 in the constraints, the given LP problem becomes:

Maximize $Z^* = -x_1 + 2x_2 + 3x_3$

subject to the constraints

(i) $-2x_1 + x_2 + 3x_3 + A_1 = 2$, (ii) $2x_1 + 3x_2 + 4x_3 + A_2 = 1$

and $x_1, x_2, x_3, A_1, A_2 \geq 0$

where $Z^* = -Z$

Phase I: This phase starts by considering the following auxiliary LP problem:

Maximize $Z^* = -A_1 - A_2$

subject to the constraints

(i) $-2x_1 + x_2 + 3x_3 + A_1 = 2$ (ii) $2x_1 + 3x_2 + 4x_3 + A_2 = 1$

and $x_1, x_2, x_3, A_1, A_2 \geq 0$

The initial solution is presented in Table 4.19.

		$c_j \rightarrow$	0	0	0	– 1	– 1
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\,(= x_B)$	x_1	x_2	x_3	A_1	A_2
– 1	A_1	2	– 2	1	3	1	0
– 1	A_2	1	2	3	(4)	0	1 →
$Z^* = -3$		z_j	0	– 4	– 7	– 1	– 1
		$c_j - z_j$	0	4	7 ↑	0	0

Table 4.19 Initial Solution

To first remove the artificial variable A_2 from the solution shown in Table 4.19, introduce variable x_3 into the basis by applying the following row operations:

$$R_1 \text{ (new)} \rightarrow R_2 \text{ (old)} \div 4 \text{ (}key\ element\text{)}; \quad R_1 \text{ (new)} \rightarrow R_1 \text{ (old)} - 3R_2 \text{ (new)}$$

The improved solution so obtained is given in Table 4.20. Since in Table 4.20, $c_j - z_j \leq 0$ corresponds to non-basic variables, the optimal solution is: $x_1 = 0$, $x_2 = 0$, $x_3 = 1/4$, $A_1 = 5/4$ and $A_2 = 0$ with Max $Z^* = -5/4$. But at the same time, the value of $Z^* < 0$ and the artificial variable A_1 appears in the basis with positive value 5/4. Hence, feasible solution to the given original LP problem does not exist.

		$c_j \rightarrow$	0	0	0	–1
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b(= x_B)$	x_1	x_2	x_3	A_1
– 1	A_1	5/4	–7/2	–5/4	0	1
0	x_3	1/4	1/2	3/4	1	0
$Z^* = -5/4$		z_j	7/2	5/4	0	– 1
		$c_j - z_j$	– 7/2	– 5/4	0	0

Table 4.20 Optimal but not Feasible Solution

Example 4.6 Use two-phase simplex method to solve following LP problem.

Maximize $Z = 3x_1 + 2x_2 + 2x_3$

subject to the constraints

(i) $5x_1 + 7x_2 + 4x_3 \leq 7$, (ii) $-4x_1 + 7x_2 + 5x_3 \geq -2$, (iii) $3x_1 + 4x_2 - 6x_3 \geq 29/7$

and $x_1, x_2, x_3 \geq 0$. [*Punjab Univ., BE (E & C) 2006 ; BE (IT) 2006*]

Solution Since RHS of constraint 2 is negative, multiplying it by – 1 on both sides and express it as: $4x_1 - 7x_2 - 5x_3 \leq 2$.

Phase I: Introducing slack, surplus and artificial variables in the constraints, the standard form of LP problem becomes:

Minimize $Z^* = A_1$

subject to the constraints

(i) $5x_1 + 7x_2 + 4x_3 + s_1 = 7$, (ii) $4x_1 - 7x_2 - 5x_3 + s_2 = 2$, (iii) $3x_1 + 4x_2 - 6x_3 - s_3 + A_1 = 29/7$

and $x_1, x_2, x_3, s_1, s_2, s_3, A_1 \geq 0$.

Setting variables $x_1 = x_2 = x_3 = s_3 = 0$, the basic feasible solution shown in Table 4.21 to the auxiliary LP problem is: $s_1 = 7, s_2 = 2, A_1 = 29/7$

Table 4.21 Initial Solution

		$c_j \rightarrow$	0	0	0	0	0	0	1	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3	A_1	*Min Ratio* x_B/x_2
0	s_1	7	5	(7)	4	1	0	0	0	1 →
0	s_2	2	4	−7	−5	0	1	0	0	—
1	A_1	29/7	3	4	−6	0	0	−1	1	29/28
$Z = 29/7$		z_j	3	4	−6	0	0	−1	1	
		$c_j - z_j$	−3	−4 ↑	6	0	0	1	0	

Since $c_2 - z_2 = -4$ is the largest negative value under x_2-column in Table 4.21, replacing basic variable s_1 with the non-basic variable x_2 into the basis. For this, apply following row operations:

$R_1(\text{new}) \rightarrow R_1(\text{old}) \div 7(\textit{key element})$; $R_2(\text{new}) \rightarrow R_2(\text{old}) + 7R_1(\text{new})$;

$R_3(\text{new}) \rightarrow R_3(\text{old}) - 4R_1(\text{new})$

to get an improved solution as shown in Table 4.22.

Table 4.22 An Improved Solution

		$c_j \rightarrow$	0	0	0	0	0	0	1	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3	A_1	*Min Ratio* x_B/x_1
0	x_2	1	5/7	1	4/7	1/7	0	0	0	7/5
0	s_2	9	9	0	−1	1	1	0	0	1 } Tie
1	A_1	1/7	(1/7)	0	−58/7	−4/7	0	−1	1	1 } Tie
$Z = 1/7$		z_j	1/7	0	−58/7	−4/7	0	−1	1	
		$c_j - z_j$	−1/7 ↑	0	58/7	4/7	0	1	0	

Solution shown in Table 4.22 is not optimal and there is a tie between the s_2 and A_1-rows for the key row. The A_1-row is selected as the key row because preference is given to the artificial variable. Thus (1/7) is the key element. Replace basic variable A_1 in the basis with the non-basic variable x_1. Applying following row operations to get the new solution as shown in Table 4.23.

$R_3(\text{new}) \rightarrow R_3(\text{old}) \div (1/7)$ key element; $R_1(\text{new}) \rightarrow R_1(\text{old}) - (5/7)\, R_3\,(\text{new})$;

$R_2(\text{new}) \rightarrow R_2(\text{old}) - 9\, R_3(\text{new})$

Table 4.23 Basic Feasible Solution

		$c_j \rightarrow$	0	0	0	0	0	0
Basic Variables Coefficient c_B	*Variables in Basis* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3
0	x_2	2/7	0	1	42	3	0	5
0	s_2	0	0	0	521	37	1	63
0	x_1	1	1	0	−58	−4	0	−7
Z = 0		z_j	0	0	0	0	0	0
		$c_j - z_j$	0	0	0	0	0	0

In Table 4.23, all $c_j - z_j = 0$ under non-basic variable columns. Thus, the current solution is optimal. Also Min $Z^* = 0$ and no artificial variable appears in the basis, and hence this solution is the basic feasible solution to the original LP problem.

Phase II: Using the solution shown in Table 4.23 as the initial solution for Phase II and carrying out computations to get optimal solution as shown in Table 4.24.

		$c_j \rightarrow$	3	2	2	0	0	0	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3	*Min Ratio* x_B/x_3
2	x_2	2/7	0	1	42	3	0	5	1/147
0	s_2	0	0	0	521	37	1	63	0 →
3	x_1	1	1	0	– 58	– 6	0	– 7	—
$Z = 25/7$		z_j	3	2	– 90	– 6	0	– 11	
		$c_j - z_j$	0	0	92 ↑	6	0	11	

Table 4.24 Initial Solution

In Table 4.24, $c_3 - z_3 = 92$ is the largest positive value corresponding non-basic variable x_3, replacing basic variable s_2 with non-basic variable x_3 into the basis. For this apply necessary row operations as usual. The new solution is shown in Table 4.25.

		$c_j \rightarrow$	3	2	2	0	0	0
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3
2	x_2	2/7	0	1	0	9/521	– 42/521	– 41/521
2	x_3	0	0	0	1	37/521	1/521	63/521
3	x_1	1	1	0	0	62/521	58/521	7/521
Z = 25/7		z_j	3	2	2	278/521	92/521	65/521
		$c_j - z_j$	0	0	0	–278/521	– 92/521	– 65/521

Table 4.25 Optimal Solution

Since all $c_j - z_j \leq 0$ in Table 4.25, the current solution is the optimal basic feasible solution: $x_1 = 1, x_2 = 2/7$, $x_3 = 0$ and Max Z = 25/7.

4.4.2 Big-M Method

Big-M method is another method of removing artificial variables from the basis. In this method, large undesirable (unacceptable penalty) coefficients to artificial variables are assigned from the point of view of the objective function. If the objective function Z is to be minimized, then a very large positive price (called *penalty*) is assigned to each artificial variable. Similarly, if Z is to be maximized, then a very large negative price (also called *penalty*) is assigned to each of these variables. The penalty is supposed to be designated by $– M$, for a maximization problem, and $+ M$, for a minimization problem, where $M > 0$. The Big-M method for solving an LP problem can be summarized in the following steps:

Step 1: Express the LP problem in the standard form by adding slack variables, surplus variables and/or artificial variables. Assign a *z*ero coefficient to both slack and surplus variables. Then assign a very large coefficient $+ M$ (minimization case) and $– M$ (maximization case) to artificial variable in the objective function.

Step 2: The initial basic feasible solution is obtained by assigning zero value to decision variables, x_1, x_2, ..., etc.

Step 3: Calculate the values of $c_j - z_j$ in last row of the simplex table and examine these values.

(i) If all $c_j - z_j \geq 0$, then the current basic feasible solution is optimal.

(ii) If for a column, k, $c_k - z_k$ is most negative and all entries in this column are negative, then the problem has an unbounded optimal solution.

(iii) If one or more $c_j - z_j < 0$ (minimization case), then select the variable to enter into the basis (solution mix) with the largest negative $c_j - z_j$ value (largest per unit reduction in the objective function value). This value also represents the opportunity cost of not having one unit of the variable in the solution. That is,

$$c_k - z_k = \text{Min } \{c_j - z_j : c_j - z_j < 0\}$$

Step 4: Determine the key row and key element in the same manner as discussed in the simplex algorithm for the maximization case.

Step 5: Continue with the procedure to update solution at each iteration till optimal solution is obtained.

Remarks At any iteration of the simplex algorithm one of the following cases may arise:

1. If at least one artificial variable is a basic variable (i.e., variable that is present in the basis) with zero value and the coefficient it M in each $c_j - z_j$ ($j = 1, 2, \ldots, n$) values is non-negative, then the given LP problem has no solution. That is, the current basic feasible solution is degenerate.
2. If at least one artificial variable is present in the basis with a positive value and the coefficients M in each $c_j - z_j$ ($j = 1, 2, \ldots, n$) values is non-negative, then the given LP problem has no optimum basic feasible solution. In this case, the given LP problem has a *pseudo optimum* basic feasible solution.

Example 4.7 Use penalty (Big-M) method to solve the following LP problem.

Minimize $Z = 5x_1 + 3x_2$

subject to the constraints

$$\text{(i) } 2x_1 + 4x_2 \le 12, \quad \text{(ii) } 2x_1 + 2x_2 = 10, \quad \text{(iii) } 5x_1 + 2x_2 \ge 10$$

and $x_1, x_2 \ge 0$.

Solution Adding slack variable, s_1; surplus variable, s_2 and artificial variables, A_1 and A_2 in the constraints of the given LP problem, the standard form of the LP problem becomes.

Minimize $Z = 5x_1 + 3x_2 + 0s_1 + 0s_2 + MA_1 + MA_2$

subject to the constraints

$$\text{(i) } 2x_1 + 4x_2 + s_1 = 12, \quad \text{(ii) } 2x_1 + 2x_2 + A_1 = 10 \quad \text{(iii) } 5x_1 + 2x_2 - s_2 + A_2 = 10$$

and $x_1, x_2, s_1, s_2, A_1, A_2 \ge 0$

An initial basic feasible solution: $s_1 = 12$, $A_1 = 10$, $A_2 = 10$ and Min $Z = 10M + 10M = 20M$ is obtained by putting $x_1 = x_2 = s_2 = 0$. It may be noted that the columns that correspond to the current basic variables and form the basis (identity matrix) are s_1 (slack variable), A_1 and A_2 (both artificial variables). The initial basic feasible solution is given in Table 4.26.

Since the value $c_1 - z_1 = 5 - 7M$ is the smallest value, therefore variable x_1 is chosen to enter into the basis (solution mix). To decide a current basic variable to leave the basis, calculate minimum ratio as shown in Table 4.26.

		$c_j \rightarrow$	5	3	0	0	M	M	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b (= x_B)$	x_1	x_2	s_1	s_2	A_1	A_2	*Min Ratio* x_B/x_1
0	s_1	12	2	4	1	0	0	0	12/2 = 6
M	A_1	10	2	2	0	0	1	0	10/2 = 5
M	A_2	10	(5)	2	0	– 1	0	1	10/5 = 2 →
$Z = 20M$		z_j	$7M$	$4M$	0	$-M$	M	M	
		$c_j - z_j$	$5 - 7M$ ↑	$3 - 4M$	0	M	0	0	

Table 4.26 Initial Solution

Iteration 1: Introduce variable x_1 into the basis and remove A_2 from the basis by applying the following row operations.

R_3 (new) $\rightarrow R_3$ (old) ÷ 5 (*key element*); R_2 (new) $\rightarrow R_2$ (old) $- 2R_3$ (new).

R_1 (new) $\rightarrow R_1$ (old) $- 2R_3$ (new).

The improved basic feasible solution is shown in Table 4.27.

		$c_j \rightarrow$	5	3	0	0	M	
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b(= x_B)$	x_1	x_2	s_1	s_2	A_1	Min Ratio x_B/x_2
0	s_1	8	0	(16/5)	1	2/5	0	8/(16/5) = 5/2 →
M	A_1	6	0	6/5	0	2/5	1	6/(6/5) = 5
5	x_1	2	1	2/5	0	– 1/5	0	2/(2/5) = 5
$Z = 10 + 6M$		z_j	5	$(6M/5) + 2$	0	$(2M/5) - 1$	M	
		$c_j - z_j$	0	$(-6M/5) + 1$ ↑	0	$(-2M/5) + 1$	0	

Table 4.27 Improved Solution

Iteration 2: Since the value of $c_2 - z_2$ in Table 4.27 is the largest negative value, variable x_2 is chosen to replace basic variable s_1 in the basis. Thus, to get an improved basic feasible solution, apply, the following row operations:

R_1 (new) $\rightarrow R_1$ (old) × 5/16 (*key element*); R_2 (new) $\rightarrow R_2$ (old) – (6/5) R_1 (new).

R_3 (new) $\rightarrow R_3$ (old) – (2/5) R_1 (new).

The new solution is shown in Table 4.28.

		$c_j \rightarrow$	5	3	0	0	M	
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b(= x_B)$	x_1	x_2	s_1	s_2	A_1	Min Ratio x_B/s_2
3	x_2	5/2	0	1	5/16	1/8	0	(5/2)/(1/8) = 40
M	A_1	3	0	0	– 3/8	(1/4)	1	3/(1/4) = 12 →
5	x_1	1	1	0	– 1/8	– 1/4	0	
$Z = 25/2 + 3M$		z_j	5	$3 - 3M/8 + 5/16$		$M/4 - 7/8$	M	
		$c_j - z_j$	0	$03M/8 - 5/16$		$-M/4 + 7/8$ ↑	0	

Table 4.28 Improved Solution

Iteration 3: Since $c_4 - z_4 < 0$ (negative) in s_2-column, the current solution is not optimal. Thus, non-basic variable s_2 is chosen to replace artificial variable A_1 in the basis. To get an improved basic feasible solution, apply the following row operations:

R_2 (new) $\rightarrow R_2$ (old) × 4 (*key element*); R_1 (new) $\rightarrow R_1$ (old) – (1/8) R_2 (new)

R_3 (new) $\rightarrow R_3$ (old) + (1/4) R_2 (new).

The improved basic feasible solution is shown in Table 4.29.

		$c_j \rightarrow$	5	3	0	0
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b(= x_B)$	x_1	x_2	s_1	s_2
3	x_2	1	0	1	1/2	0
0	s_2	12	0	0	– 3/2	1
5	x_1	4	1	0	– 1/2	0
$Z = 23$		z_j	5	3	– 1	0
		$c_j - z_j$	0	0	1	0

Table 4.29 Optimal Solution

In Table 4.24, all $c_j - z_j \geq 0$. Thus an optimal solution is arrived at with the value of variables as: $x_1 = 4$, $x_2 = 1$, $s_1 = 0$, $s_2 = 12$ and Min $Z = 23$.

Example 4.8 Use penalty (Big-M) method to solve the following LP problem.

Maximize $Z = x_1 + 2x_2 + 3x_3 - x_4$

subject to the constraints

(i) $x_1 + 2x_2 + 3x_3 = 15$, (ii) $2x_1 + x_2 + 5x_3 = 20$, (iii) $x_1 + 2x_2 + x_3 + x_4 = 10$

and $x_1, x_2, x_3, x_4 \geq 0$ [*Calicut, BTech. (Engg), 2000; Bangalore BE (Mech.), 2000; AMIE, 2009*]

Solution Since all constraints of the given LP problem are equations, therefore only artificial variables A_1 and A_2 are added in the constraints to convert given LP problem to its standard form. The standard form of the problem is stated as follows:

Maximize $Z = x_1 + 2x_2 + 3x_3 - x_4 - MA_1 - MA_2$

subject to the constraints

(i) $x_1 + 2x_2 + 3x_3 + A_1 = 15$, (ii) $2x_1 + x_2 + 5x_3 + A_2 = 20$

(iii) $x_1 + 2x_2 + x_3 + x_4 = 10$

and $x_1, x_2, x_3, x_4, A_1, A_2 \geq 0$

An initial basic feasible solution is given in Table 4.30.

Table 4.30 Initial Solution

		$c_j \rightarrow$	1	2	3	–1	– M	– M	
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b(= x_B)$	x_1	x_2	x_3	x_4	A_1	A_2	Min Ratio x_B/x_3
– M	A_1	15	1	2	3	0	1	0	15/3 = 5
– M	A_2	20	2	1	(5)	0	0	1	20/5 = 4 →
– 1	x_4	10	1	2	1	1	0	0	10/1 = 10
$Z = -35M - 10$		z_j	$-3M-1$	$-3M-2$	$-8M-1$	–1	$-M$	$-M$	
		$c_j - z_j$	$3M+2$	$3M+4$	$8M+4$ ↑	0	0	0	

Since value of $c_3 - z_3$ in Table 4.30 is the largest positive value, the non-basic variable x_3 is chosen to replace the artificial variable A_2 in the basis. Thus, to get an improved solution, apply the following row operations.

R_2 (new) $\rightarrow R_1$ (old) ÷ 5 (*key element*) ; R_1 (new) $\rightarrow R_1$ (old) – 3 R_2 (new).

R_3 (new) $\rightarrow R_3$ (old) – R_2 (new)

The improved basic feasible solution is shown in Table 4.31.

Table 4.31 Improved Solution

		$c_j \rightarrow$	1	2	3	– 1	– M	
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b(= x_B)$	x_1	x_2	x_3	x_4	A_1	Min Ratio x_B/x_2
– M	A_1	3	– 1/5	(7/5)	0	0	1	$\frac{3}{7/5} = \frac{15}{7}$ →
3	x_3	4	2/5	1/5	1	0	0	$\frac{4}{1/5} = 20$
– 1	x_4	6	3/5	9/5	0	1	0	$\frac{6}{9/5} = \frac{30}{9}$
$Z = -3M + 6$		z_j	$M/5 + 3/5$	$-7M/5 - 6/5$	3	– 1	$-M$	
		$c_j - z_j$	$-M/5 - 2/5$	$7M/5 + 16/5$ ↑	0	0	0	

Since value of $c_2 - z_2 > 0$ (positive) in Table 4.31, the non-basic variable x_2 is chosen to replace the artificial variable A_1 in the basis. Thus, to get an improved basic feasible solution, apply the following row operations:

R_1 (new) $\rightarrow R_1$ (old) $\times$ (5/7) (*key element*); R_2 (new) $\rightarrow R_2$ (old) – (1/5) R_1 (new).

R_3 (new) $\rightarrow R_3$ (old) – (9/5) R_1 (new).

The improved basic feasible solution is shown in Table 4.32.

		$c_j \rightarrow$	*1*	*2*	*3*	*– 1*	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b(= x_B)$	x_1	x_2	x_3	x_4	*Min Ratio* x_B/x_1
2	x_2	15/7	– 1/7	1	0	0	–
3	x_3	25/7	3/7	0	1	0	25/7 × 7/3 = 25/3
– 1	x_4	15/7	(6/7)	0	0	1	15/7 × 7/6 = 15/6 →
Z = 90/7		z_j	1/7	2	3	–1	
		$c_j - z_j$	6/7 ↑	0	0	0	

Table 4.32 Improved Solution

Once again, the solution shown in Table 4.32 is not optimal as $c_1 - z_1 > 0$ in x_1-column. Thus, applying the following row operations for replacing non-basic variable x_1 in the basis with basic variable x_4:

R_3 (new) $\rightarrow R_3$ (old) $\times$ (7/6) (*key element*) ; R_1 (new) $\rightarrow R_1$ (old) + (1/7) R_3 (new).

R_2 (new) $\rightarrow R_2$ (old) – (3/7) R_3 (new).

The improved basic feasible solution is shown in Table 4.33.

		$c_j \rightarrow$	*1*	*2*	*3*	*– 1*
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b(= x_B)$	x_1	x_2	x_3	x_4
2	x_2	15/6	0	1	0	1/6
3	x_3	15/6	0	0	1	– 3/6
1	x_1	15/6	1	0	0	7/6
Z = 15		z_j	1	2	3	0
		$c_j - z_j$	0	0	0	– 1

Table 4.33 Optimal Solution

In Table 4.28, since all $c_j - z_j \leq 0$, therefore, an optimal solution is arrived at with value of variables as: x_1 = 15/6, x_2 = 15/6, x_3 = 15/6 and Max Z = 15.

Example 4.9 ABC Printing Company is facing a tight financial squeeze and is attempting to cut costs wherever possible. At present it has only one printing contract, and luckily, the book is selling well in both the hardcover and the paperback editions. It has just received a request to print more copies of this book in either the hardcover or the paperback form. The printing cost for the hardcover books is Rs 600 per 100 books while that for paperback is only Rs 500 per 100. Although the company is attempting to economize, it does not wish to lay off any employee. Therefore, it feels obliged to run its two printing presses – I and II, at least 80 and 60 hours per week, respectively. Press I can produce 100 hardcover books in 2 hours or 100 paperback books in 1 hour. Press II can produce 100 hardcover books in 1 hour or 100 paperbacks books in 2 hours. Determine how many books of each type should be printed in order to minimize costs.

Solution Let x_1 and x_2 be the number of batches containing 100 hard cover and paperback books, to be printed respectively. The LP problem can then be formulated as follows.

Minimize $Z = 600x_1 + 500x_2$

subject to the constraints

(i) $2x_1 + x_2 \geq 80$, (ii) $x_1 + 2x_2 \geq 60$ (Printing press hours)

and $x_1, x_2 \geq 0$

Standard form Adding surplus variables s_1, s_2 and artificial variables A_1, A_2 in the the constraints, the standard form of the LP problem becomes

Minimize $Z = 600x_1 + 500x_2 + 0s_1 + 0s_2 + MA_1 + MA_2$
subject to the constraints

(i) $2x_1 + x_2 - s_1 + A_1 = 80$, (ii) $x_1 + 2x_2 - s_2 + A_2 = 60$

and $x_1, x_2, s_1, s_2, A_1, A_2 \geq 0$

Solution by simplex method The initial basic feasible solution: $A_1 = 80, A_2 = 60$ and Min $Z = 80M + 60M = 140M$ obtained by putting $x_1 = x_2 = s_1 = s_2 = 0$ is shown in Table 4.34.

Table 4.34 Initial Solution

		$c_j \rightarrow$	600	500	0	0	M	M	
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b(= x_B)$	x_1	x_2	s_1	s_2	A_1	A_2	Min Ratio x_B/x_2
M	A_1	80	2	1	− 1	0	1	0	80/1
M	A_2	60	1	(2)	0	− 1	0	1	60/2 →
$Z = 140M$		z_j	$3M$	$3M$	$-M$	$-M$	M	M	
		$c_j - z_j$	$600 - 3M$	$500 - 3M$ ↑	M	M	0	0	

Since $c_2 - z_2$ value in x_2-column of Table 4.34 is the largest negative, therefore non-basic variable x_2 is chosen to replace basic variable A_2 in the basis. For this, apply following row operations.

R_2 (new) = R_2 (old) ÷ 2 (*key element*); R_1 (new) → R_1 (old) − R_2 (new)

to get an improved basic feasible solution as shown in Table 4.35.

Table 4.35 Improved Solution

		$c_j \rightarrow$	600	500	0	0	M	
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b(= x_B)$	x_1	x_2	s_1	s_2	A_1	Min Ratio x_B/x_1
M	A_1	50	(3/2)	0	− 1	1/2	1	100/3 →
500	x_2	30	1/2	1	0	− 1/2	0	60
$Z = 15{,}000 + 50M$		z_j	$3M/2 + 250$	500	$-M$	$M/2 - 250$	M	
		$c_j - z_j$	$350 - 3M/2$ ↑	0	M	$250 - M/2$	0	

Again, since value $c_1 - z_1$ of in x_1-column of Table 4.35 is the largest negative, non-basic variable x_1 is chosen to replace basic variable A_1 in the basis. For this, apply following row operations:

R_1 (new) → R_1 (old) × (2/3) (*key element*) ; R_2 (new) → R_2 (old) − (1/2) R_1 (new)

to get an improved basic feasible solution as shown in Table 4.36.

Table 4.36 Optimal Solution

		$c_j \rightarrow$	600	500	0	0
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b(= x_B)$	x_1	x_2	s_1	s_2
600	x_1	100/3	1	0	− 2/3	1/3
500	x_2	40/3	0	1	1/3	− 2/3
$Z = 80{,}000/3$		z_j	600	500	− 700/3	− 400/3
		$c_j - z_j$	0	0	700/3	400/3

In Table 4.36, all $c_j - z_j \geq 0$ and no artificial variable is present in the basis (solution mix). Hence, an optimum solution is arrived at with x_1 = 100/3 batches of hardcover books, x_2 = 40/3 batches of paperback books, at a total minimum cost, Z = Rs. 80,000/3.

Example 4.10 An advertising agency wishes to reach two types of audiences: Customers with annual income greater than Rs 15,000 (target audience A) and customers with annual income less than Rs 15,000 (target audience B). The total advertising budget is Rs 2,00,000. One programme of TV advertising costs Rs 50,000; one programme on radio advertising costs Rs 20,000. For contract reasons, at least three programmes ought to be on TV, and the number of radio programmes must be limited to five. Surveys indicate that a single TV programme reaches 4,50,000 customers in target audience A and 50,000 in target

audience B. One radio programme reaches 20,000 in target audience A and 80,000 in target audience *B*. Determine the media mix to maximize the total reach. [*Delhi Univ*., *MBA*, *2000*]

Solution Let x_1 and x_2 be the number of insertions in TV and radio, respectively. The LP problem can then be formulated as follows:

Maximize (total reach) $Z = (4{,}50{,}000 + 50{,}000)\,x_1 + (20{,}000 + 80{,}000)\,x_2$

$= 5{,}00{,}000\,x_1 + 1{,}00{,}000\,x_2 = 5x_1 + x_2$

subject to the constraints

(i) $50{,}000\,x_1 + 20{,}000\,x_2 \le 2{,}00{,}000$ or $5x_1 + 2x_2 \le 20$ (Advt. budget)

(ii) $x_1 \ge 3$ (Advt. on TV) (iii) $x_2 \le 5$ (Advt. on Radio)

and $x_1, x_2 \ge 0$.

Standard form Adding slack/surplus and/or artificial variables in the constraints of LP problem. Then standard form of given LP problem becomes

Maximize $Z = 5x_1 + x_2 + 0s_1 + 0s_2 + 0s_3 - MA_1$

subject to the constraints

(i) $5x_1 + 2x_2 + s_1 = 20$, (ii) $x_1 - s_2 + A_1 = 3$, (iii) $x_2 + s_3 = 5$

and $x_1, x_2, s_1, s_2, s_3, A_1 \ge 0$

Solution by simplex method An initial basic feasible solution: $s_1 = 20$, $A_1 = 3$, $s_3 = 5$ and Max $Z = -3M$ obtained by putting $x_1 = x_2 = s_2 = 0$ is shown in simplex Table 4.37.

	$c_j \rightarrow$		*5*	*1*	*0*	*0*	*0*	*– M*	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b(= x_B)$	x_1	x_2	s_1	s_2	s_3	A_1	*Min Ratio* x_B/x_1
0	s_1	20	5	2	1	0	0	0	20/5 = 4
$-M$	A_2	3	(1)	0	0	– 1	0	1	3/1 = 3 →
0	s_3	5	0	1	0	0	1	0	—
$Z = -3M$		z_j	$-M$	0	0	M	0	$-M$	
		$c_j - z_j$	$M + 5$ ↑	1	0	$-M$	0	0	

Table 4.37 Initial Solution

The $c_1 - z_1$ value in x_1-column of Table 4.37 is the largest positive value. The non-basic variable x_1 is chosen to replace basic variable A_1 in the basis. For this apply following row operations

R_2 (new) $\rightarrow R_2$ (old) ÷ 1 (*key element*); R_1 (new) $\rightarrow R_1$ (old) $- 5R_2$ (new)

to get the improved basic feasible solution as shown in Table 4.38.

	$c_j \rightarrow$		*5*	*1*	*0*	*0*	*0*	
Basic Variables Coefficient c_B	*Basic Variable* B	*Basic Variables Value* $b(= x_B)$	x_1	x_2	s_1	s_2	s_3	*Min Ratio* x_B/s_2
0	s_1	5	0	2	1	(5)	0	5/5 = 1 →
5	x_1	3	1	0	0	– 1	0	—
0	s_3	5	0	1	0	0	1	—
$Z = 15$		z_j	5	0	0	– 5	0	
		$c_j - z_j$	0	1	0	5 ↑	0	

Table 4.38 Improved Solution

Again in Table 4.38, $c_4 - z_4$ value in s_2-column is the largest positive. Thus non-basic variable s_2 is chosen to replace basic variable s_1 into the basis. For this apply the following row operations:

R_1 (new) $\rightarrow R_1$ (old) ÷ 5 (*key element*); R_2 (new) $\rightarrow R_2$ (old) + R_1 (new)

to get the improved basic feasible solution as shown in Table 4.39.

Table 4.39
Optimal Solution

		$c_j \rightarrow$	5	1	0	0	0
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b(= x_B)$	x_1	x_2	s_1	s_2	s_3
0	s_2	1	0	2/5	1/5	1	0
5	x_1	4	1	2/5	1/5	0	0
0	s_3	5	0	1	0	0	1
$Z = 20$		z_j	5	2	1	0	0
		$c_j - z_j$	0	− 1	− 1	0	0

Since all $c_j - z_j \geq 0$ in Table 4.39, the total reach of target audience cannot be increased further. Hence, the optimal solution is: $x_1 = 4$ insertions in TV and $x_2 = 0$ in radio with Max (total audience) $Z = 20{,}00{,}000$.

Example 4.11 An Air Force is experimenting with three types of bombs P, Q and R in which three kinds of explosives, viz., A, B and C will be used. Taking the various factors into account, it has been decided to use the maximum 600 kg of explosive A, at least 480 kg of explosive B and exactly 540 kg of explosive C. Bomb P requires 3, 2, 2 kg, bomb Q requires 1, 4, 3 kg and bomb R requires 4, 2, 3 kg of explosives A, B and C respectively. Bomb P is estimated to give the equivalent of a 2 ton explosion, bomb Q, a 3 ton explosion and bomb R, a 4 ton explosion respectively. Under what production schedule can the Air Force make the biggest bang?

Solution Let x_1, x_2 and x_3 be the number of bombs of type P, Q and R to be experimented, respectively.

Then the LP problem can be formulated as:

Maximize $Z = 2x_1 + 3x_2 + 4x_3$

subject to the constraints

(i) Explosive A : $3x_1 + x_2 + 4x_3 \leq 600$, (ii) Explosive B : $2x_1 + 4x_2 + 2x_3 \geq 480$,

(iii) Explosive C : $2x_1 + 3x_2 + 3x_3 = 540$,

and $x_1, x_2, x_3 \geq 0$.

Standard form Adding slack, surplus and artificial variables in the constraints of LP problem, the standard form of LP problem becomes:

Maximize $Z = 2x_1 + 3x_2 + 4x_3, + 0s_1 + 0s_2 - MA_1 - MA_2$

subject to the constraints

(i) $3x_1 + x_2 + 4x_3 + s_1 = 600$, (ii) $2x_1 + 4x_2 + 2x_3 - s_2 + A_1 = 480$, (iii) $2x_1 + 3x_2 + 3x_3 + A_2 = 540$,

and $x_1, x_2, x_3 s_1, s_2, A_1, A_2 \geq 0$.

Solution by simplex method The initial basic feasible solution: $s_1 = 600$, $A_1 = 480$, $A_2 = 540$, Max $Z = -1{,}020$ M obtained by putting basic variables $x_1 = x_2 = s_2 = 0$ is shown in Table 4.40.

Table 4.40
Initial Solution

		$c_j \rightarrow$	2	3	4	0	0	–M	–M	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2	A_1	A_2	*Min Ratio* x_B/x_2
0	s_1	600	3	1	4	1	0	0	0	600/1
$-M$	A_1	480	2	(4)	2	0	–1	1	0	480/4 →
$-M$	A_2	540	2	3	3	0	0	0	1	540/3
$Z = 1{,}020\ M$		z_j	$-4M$	$-7M$	$-5M$	0	M	$-M$	$-M$	
		$c_j - z_j$	$2 + 4M$	$3 + 7M$ ↑	$4 + 5M$	0	$-M$	0	0	

Since in Table 4.40, $c_2 - z_2$ value in x_2–column is largest positive, non-basic variable x_2 is chosen to replace basic variable A_1 in the basis. For this, apply the following row operations:

$R_2(\text{new}) \rightarrow R_2(\text{old}) \times (1/4)$ (*key element*); $R_1(\text{new}) \rightarrow R_1(\text{old}) - R_2(\text{new})$;

$R_3(\text{new}) \rightarrow R_3(\text{old}) - 3R_2(\text{new})$

to get the improved basic feasible solution as shown in Table 4.41.

	$c_j \rightarrow$	2	3	4	0	0	–M	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (=x_B)$	x_1	x_2	x_3	s_1	s_2	A_1	*Min Ratio* x_B/x_3
0	s_1	480	5/2	0	7/2	1	1/4	0	960/7
3	x_2	120	1/2	1	1/2	0	–1/4	0	240
–M	A_1	180	1/3	0	(3/2)	0	3/4	1	120 →
$Z = 360 - 180M$		z_j	3/2 – M/2	3	3/2 – 3M/2	0	–3/4 – 3M/4	–M	
		$c_j - z_j$	1/2 + M/2	0	5/2 + 3M/2 ↑	0	3/4 + 3M/4	0	

Table 4.41 Improved Solution

In Table 4.41, $c_3 - z_3$ value in x_3-column is largest positive, non-basic variable x_3 is chosen to replace basic variable A_2 into the basis. For this, apply following row operations:

$R_3(\text{new}) \rightarrow R_3(\text{old}) \times (2/3)$ (*key element*); $R_1(\text{new}) \rightarrow R_1(\text{old}) - (7/2)\, R_3(\text{new})$;

$R_2(\text{new}) \rightarrow R_2(\text{old}) - (1/2)\, R_3(\text{new})$.

to get the improved basic feasible solution as shown in Table 4.42.

	$c_j \rightarrow$	2	3	4	0	0
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (=x_B)$	x_1	x_2	x_3	s_1	s_2
0	s_1	60	4/3	0	0	1	–3/2
3	x_2	60	1/3	1	0	0	–1/2
4	x_3	120	1/3	0	1	0	1/2
$Z = 660$		z_j	7/3	3	4	0	1/2
		$c_j - z_j$	–1/3	0	0	0	–1/2

Table 4.42 Optimal Solution

In Table 4.42, all $c_j - z_j \le 0$ and artificial variables A_1 and A_2 have been removed from the basis (solution mix). Thus, an optimal solution is arrived at with $x_1 = 0$ bombs of type P, $x_2 = 60$ bombs of type Q, $x_3 = 120$ bombs of type R, at largest benefit of $Z = 660$.

SELF PRACTICE PROBLEMS A

1. A television company has three major departments for manufacturing two of its models – A and B. The monthly capacities of the departments are given as follows:

	Per Unit Time Requirement (hours)		*Hours Available this Month*
	Model A	*Model B*	
Department *I*	4.0	2.0	1,600
Department *II*	2.5	1.0	1,200
Department *III*	4.5	1.5	1,600

The marginal profit per unit from model A is Rs 400 and from model B is Rs 100. Assuming that the company can sell any quantity of either product due to favourable market conditions, determine the optimum output for both the models, the highest possible profit for this month and the slack time in the three departments.

2. A manufacturer of leather belts makes three types of belts A, B and C which are processed on three machines M_1, M_2 and M_3. Belt A requires 2 hours on machine M_1 and 3 hours on machine M_2 and 2 hours on machine M_3. Belt B requires 3 hours on machine M_1, 2 hours on machine M_2 and 2 hours on machine M_3 and Belt C requires 5 hours on machine M_2 and 4 hours on machine M_3. There are 8 hours of time per day available on machine M_1, 10 hours of time per day available on machine M_2 and 15 hours of time per day available on machine M_3. The profit gained from belt A is Rs 3.00 per unit, from Belt B is Rs 5.00 per unit, from belt C is Rs 4.00 per unit. What should be the daily production of each type of belt so that the products yield the maximum profit?

3. A company produces three products A, B and C. These products require three ores O_1, O_2 and O_3. The maximum quantities of the ores O_1, O_2 and O_3 available are 22 tonnes, 14 tonnes and 14 tonnes, respectively. For one tonne of each of these products, the ore requirements are:

	A	*B*	*C*
O_1	3	–	3
O_2	1	2	3
O_3	3	2	3
Profit per tonne (Rs in thousand)	1	4	5

The company makes a profit of Rs 1,000, 4,000 and 5,000 on each tonne of the products A, B and C, respectively. How many

tonnes of each product should the company produce in order to maximize its profits.

4. A manufacturing firm has discontinued the production of a certain unprofitable product line. This has created considerable excess production capacity. Management is considering to devote this excess capacity to one or more of three products; call them product 1, 2 and 3. The available capacity on the machines that might limit output is summarized in the following table:

Machine Type	*Available Time (in Machine-hours per Week)*
Milling Machine	250
Lathe	150
Grinder	50

The number of machine-hours required for each unit of the respective product is as follows:

Machine Type	*Productivity (in Machine-hours per Unit)*		
	Product 1	*Product 2*	*Product 3*
Milling Machine	8	2	3
Lathe	4	3	0
Grinder	2	–	1

The profit per unit would be Rs 20, Rs 6 and Rs 8, respectively for product 1, 2 and 3. Find how much of each product the firm should produce in order to maximize its profit.

[*Delhi Univ., MBA 2006*]

5. A farmer has 1,000 acres of land on which he can grow corn, wheat or soyabean. Each acre of corn costs Rs 100 for preparation, requires 7 men-days of work and yields a profit of Rs 30. An acre of wheat costs Rs 120 to prepare, requires 10 men-days of work and yields a profit of Rs 40. An acre of soyabean costs Rs 70 to prepare, requires 8 men-days of work and yields a profit of Rs 20. If the farmer has Rs 1,00,000 for preparation and can count on 8,000 men-days of work, determine how many acres should be allocated to each crop in order to maximize profits? [*Delhi Univ., MBA, 2004*]

6. The annual handmade furniture show and sale is supposed to take place next month and the school of vocational studies is also planning to make furniture for this sale. There are three wood-working classes – I year, II year and III year, at the school and they have decided to make styles of chairs – *A*, *B* and *C*. Each chair must receive work in each class. The time in hours required for each chair in each class is:

Chair	*I Year*	*II Year*	*III Year*
A	2	4	3
B	3	3	2
C	2	1	4

During the next month there will be 120 hours available to the I year class, 160 hours to the II year class, and 100 hours to the III year class for producing the chairs. The teacher of the wood-working classes feels that a maximum of 40 chairs can be sold at the show. The teacher has determined that the profit from each type of chair will be: A, Rs 40; B, Rs 35 and C, Rs. 30. How many chairs of each type should be made in order to maximize profits at the show and sale?

7. Mr Jain, the marketing manager of ABC Typewriter Company is trying to decide how he should allocate his salesmen to the company's three primary markets. Market I is in the urban area and the salesman can sell, on the average, 40 typewriters a week. Salesmen in the other two markets, II and III can sell, on the average, 36 and 25 typewriters per week, respectively. For the coming week, three of the salesmen will be on vacation, leaving only 12 men available for duty. Also because of lack of company care, a maximum of 5 salesmen can be allocated to market area I. The selling expenses per week for salesmen in each area are Rs 800 per week for area I, Rs 700 per week for area II, and Rs 500 per week for area III. The budget for the next week is Rs 7,500. The profit margin per typewriter is Rs 150. Determine how many salesmen should be assigned to each area in order to maximize profits?

8. Three products – A, B and C – are produced in three machine centres X, Y and Z. All three products require a part of their manufacturing operation at each of the machine centres. The time required for each operation on various products is indicated in the following table. Only 100, 77 and 80 hours are available at machine centres X, Y and *Z*, respectively. The profit per unit from A, B and C is Rs 12, Rs 3 and Re 1, respectively.

Products	*Machine Centres*			*Profit per Unit (Rs)*
	X	*Y*	*Z*	
A	10	7	2	12
B	2	3	4	3
C	1	2	1	1
Available hours	100	77	80	

(a) Determine suitable product mix so as to maximize the profit. Comment on the queries (b) and (c) from the solution table obtained.

(b) Satisfy that full available hours of X and Y have been utilized and there is surplus hours of Z. Find out the surplus hours of Z.

(c) Your aim is to utilize surplus capacity of Z. Can you say from the table that the introduction of more units of Y is required?

9. A certain manufacturer of screw fastenings found that there is a market for packages of mixed screw sizes. His market research data indicated that two mixtures of three screw types (1, 2 and 3), properly priced, could be most acceptable to the public. The relevant data is:

Mixture	*Specifications*	*Selling Price (Rs/kg)*
A	≥ 50% type 1 ≤ 30% type 2 and quantity of type 3	5
B	≥ 35% type 1 ≤ 45% type 2 and quantity of type 3	4

For these screws, the plant capacity and manufacturing cost are as follows:

Screw Type	*Plant Capacity (kg/day × 100)*	*Manufacturing Cost (Rs/kg)*
1	10	4.50
2	10	3.50
3	6	2.70

What production shall this manufacturer schedule for greatest profit, assuming that he can sell all that he manufactures?

10. A blender of whisky imports three grades A, B and C. He mixes them according to the recipes that specify the maximum or minimum percentages of grades A and C in each blend. These are shown in the table below.

Blend	*Specification*	*Price per Unit* (*Rs*)
Blue Dot	Not less than 60% of A Not more than 20% of C	6.80
Highland Fling	Not more than 60% of C Not less than 15% of A	5.70
Old Frenzy	Not more than 50% of C	4.50

Following are the supplies of the three whiskies along with their cost.

Whisky	*Maximum Quantity Available per Day*	*Cost per Unit* (*Rs*)
A	2,000	7.00
B	2,500	5.00
C	1,200	4.00

Show how to obtain the first matrix in a simplex computation of a production policy that will maximize profits.

11. An animal feed company must produce on a daily basis 200 kg of a mixture that consists ingredients x_1 and x_2 ingredient. x_1 costs Rs 3 per kg and x_2 costs Rs 8 per kg. Not more than 80 kg of x_1 can be used and at least 60 kg of x_2 must be used. Find out how much of each ingredient should be used if the company wants to minimize costs.

12. A diet is to contain at least 20 ounces of protein and 15 ounces of carbohydrate. There are three foods A, B and C available in the market, costing Rs 2, Re 1 and Rs 3 per unit, respectively. Each unit of A contains 2 ounces of protein and 4 ounces of carbohydrate; each unit of B contains 3 ounces of protein and 2 ounces of carbohydrate; and each unit of C contains 4 ounces of protein and 2 ounces of carbohydrate. How many units of each food should the diet contain so that the cost per unit diet is minimum?

13. A person requires 10, 12 and 12 units of chemicals A, B and C, respectively for his garden. A typical liquid product contains 5, 2 and 1 unit of A, B and C, respectively per jar. On the other hand a typical dry product contains 1, 2 and 4 units of A, B and C per unit. If the liquid product sells for Rs 3 per jar and the dry product for Rs 2 per carton, how many of each should be purchased in order to minimize the cost and meet the requirement?

14. A scrap metal dealer has received an order from a customer for a minimum of 2,000 kg of scrap metal. The customer requires that at least 1,000 kg of the shipment of metal be of high quality copper that can be melted down and used to produce copper tubing. Furthermore, the customer will not accept delivery of the order if it contains more than 175 kg metal that he deems unfit for commercial use, i.e. metal that contains an excessive amount of impurities and cannot be melted down and refined profitably.

    The dealer can purchase scrap metal from two different suppliers in unlimited quantities with following percentages (by weight) of high quality copper and unfit scrap.

	Supplier A	*Supplier B*
Copper	25%	75%
Unfit scrap	5%	10%

The cost per kg of metal purchased from supplier A and B are Re 1 and Rs 4, respectively. Determine the optimal quantities of metal to be purchased for the dealer from each of the two suppliers.

15. A marketing manager wishes to allocate his annual advertising budget of Rs 2,00,000 to two media vehicles – A and B. The unit cost of a message in media A is Rs 1,000 and that of B is Rs 1,500. Media A is a monthly magazine and not more than one insertion is desired in one issue, whereas at least five messages should appear in media B. The expected audience for unit messages in media A is 40,000 and that of media B is 55,000. Develop an LP model and solve it for maximizing the total effective audience.

16. A transistor radio company manufactures four models A, B, C and D. Models A, B and C, have profit contributions of Rs 8, Rs 15 and Rs 25 respectively and has model D a loss of Re 1. Each type of radio requires a certain amount of time for the manufacturing of components, for assembling and for packing. A dozen units of model A require one hour for manufacturing, two hours for assembling and one hour for packing. The corresponding figures for a dozen units of model B are 2, 1 and 2, and for a dozen units of C are 3, 5 and 1. A dozen units of model D however, only require 1 hour of packing. During the forthcoming week, the company will be able to make available 15 hours of manufacturing, 20 hours of assembling and 10 hours of packing time. Determine the optimal production schedule for the company.

17. A transport company is considering the purchase of new vehicles for providing transportation between the Delhi Airport and hotels in the city. There are three vehicles under consideration: Station wagons, minibuses and large buses. The purchase price would be Rs 1,45,000 for each station wagon, Rs 2,50,000 for each minibus and Rs 4,00,000 for each large bus. The board of directors has authorized a maximum amount of Rs 50,00,000 for these purchases. Because of the heavy air travel, the new vehicles would be utilized at maximum capacity, regardless of the type of vehicles purchased. The expected net annual profit would be Rs 15,000 for the station wagon, Rs 35,000 for the minibus and Rs 45,000 for the large bus. The company has hired 30 new drivers for the new vehicles. They are qualified drivers for all three types of vehicles. The maintenance department has the capacity to handle an additional 80 station wagons. A minibus is equivalent to 1.67 station wagons and each large bus is equivalent to 2 station wagons in terms of their use of the maintenance department. Determine the number of each type of vehicle that should be purchased in order to maximize profit.

18. Omega Data Processing Company performs three types of activities: Payroll, accounts receivables, and inventories. The profit and time requirement for keypunch, computation and office printing, for a standard job, are shown in the following table:

Job	*Profit/ Standard Job* (*Rs*)	*Time Requirement* (*Min*)		
		Keypunch	*Computation*	*Printing*
Payroll	275	1,200	20	100
A/c Receivable	125	1,400	15	60
Inventory	225	800	35	80

Omega guarantees overnight completion of the job. Any job scheduled during the day can be completed during the day or night. Any job scheduled during the night, however, must be completed during the night. The capacities for both day and night are shown in the following table:

Capacity (*Min*)	*Keypunch*	*Computation*	*Print*
Day	4,200	150	400
Night	9,200	250	650

Determine the mixture of standard jobs that should be accepted during the day and night.

19. A furniture company can produce four types of chairs. Each chair is first made in the carpentry shop and then furnished, waxed and polished in the finishing shop. The man-hours required in each are:

	Chair Type			
	1	*2*	*3*	*4*
Carpentry shop	4	9	7	10
Finishing shop	1	1	3	40
Contribution per chair (Rs)	12	20	18	40

The total number of man-hours available per month in carpentry and finishing shops are 6,000 and 4,000, respectively.

Assuming an abundant supply of raw material and an abundant demand for finished products, determine the number of each type of chairs that should be produced for profit maximization.

20. A metal products company produces waste cans, filing cabinets, file boxes for correspondence, and lunch boxes. Its inputs are sheet metal of two different thickness, called A and B, and manual labour. The input-output relationship for the company are shown in the table given below:

	Waste Cans	*Filing Cabinets*	*Correspondence Boxes*	*Lunch Boxes*
Sheet metal A	6	0	2	3
Sheet metal B	0	10	0	0
Manual labour	4	8	2	3

The sales revenue per unit of waste cans, filing cabinets, correspondence boxes and lunch boxes are Rs 20, Rs 400, Rs 90 and Rs 20, respectively. There are 225 units of sheet metal A available in the company's inventory, 300 of sheet metal B, and a total of 190 units of manual labour. What is the company's optimal sales revenue?

[*Delhi Univ., MBA, 2003, 2005*]

21. A company mined diamonds in three locations in the country. The three mines differed in terms of their capacities, number, weight of stones mined, and costs. These are shown in the table below:

Due to marketing considerations, a monthly production of exactly 1,48,000 stones was required. A similar requirement called for at least 1,30,000 carats (The average stone size was at least 130/148 = 0.88 carats). The capacity of each mine is measured in cubic meter. The mining costs are not included from the treatment costs and assume to be same at each mine. The problem for the company was to meet the marketing requirements at the least cost.

Mine	*Capacity (M^3 of earth processed)*	*Treatment Costs (Rs. per M^3)*	*Crude (Carats per M^3)*	*Stone Count (Number of stone per M^3)*
Plant 1	83,000	0.60	0.360	0.58
Plant 2	3,10,000	0.36	0.220	0.26
Plant 3	1,90,000	0.50	0.263	0.21

Formulate a linear programming model to determine how much should be mined at each location.

HINTS AND ANSWERS

1. Let x_1 and x_2 = units of models A and B to be manufactured, respectively.

$$\text{Max } Z = 400x_1 + 400x_2$$
$$\text{subject to } 4x_1 + 2x_2 \le 1{,}600$$
$$5x_1/2 + x_2 \le 1{,}200$$
$$9x_1/2 + 3x_2/2 \le 1{,}600$$
$$\text{and } x_1, x_2 \ge 0$$

Ans. $x_1 = 355.5$, $x_2 = 0$ and Max Z = 1,42,222.2.

2. Let x_1, x_2 and x_3 = units of types A, B and C belt to be manufactured, respectively.

$$\text{Max } Z = 3x_1 + 5x_2 + 4x_3$$
$$\text{subject to } 2x_1 + 3x_2 \le 8$$
$$2x_2 + 5x_3 \le 10$$
$$3x_1 + 2x_2 + 4x_3 \le 15$$
$$\text{and } x_1, x_2, x_3 \ge 0$$

Ans. $x_1 = 89/41$, $x_2 = 50/41$ $x_3 = 64/41$ and Max Z = 775/41.

3. Let x_1, x_2 and x_3 = quantity of products A, B and C to be produced, respectively.

$$\text{Max } Z = x_1 + 4x_2 + 5x_3$$
$$\text{subject to } 3x_1 + 3x_2 \le 22$$
$$x_1 + 2x_2 + 3x_3 \le 14$$
$$3x_1 + 2x_2 \le 14$$
$$\text{and } x_1, x_2, x_3 \ge 0$$

Ans. $x_1 = 0$, $x_2 = 7$, $x_3 = 0$ and Max Z = Rs 28,000.

4. Let x_1, x_2 and x_3 = number of units of products 1, 2 and 3 to be produced per week, respectively.

$$\text{Max } Z = 20x_1 + 6x_2 + 8x_3$$
$$\text{subject to } 8x_1 + 2x_2 + 3x_3 \le 250$$
$$4x_1 + 3x_2 \le 150$$
$$2x_1 + x_3 \le 50$$
$$\text{and } x_1, x_2, x_3 \ge 0$$

Ans. $x_1 = 0$, $x_2 = 50$, $x_3 = 50$ and Max Z = 700.

5. Let x_1, x_2 and x_3 = acreage of corn, wheat and soyabean, respectively.

$$\text{Max } Z = 30x_1 + 40x_2 + 20x_3$$
$$\text{subject to } 10x_1 + 12x_2 + 7x_3 \le 10{,}000$$
$$7x_1 + 10x_2 + 8x_3 \le 8{,}000$$
$$x_1 + x_2 + x_3 \le 1{,}000$$
$$\text{and } x_1, x_2, x_3 \ge 0$$

Ans. $x_1 = 250$, $x_2 = 625$, $x_3 = 0$ and Max Z = Rs 32,500.

6. Let x_1, x_2 and x_3 = number of units of chair of styles A, B and C, respectively.

$$\text{Max } Z = 40x_1 + 35x_2 + 30x_3$$
$$\text{subject to } 2x_1 + 3x_2 + 2x_3 \le 120$$
$$4x_1 + 3x_2 + x_3 \le 160$$
$$3x_1 + 2x_2 + 4x_3 \le 100$$
$$x_1 + x_2 + x_3 \le 40$$
$$\text{and } x_1, x_2, x_3 \ge 0$$

Ans. $x_1 = 20$, $x_2 = 20$, $x_3 = 0$ and Max Z = Rs 1,500.

7. Let x_1, x_2 and x_3 = salesman assigned to area, 1, 2 and 3, respectively.

Max. $Z = 40 \times 150x_1 + 36 \times 150x_2 + 25 \times 150x_3 - (800x_1 + 700x_2 + 500x_3)$

subject to (i) $x_1 + x_2 + x_3 \le 12$; (ii) $x_1 \le 5$;
(iii) $800x_1 + 700x_2 + 500x_3 \le 7{,}500$

and $x_1, x_2, x_3 \ge 0$

11. Let x and y = number of kg of ingredients x_1, x_2, respectively.

Min (total cost) $Z = 3x + 8y$

subject to (i) $x + y = 200$; (ii) $x \le 80$; (iii) $y \ge 60$

and $x, y \ge 0$

Ans. $x = 80$, $y = 120$ and Min Z = Rs 1,200.

12. Let x_1, x_2 and x_3 = number of units of food A, B and C, respectively which a diet must contain.

Min (total cost) $Z = 2x_1 + x_2 + x_3$

subject to $2x_1 + 2x_2 + 4x_3 \ge 20$
$4x_1 + 2x_2 + 2x_3 \ge 15$

and $x_1, x_2, x_3 \ge 0$

13. Let x_1 and x_2 = number of units of liquid and dry product produced, respectively.

Min (total cost) $Z = 3x_1 + 2x_2$

subject to (i) $5x_1 + x_2 \ge 10$; (ii) $2x_1 + 2x_2 \ge 12$;
(iii) $x_1 + 4x_2 \ge 12$

and $x_1, x_2 \ge 0$

Ans. $x_1 = 1$, $x_2 = 5$ and Min $Z = 13$.

14. Let x_1 and x_2 = number of scrap (in kg) purchased from suppliers A and B, respectively.

Min (total cost) $Z = x_1 + 4x_2$

subject to $2.25x_1 + 0.75x_2 \ge 1{,}000$
$0.05x_1 + 0.10x_2 \ge 175$
$x_1 + x_2 \ge 2{,}000$

and $x_1, x_2 \ge 0$

Ans. $x_1 = 2{,}500$, $x_2 = 500$ and Min $Z = 4{,}500$.

15. Let x_1 and x_2 = number of insertions messages for media A and B, respectively.

Min (total effective audience) $Z = 40{,}000x_1 + 55{,}000x_2$

subject to (i) $1{,}000x_1 + 1500x_2 \le 2{,}00{,}000$;
(ii) $x_1 \le 12$; (iii) $x_2 \ge 5$

and $x_1, x_2 \ge 0$

Ans. $x_1 = 12$, $x_2 = 16/3$ and Max $Z = 77{,}3333.33$.

16. Let x_1, x_2, x_3 and x_4 = unit of models A, B, C and D to be produce, respectively.

Max (total income) $Z = 8x_1 + 15x_2 + 25x_3 - x_4$

subject to (i) $x_1 + 2x_2 + 3x_3 = 15$; (ii) $2x_1 + x_2 + 5x_3 = 20$
(iii) $x_1 + 2x_2 + x_3 + x_4 = 10$

and $x_1, x_2, x_3, x_4 \ge 0$

Ans. $x_1 = 5/2$, $x_2 = 5/2$, $x_3 = 5/2$, $x_4 = 0$ and Max $Z = 120$.

17. Let x_1, x_2 and x_3 = number of station wagons, minibuses and large buses to be purchased, respectively.

Max $Z = 15{,}000x_1 + 35{,}000x_2 + 45{,}000x_3$

subject to $x_1 + x_2 + x_3 \le 30$
$1{,}45{,}000x_1 + 2{,}50{,}000x_2 + 4{,}00{,}000x_3 \le 50{,}00{,}000$
$x_1 + 1.67x_2 + 0.5x_3 \le 80$

and $x_1, x_2, x_3 \ge 0$

18. Let x_{ij} represents ith job and jth activity

Max $Z = 275\,(x_{11} + x_{12}) + 125(x_{21} + x_{22}) + 225\,(x_{31} + x_{32})$

subject to

$1{,}200\,(x_{11} + x_{12}) + 1{,}400\,(x_{21} + x_{22}) + 800\,(x_{31} + x_{32}) \le 13{,}400$
$20\,(x_{11} + x_{12}) + 15\,(x_{21} + x_{22}) + 35\,(x_{31} + x_{32}) \le 400$
$100\,(x_{11} + x_{12}) + 60\,(x_{21} + x_{22}) + 80\,(x_{31} + x_{32}) \le 1{,}050$
$1{,}200x_{12} + 1{,}400x_{22} + 800x_{32} \le 9{,}200$
$20x_{12} + 15x_{22} + 35x_{32} \le 250$
$100x_{12} + 60x_{22} + 80x_{32} \le 650$

and $x_{ij} \ge 0$ for all i, j

19. Let x_1, x_2, x_3 and x_4 = chair types 1, 2, 3 and 4 to be produced, respectively.

Max $Z = 12x_1 + 20x_2 + 18x_3 + 40x_4$

subject to $4x_1 + 9x_2 + 7x_3 + 10x_4 \le 6{,}000$
$x_1 + x_2 + 3x_3 + 40x_4 \le 4{,}000$

and $x_1, x_2, x_3, x_4 \ge 0$

Ans. $x_1 = 4{,}000/3$, $x_2 = x_3 = 0$, $x_4 = 200/3$ and Max Z = Rs 56,000/3.

21. Let x_1, x_2 and x_3 = cubic metric of earth processed at plants 1, 2 and 3, respectively.

Min $Z = 0.60x_1 + 0.36x_2 + 0.50x_3$

subject to

$0.58x_1 + 0.26x_2 + 0.21x_3 = 1{,}48{,}000$ (Stone count requirement)

$0.36x_1 + 0.22x_2 + 0.263x_3 \le 1{,}30.000$ (Carat requirement)

$x_1 \le 83{,}000$; $x_2 \le 3{,}10{,}000$;

$x_3 \le 1{,}90{,}000$ (Capacity requirement)

Ans. $x_1 = 61{,}700$; $x_2 = 3{,}10{,}000$; $x_3 = 1{,}50{,}500$ and Min Z = Rs 2,23,880.

4.5 SOME COMPLICATIONS AND THEIR RESOLUTION

In this section, some of the complications that may arise in applying the simplex method for solving both maximization and minimization LP problems and their resolution are discussed.

4.5.1 Unrestricted Variables

In actual practice, decision variables, x_j ($j = 1, 2, \ldots, n$) should have non-negative values. However, in many situations, one or more of these variables may have either positive, negative or zero value. Variables that can

assume positive, negative or zero value are called *unrestricted variables.* Since use of the simplex method requires that all the decision variables must have non-negative value at each iteration, therefore in order to convert an LP problem involving unrestricted variables into an equivalent LP problem having only non-negative variables, each of unrestricted variable is expressed as the difference of two non-negative variables.

Let variable x_r be unrestricted in sign. We define two new variables say x_r' and x_r'' such that

$$x_r = x_r' - x_r''; \qquad x_r', x_r'' \geq 0$$

If $x_r' \geq x_r''$, then $x_r \geq 0$, but and if $x_r' \leq x_r''$, then $x_r \leq 0$. Also if $x_r' = x_r''$, then $x_r = 0$. Hence depending on the value of x_r' and x_r'', the variable x_r can have either positive or negative sign. For example, the following LP problem

An unrestricted variable in an LP model can have either positive, negative or zero value.

$$\text{Maximize } Z = \sum_{j \neq r}^{n} c_j x_j + c_r x_r$$

subject to the constraints

$$\sum_{j \neq r}^{n} a_{ij} x_j + a_{ir} x_r = b_i, \quad i = 1, 2, \ldots, m$$

and $x_j \geq 0$, and x_r unrestricted in sign; $j = 1, 2, \ldots, n, \quad j \neq r$;

can be converted into its equivalent standard form as follows:

$$\text{Maximize } Z = \sum_{j \neq r}^{n} c_j x_j + c_r (x_r' + x_r'')$$

subject to the constraints

$$\sum_{j \neq r}^{n} a_{ij} x_j + a_{ir} (x_r' - x_r'') = b_i, \quad i = 1, 2, \ldots, m$$

and $x_j, x_r', x_r'' \geq 0; \quad j = 1, 2, \ldots, n, \quad j \neq r.$

Variables x_r' and x_r'' simultaneously cannot appear in the basis (since the column vectors corresponding to these variables are linearly dependent). Thus any of the following three cases may arise at the optimal solution:

(i) $x_r' = 0 \Rightarrow x_r = -x_r''$ (ii) $x_r'' = 0 \Rightarrow x_r = x_r'$ (iii) $x_r' = x_r'' = 0 \Rightarrow x_r = 0$

This indicates that the value of x_r' and x_r'' uniquely determines the values of the variable x_r.

Example 4.12 Use the simplex method to solve the following LP problem.

Maximize $Z = 3x_1 + 2x_2 + x_3$

subject to the constraints

(i) $2x_1 + 5x_2 + x_3 = 12$, (ii) $3x_1 + 4x_2 = 11$

and $x_2, x_3 \geq 0$, and x_1 unrestricted [*Bombay, BSc (Maths), 2001*]

Solution Introducing an artificial variable A_1 in the second constraint of the given LP problem in order to obtain the basis matrix as shown in Table 4.43. Since x_1 is unrestricted in sign, introducing the non-negative variables x_1' and x_2'' so that $x_1 = x_1' - x_1''$, where $x_1', \quad x_1'' \geq 0$. The standard form of the LP problem now becomes:

Maximize $Z = 3(x_1' - x_1'') + 2x_2 + x_3 - MA_1$

subject to the constraints

(i) $2(x_1' - x_1'') + 5x_2 + x_3 = 12$, (ii) $3(x_1' - x_1'') + 4x_2 + A_1 = 11$

and $x_1', x_1'', x_2, x_3, A_1 \geq 0$

The initial solution is shown in Table 4.43.

		$c_j \rightarrow$	3	– 3	2	1	– M	
Basic Variables Coefficient	Basic Variables **B**	Basic Variables Value **b**(= $\boldsymbol{x_B}$)	x_1'	x_1''	x_2	x_3	A_1	Min Ratio $\boldsymbol{x_B}/x_2$
1	x_3	12	2	– 2	(5)	1	0	12/5 →
– M	A_1	11	3	– 3	4	0	1	11/4
$Z = -11M + 12$		z_j	$-3M+2$	$3M-2$	$-4M+5$	1	– M	
		$c_j - z_j$	$3M+1$	$-3M-1$	$4M-3$ ↑	0	0	

Table 4.43 Initial Solution

In Table 4.43, $c_3 - z_3$ is the largest positive value, the non-basic variable x_2 is chosen to enter into the basis to replace basic variable x_3 in the basis. For this, apply the following row operations:

R_1 (new) $\rightarrow R_1$ (old)/5 (*key element*) $\quad R_2$ (new) $\rightarrow R_2$ (old) – $4R_1$ (new).

to get an improved solution shown in Table 4.44.

		$c_j \rightarrow$	3	– 3	2	1	– M	
Basic Variables Coefficient	Basic Variables **B**	Basic Variables Value **b**(= $\boldsymbol{x_B}$)	x_1'	x_1''	x_2	x_3	A_1	Min Ratio $\boldsymbol{x_B}/x_1'$
2	x_2	12/5	2/5	–2/5	1	1/5	0	$\frac{12}{5} \times \frac{5}{2} = 6$
– M	A_1	7/5	(7/5)	–7/5	0	– 4/5	1	$\frac{7}{5} \times \frac{5}{7} = 1$ →
$Z = -7M/5 + 24/5$		z_j	$-7M/5+4/5$	$7M/5-4/5$	2	$4M/5+2/5$	– M	
		$c_j - z_j$	$7M/5+11/5$ ↑	$-7M/5-11/5$	0	$-4M/5+3/5$	0	

Table 4.44 Improved Solution

In Table 4.44, as $c_1 - z_1$ in x'_1-column is positive, the non-basic variable x'_1 is chosen to enter into the basis to replace basic variable A_1 in the basis. For this, apply the following row operations:

R_2 (new) $\rightarrow R_2$ (old) × (5/7) (*key element*); $\quad R_1$ (new) $\rightarrow R_1$ (old) – (2/5) R_2 (new).

to get an improved solution shown in Table 4.45.

		$c_j \rightarrow$	3	– 3	2	1	
Basic Variables Coefficient	Basic Variables **B**	Basic Variables Value **b**(= $\boldsymbol{x_B}$)	x_1'	x_1''	x_2	x_3	Min Ratio $\boldsymbol{x_B}/x_3$
2	x_2	2	0	0	1	(3/7)	2/(3/7) = 14/3 →
3	x_1'	1	1	–1	0	– 4/7	—
$Z = 7$		z_j	3	– 3	2	– 6/7	
		$c_j - z_j$	0	0	0	13/7 ↑	

Table 4.45 Improved Solution

Further, in order to improve the solution given in Table 4.45, the non-basic variable x_3 is chosen to enter into the basis and remove basic variable x_2 from the basis. For this, apply the following row operations:

R_1 (new) $\rightarrow R_1$ (old) × (7/3) (*key element*); $\quad R_2$ (new) $\rightarrow R_2$ (old) + (4/7) R_1 (new)

to get an improved solution given in Table 4.46.

Table 4.46
Optimal Solution

		$c_j \rightarrow$	3	– 3	2	1
Basic Variables Coefficient	*Basic Variables* $\boldsymbol{B}$	*Basic Variables Value* $\boldsymbol{b}(= \boldsymbol{x_B})$	x_1'	x_1''	x_2	x_3
1	x_3	14/3	0	0	7/3	1
3	x_1'	11/3	1	– 1	4/3	0
$Z = 47/3$		z_j	3	– 3	19/3	1
		$c_j - z_j$	0	0	– 13/3	0

Since in Table 4.46 all $c_j - z_j \leq 0$, an optimal solution: $x_1' = 11/3$ or $x_1 = x_1' - x_1'' = 11/3 - 0 = 11/3$; $x_3 = 14/3$ is arrived with Max $Z = 47/3$.

4.5.2 Tie for Entering Basic Variable (Key Column)

While solving an LP problem using simplex method two or more columns of simplex table may have same $c_j - z_j$ value (positive or negative depending upon the type of LP problem). In order to break this tie, the selection for key column (entering variable) can be made arbitrary. However, the number of iterations required to arrive at the optimal solution can be minimized by adopting the following rules:

(i) If there is a tie between two decision variables, then the selection can be made arbitrarily.

(ii) If there is a tie between a decision variable and a slack (or surplus) variable, then select the decision variable to enter into basis.

(iii) If there is a tie between two slack (or surplus) variables, then the selection can be made arbitrarily.

4.5.3 Tie for Leaving Basic Variable (Key Row) – Degeneracy

While solving an LP problem a situation may arise where either the minimum ratio to identify the basic variable to leave the basis is not unique or value of one or more basic variables in the $\boldsymbol{x_B}$ becomes zero. This causes the problem of degeneracy.

Usually, the problem of degeneracy arises due to redundant constraint, i.e. one or more of the constraints of LP problem are not part of feasible solution space. For example, constraints such as $x_1 \leq 5$, $x_2 \leq 5$ and $x_1 + x_2 \leq 5$ in the LP problem make constraint $x_1 + x_2 \leq 5$ unnecessary (redundant).

Degeneracy may occur at any iteration of the simplex method. In order to break the tie in the minimum ratios, the selection can be made arbitrarily. However, the number of iterations required to arrive at the optimal solution can be minimized by adopting the following rules.

(i) Divide the coefficients of slack variables in the simplex table where degeneracy is seen by the corresponding positive numbers of the key column in the row, starting from left to right.

(ii) Compare the ratios in step (i) from left to right columnwise, select the row that contains the smallest ratio.

Remark When there is a tie between a slack and artificial variable to leave the basis, preference should be given to the artificial variable for leaving the basis.

Example 4.13 Solve the following LP problem

Maximize $Z = 3x_1 + 9x_2$

subject to the constraints

(i) $x_1 + 4x_2 \leq 8$, (ii) $x_1 + 2x_2 \leq 4$

and $x_1, x_2 \geq 0$

Solution Adding slack variables s_1 and s_2 to the constraints, the problem can be expressed as

Maximize $Z = 3x_1 + 9x_2 + 0s_2 + 0s_2$

subject to the constraints

(i) $x_1 + 4x_2 + s_1 = 8$, (ii) $x_1 + 2x_2 + s_2 = 4$

and $x_1, x_2, s_1, s_2 \geq 0$

The initial basic feasible solution is given in Table 4.47. Since $c_2 - z_2 = 9$ is the largest positive value, therefore, variable x_2 is selected to be entered into the basis. However, both variables s_1 and s_2 are eligible to leave the basis as the minimum ratio is same, i.e. 2, so there is a tie among the ratio in rows s_1 and s_2. This causes the problem of degeneracy. To obtain the key row for resolving degeneracy, apply the following procedure:

(i) Write the coefficients of the slack variables as shown in Table 4.47.

		$c_j \rightarrow$	3	9	0	0	
Basic Variables Coefficient	*Basic Variables* $\boldsymbol{B}$	*Basic Variables Value* $\boldsymbol{b}(= \boldsymbol{x_B})$	x_1	x_2	s_1	s_2	*Min Ratio* $\boldsymbol{x_B}/x_2$
0	s_1	8	1	4	1	0	8/4 = 2
0	s_2	4	1	(2)	0	1	4/2 = 2
$Z = 0$		z_j	0	0	0	0	
		$c_j - z_j$	3	9 ↑	0	0	

Table 4.47 Initial Solution

Row	x_2-*column (Key Column)*	*Column* s_1	*Column* s_2
s_1	4	1	0
s_2	2	0	1

(ii) Dividing the coefficients of slack variables s_1 and s_2 by the corresponding elements in the key column as shown below in the table.

Row	x_2-*column (Key Column)*	*Column* s_1	*Column* s_2
s_1	4	1/4 = 1/4	0/4 = 0
s_2	2	0/2 = 0	1/2 = 1/2

(iii) Comparing the ratios of Step (ii) from left to right columnwise, the minimum ratio (i.e., 0/2 = 0) occurs in the s_2-row. Thus, variable s_2 is selected to leave the basis. The new solution is shown in Table 4.48.

		$c_j \rightarrow$	3	9	0	0
Basic Variables Coefficient	*Basic Variables* $\boldsymbol{B}$	*Basic Variables Value* $\boldsymbol{b}(= \boldsymbol{x}_B)$	x_1	x_2	s_1	s_2
0	s_1	0	− 1	0	1	− 2
9	x_2	2	1/2	1	0	1/2
$Z = 18$		z_j	9/2	9	0	9/2
		$c_j - z_j$	− 3/2	0	0	− 9/2

Table 4.48 Optimal Solution

In Table 4.48, all $c_j - z_j \leq 0$. Hence, an optimal solution is arrived at. The optimal basic feasible solution is: $x_1 = 0$, $x_2 = 2$ and Max $Z = 18$.

4.6 TYPES OF LINEAR PROGRAMMING SOLUTIONS

While solving any LP problem using simplex method, at the stage of optimal solution, the following three types of solutions may exist:

4.6.1 Alternative (Multiple) Optimal Solutions

The $c_j - z_j$ values in the simplex table indicates the contribution in the objective function value by each unit of a variable chosen to enter into the basis. Also, an optimal solution to a maximization LP problem is reached when all $c_j - z_j \leq 0$. But, if $c_j - z_j = 0$ for a non-basic variable column in the optimal simplex table and such non-basic variable is chosen to enter into the basis, then another optimal solution so obtained will show no improvement in the value of objective function.

Alternative optimal solutions arise when $c_j - z_j = 0$ for non-basic variable columns in the simplex table.

Example 4.14 Solve the following LP problem.

Maximize $Z = 6x_1 + 4x_2$

subject to the constraints

(i) $2x_1 + 3x_2 \leq 30$, (ii) $3x_1 + 2x_2 \leq 24$, (iii) $x_1 + x_2 \geq 3$

and $x_1, x_2 \geq 0$.

Solution Adding slack variables s_1, s_2, surplus variable s_3 and artificial variable A_1 in the constraint set, the LP problem becomes

Maximize $Z = 6x_1 + 4x_2 + 0s_1 + 0s_2 + 0s_3 - MA_1$

subject to the constraints

(i) $2x_1 + 3x_2 + s_1 = 30$, (ii) $3x_1 + 2x_2 + s_2 = 24$ (iii) $x_1 + x_2 - s_3 + A_1 = 3$

and $x_1, x_2, s_1, s_2, s_3, A_1 \geq 0$

The optimal solution: $x_1 = 8$, $x_2 = 0$ and Max $Z = 48$ for this LP problem is shown in Table 4.49.

Table 4.49 Optimal Solution

		$c_j \rightarrow$	6	4	0	0	0	
Basic Variables Coefficient	*Basic Variables* $\boldsymbol{B}$	*Basic Variables Value* $\boldsymbol{b}(= \boldsymbol{x_B})$	x_1	x_2	s_1	s_2	s_3	*Min Ratio* $\boldsymbol{x_B}/x_2$
0	s_1	14	0	(5/3)	1	− 2/3	0	14/(15/3) = 42/5 →
0	s_3	5	0	− 1/3	0	1/3	1	—
6	x_1	8	1	2/3	0	1/3	0	8/(2/3) = 12
Z = 48		z_j	6	4	0	2	0	
		$c_j - z_j$	0	0 ↑	0	− 2	0	

In Table 4.49, $c_2 - z_2 = 0$ corresponds to a non-basic variable, x_2. Thus, an alternative optimal solution can also be obtained by entering variable x_2 into the basis and removing basic variable, s_1 from the basis. The new solution is shown in Table 4.50.

Table 4.50 Alternative Solution

		$c_j \rightarrow$	6	4	0	0	0
Basic Variables Coefficient	*Basic Variables* $\boldsymbol{B}$	*Basic Variables Value* $\boldsymbol{b}(= \boldsymbol{x_B})$	x_1	x_2	s_1	s_2	s_3
4	x_2	42/5	0	1	3/5	− 2/5	0
0	s_3	39/5	0	0	1/5	1/5	1
6	x_1	12/5	1	0	− 2/5	3/5	0
Z = 48		z_j	6	4	0	2	0
		$c_j - z_j$	0	0	0	− 2	0

The optimal solution shown in Table 4.50 is: $x_1 = 12/5$, $x_2 = 42/5$ and Max $Z = 48$. Since this optimal solution shows no change in the value of objective function, it is an alternative solution.

Once again, $c_3 - z_3 = 0$ corresponds to non-basic variable, s_1. This again indicates that an alternative optimal solution exists. The infinite number of solutions that can be obtained for this LP problem are as follows:

Variables	*Solution Values*		*General Solution*
	1	*2*	
x_1	8	12/5	$x_1 = 8\lambda + (12/5)(1-\lambda)$
x_2	0	42/5	$x_2 = 0\lambda + (42/5)(1-\lambda)$
s_1	14	0	$s_1 = 14\lambda + (0)(1-\lambda)$
s_3	5	39/5	$s_3 = 5\lambda + (39/5)(1-\lambda)$

For each arbitrary value of λ, the value of objective function will remain same.

4.6.2 Unbounded Solution

In a maximization LP problem, if $c_j - z_j > 0$ ($c_j - z_j < 0$ for a minimization case) corresponds to a non-basic variable column in simplex table, and all a_{ij} values in this column are negative, then minimum ratio to decide basic variable to leave the basis can not be calculated. It is because negative value in denominator would indicate the entry of a non-basic variable in the basis with a negative value (an infeasible solution). Also, a zero value in the denominator would result in a ratio having an infinite value and would indicate that the value of non-basic variable could be increased infinitely with any of the current basic variables being removed from the basis.

Example 4.15 Solve the following LP problem.

Maximize $Z = 3x_1 + 5x_2$

subject to the constraints

(i) $x_1 - 2x_2 \le 6$, (ii) $x_1 \le 10$, (iii) $x_2 \ge 1$

and $x_1, x_2 \ge 0$

Solution Adding slack variables s_1, s_2, surplus variable s_3 and artificial variable A_1 in the constraint set. Then the standard form of LP problem becomes

Maximize $Z = 3x_1 + 5x_2 + 0s_1 + 0s_2 + 0s_3 - MA_1$

subject to the constraints

(i) $x_1 - 2x_2 + s_1 = 6$, (ii) $x_1 + s_2 = 10$, (iii) $x_2 - s_3 + A_1 = 1$

and $x_1, x_2, s_1, s_2, s_3, A_1 \ge 0$

The initial solution to this LP problem is shown in Table 4.51.

Unbounded solution occurs when value of decision variables in the solution of LP problem becomes infinitely large without violating any given constraints.

		$c_j \rightarrow$	*3*	*5*	*0*	*0*	*0*	*−M*	
Basic Variables Coefficient	*Basic Variables* **B**	*Basic Variables Value* **b** (= x_B)	x_1	x_2	s_1	s_2	s_3	A_1	*Min Ratio* x_B/x_2
0	s_1	6	1	−2	1	0	0	0	–
0	s_2	10	1	0	0	1	0	0	–
$-M$	A_1	1	0	(1)	0	0	−1	1	1 →
$Z = -M$		z_j	0	$-M$	0	0	M	$-M$	
		$c_j - z_j$	3	$5 + M$ ↑	0	0	$-M$	0	

Table 4.51 Initial Solution

Iteration 1: Since $c_2 - z_2 \ge 0$, non-basic variable x_2 is chosen to enter into the basis in place of basic variable A_1. The new solution is shown in Table 4.52.

		$c_j \rightarrow$	3	5	0	0	0	$-M$
Basic Variables Coefficient	Basic Variables $\boldsymbol{B}$	Basic Variables Value $\boldsymbol{b}$ $(= \boldsymbol{x_B})$	x_1	x_2	s_1	s_2	s_3	A_1
0	s_1	8	1	0	1	0	-2	2
0	s_2	10	1	0	0	1	0	0
5	x_2	1	0	1	0	0	-1	1
$Z = 5$		z_j	0	5	0	0	-5	5
		$c_j - z_j$	3	0	0	0	5	$-M-5$

Table 4.52

In Table 4.52, $c_5 - z_5 = 5$ is largest positive, so variable s_3 should enter into the basis. But, coefficients in the 's_3' column are all negative or zero. This indicates that s_3 cannot be entered into the basis. However, the value of s_3 can be increased infinitely without removing any one of the basic variables. Further, since s_3 is associated with x_2 in the third constraint, x_2 will also be increased infinitely because it can be expressed as $x_2 = 1 + s_3 - A_1$. Hence, the solution to the given problem is unbounded.

4.6.3 Infeasible Solution

If LP problem solution does not satisfy all of the constraints, then such a solution is called *infeasible solution*. Also, infeasible solution occurs when all $c_j - z_j$ values satisfy optimal solution condition but at least one of artificial variables appears in the basis with a positive value. This situation may occur when an LP model is either improperly formulated or more than two of the constraints are incompatible.

Example 4.16 Solve the following LP problem

Maximize $Z = 6x_1 + 4x_2$

subject to the constraints

(i) $x_1 + x_2 \leq 5$, (ii) $x_2 \geq 8$

and $x_1, x_2 \geq 0$.

Solution Adding slack, surplus and artificial variables, the standard form of LP problem becomes

Maximize $Z = 6x_1 + 4x_2 + 0s_1 + 0s_2 - MA_1$

subject to the constraints

(i) $x_1 + x_2 + s_1 = 5$, (ii) $x_2 - s_2 + A_1 = 8$

and $x_1, x_2, s_1, s_2, A_1 \geq 0$

The initial solution to this LP problem is shown in Table 4.53.

		$c_j \rightarrow$	6	4	0	0	$-M$	
Basic Variables Coefficient	Basic Variables $\boldsymbol{B}$	Basic Variables Value $\boldsymbol{b}$ $(= \boldsymbol{x_B})$	x_1	x_2	s_1	s_2	A_1	Min Ratio $\boldsymbol{x_B}/x_2$
6	s_1	5	1	(1)	1	0	0	$5/1 \rightarrow$
$-M$	A_1	8	0	1	0	-1	1	8/1
$Z = 30 - 8M$		z_j	6	$6 - M$	0	M	$-M$	
		$c_j - z_j$	0	$-2 + M$ ↑	0	$-M$	0	

Table 4.53
Initial Solution

Iteration 1: Since $c_2 - z_2 = M - 2 (\geq 0)$, non-basic variable x_2 is chosen to enter into the basis to replace basic variable x_1. The new solution is shown in Table 4.58.

		$c_j \rightarrow$	6	4	0	0	$-M$
Basic Variables Coefficients	*Basic Variables* $\boldsymbol{B}$	*Basic Variables Values* $\boldsymbol{b}\ (= \boldsymbol{x_B})$	x_1	x_2	s_1	s_2	A_1
4	x_2	5	1	1	1	0	0
$-M$	A_1	3	-1	0	-1	-1	1
$Z = 20 - 3M$		z_j	$4 + M$	4	$4 + M$	M	$-M$
		$c_j - z_j$	$2 - M$	0	$-4 - M$	$-M$	0

Table 4.54 Optimal but Infeasible Solution

In table 5.44, since all $c_j - z_j \leq 0$, the current solution is optimal. But this solution is not feasible for the given LP problem because values of decision variables are: $x_1 = 0$ and $x_2 = 5$ violates second constraint, $x_2 \geq 8$. The presence of artificial variable $A_1 = 3$ in the solution also indicates that the optimal solution violates the second constraint ($x_2 \geq 8$) by 3 units.

CONCEPTUAL QUESTIONS

1. Define slack and surplus variables in a linear programming problem.
2. Explain the various steps of the simplex method involved in the computation of an optimum solution to a linear programming problem.
3. (a) Give outlines of the simplex method in linear programming.
   (b) What is simplex? Describe the simplex method of solving linear programming problem.
   (c) What do you understand by the term two-phase method of solving linear programming problem?
   (d) Outline the simplex method in linear programming. Why is it called so?
   (e) Explain the purpose and procedure of the simplex method.
4. What do you mean by an optimal basic feasible solution to a linear programming problem?
5. Given a general linear programming problem, explain how you would test whether a basic feasible solution is an optimal solution or not. How would you proceed to change the basic feasible solution in case it is not optimal?
6. Explain the meaning of basic feasible solution and degenerate solution in a linear programming problem.
7. Explain what is meant by the terms degeneracy and cycling in linear programming? How can these problems be resolved?
8. Explain the term artificial variables and its use in linear programming.
9. What are artificial variables? Why do we need them? Describe the two-phase method of solving an LP problem with artificial variables.
10. What is the significance of $c_j - z_j$ numbers in the simplex table? Interpret their economic significance in terms of marginal worth.
11. What is meant by the term opportunity cost? How is this concept used in designing a test for optimality?
12. How do the graphical and simplex methods of solving LP problems differ from each other? In what ways are they same?
13. How do maximization and minimization problems differ when applying the simplex method?
14. What is the reason behind the use of the minimum ratio test in selecting the key row? What might happen without it?
15. What conditions must exist in a simplex table to establish the existence of an alternative solution? No feasible solution? Unbounded solution? Degeneracy?

SELF PRACTICE PROBLEMS B

1. Solve the following LP problems using the simplex method.

(i) Max $Z = 3x_1 + 2x_2$
subject to $x_1 + x_2 \leq 4$
$x_1 - x_2 \leq 2$
and $x_1, x_2 \geq 0$

(ii) Max $Z = 5x_1 + 3x_2$
subject to $x_1 + x_2 \leq 2$
$5x_1 + 2x_2 \leq 10$
$3x_1 + 8x_2 \leq 12$
and $x_1, x_2 \geq 0$

(iii) Max $Z = 3x_1 + 2x_2 + 5x_3$
subject to $x_1 + 2x_2 + x_3 \leq 430$
$3x_1 + 2x_3 \leq 460$
$x_1 + 4x_3 \leq 420$
and $x_1, x_2, x_3 \geq 0$
[*Shivaji, MSc* (*Maths*), *1995*]

(iv) Max $Z = x_1 - 3x_2 + 2x_3$
subject to $3x_1 - x_2 + 3x_3 \leq 7$
$-2x_1 + 4x_2 \leq 12$
$-4x_1 + 3x_2 + 8x_3 \leq 10$
and $x_1, x_2, x_3 \geq 0$
[*Meerut, MSc* (*Maths*), *1997; Bhathiar MSc* (*Maths*), *1996*]

(v) Max $Z = 4x_1 + 5x_2 + 9x_3 + 11x_4$
subject to
$x_1 + x_2 + x_3 + x_4 \leq 15$
$7x_1 + 5x_2 + 3x_3 + 2x_4 \leq 120$
$3x_1 + 5x_2 + 10x_3 + 15x_4 \leq 100$
and $x_1, x_2, x_3, x_4 \geq 0$

(vi) Max $Z = x_1 + x_2 + x_3$
subject to $4x_1 + 5x_2 + 3x_3 \leq 15$
$10x_1 + 7x_2 + x_3 \leq 12$
and $x_1, x_2, x_3 \geq 0$

(vii) Max $Z = x_1 + x_2 + x_3$
subject to $3x_1 + 2x_2 + x_3 \le 3$
$2x_1 + x_2 + 2x_3 \le 2$
and $x_1, x_2, x_3 \ge 0$

(viii) Max $Z = 2x_1 + 4x_2 + 3x + x_4$
subject to $x_1 + 3x_2 + x_4 \le 4$
$2x_1 + x_2 \le 3$
$x_2 + 4x_3 + x_4 \le 3$
and $x_1, x_2, x_3, x_4 \ge 0$

(ix) Max $Z = 107x_1 + x_2 + 2x_3$
subject to $14x_1 + x_2 - 6x_3 + 3x_4 \le 7$
$16x_1 + 0.5x_2 + 6x_3 \le 5$
$3x_1 - x_2 - x_3 \le 10$
and $x_1, x_2, x_3, x_4 \ge 0$
[*Sambalpur, MSc* (*Maths*), *2003*]

(x) Max $Z = 4x_1 + 10x_2$
subject to $2x_1 + x_2 \le 50$
$2x_1 + 5x_2 \le 100$
$2x_1 + 3x_2 \le 90$
and $x_1, x_2 \ge 0$

(xi) Max $Z = x_1 - x_2 + 3x_3$
subject to $x_1 + x_2 + x_3 \le 10$
$2x_1 - x_3 \le 2$
$2x_1 - 2x_2 + 3x_3 \le 0$
and $x_1, x_2 \ge 0$
[*Karnataka, MSc* (*Maths*), *2004*]

(xii) Max $Z = 4x_1 + x_2 + 3x_3 + 5x_4$
subject to
$4x_1 - 6x_2 - 5x_3 + 4x_4 \ge -20$
$3x_1 - 2x_2 + 4x_3 + x_4 \le 10$
$8x_1 - 3x_2 + 3x_3 + 2x_4 \le 20$
and $x_1, x_2, x_3, x_4 \ge 0$
has an unbound solution.

2. Use the two-phase method to solve the following LP problems:

(i) Max $Z = 3x_1 - x_2$
subject to $2x_1 + x_2 \ge 2$
$x_1 + 3x_2 \le 2$
$x_2 \le 4$
and $x_1, x_2 \ge 0$

(ii) Min $Z = 3x_1 - x_2 - x_3$
subject to $x_1 - 2x_2 + x_3 \le 11$
$-4x_1 + x_2 + 2x_3 \ge 3$
$-2x_1 + x_3 = 1$
and $x_1, x_2, x_3 \ge 0$

(iii) Min $Z = 7.5x_1 - 3x_2$
subject to $3x_1 - x_2 - x_3 \ge 3$
$x_1 - x_2 + x_3 \ge 2$
and $x_1, x_2, x_3 \ge 0$

(iv) Min $Z = 3x_1 - x_2$
subject to $2x_1 + x_2 \ge 2$
$x_1 + 3x_2 \le 2$
$x_2 \le 4$
and $x_1, x_2 \ge 0$

(v) Min $Z = 5x_1 + 8x_2$
subject to $3x_1 + 2x_2 \ge 3$
$x_1 + 4x_2 \ge 4$
$x_1 + x_2 \le 5$
and $x_1, x_2 \ge 0$

(vi) Min $Z = 3x_1 + 2x_2$
subject to $2x_1 + x_2 \ge 2$
$3x_1 + 4x_2 \ge 12$
and $x_1, x_2 \ge 0$

3. Use penalty (Big-M) method to solve the following LP problems:

(i) Min $Z = 3x_1 - x_2$
subject to $2x_1 + x_2 \ge 2$
$x_1 + 3x_2 \le 3$
$x_2 \ge 4$
$x_1, x_2 \ge 0$

(ii) Min $Z = 2x_1 + x_2$
subject to $3x_1 + x_2 = 3$
$4x_1 + 3x_2 \ge 6$
$x_1 + 2x_2 \le 4$
$x_1, x_2 \ge 0$

(iii) Max $Z = 3x_1 + 2x_2 + 3x_3$
subject to $2x_1 + x_2 + x_3 \le 2$
$3x_1 + 4x_2 + 2x_3 \le 8$
and $x_1, x_2, x_3 \ge 0$

(iv) Max $Z = x_1 + x_2 + x_4$
subject to $x_1 + x_2 + x_3 + x_4 = 4$
$x_1 + 2x_2 + x_3 + x_4 = 4$
$x_1 + 2x_2 + x_3 = 4$

(v) Min $Z = x_1 - 3x_2 + 2x_3$
subject to $3x_1 - x_2 + 2x_3 \le 7$
$-2x_1 + 4x_2 \le 12$
$-4x_1 + 3x_2 + 8x_3 \le 10$
and $x_1, x_2, x_3 \ge 0$
[*Meerut MSc* (*Maths*), *2005*]

(vi) Min $Z = 5x_1 + 2x_2 + 10x_3$
subject to $x_1 - x_3 \le 10$
$x_2 + x_3 \ge 10$
and $x_1, x_2, x_3 \ge 0$

4. Solve the following LP problems and remove the complication (if any).

(i) Max $Z = 2x_1 + 3x_2 + 10x_3$
subject to $x_1 + 2x_3 = 2$
$x_2 + x_3 = 1$
and $x_1, x_2, x_3 \ge 0$

(ii) Max $Z = 5x_1 - 2x_2 + 3x_3$
subject to $2x_1 + 2x_2 - x_3 \ge 2$
$3x_1 - 4x_2 \le 3$
$x_2 + 3x_3 \le 5$
and $x_1, x_2, x_3 \ge 0$

(iii) Max $Z = 5x_1 + 3x_2$
subect to $x_1 + x_2 \le 2$
$5x_1 + 2x_2 \le 10$
$3x_1 + 8x_2 \le 12$
and $x_1, x_2 \ge 0$

(iv) Max $Z = 22x_1 + 30x_2 + 25x_3$
subject to $2x_1 + 2x_2 + \le 100$
$2x_1 + x_2 + x_3 \ge 100$
$x_1 + 2x_2 + 2x_3 \le 100$
and $x_1, x_2, x_3 \ge 0$

(v) Max $Z = 8x_2$
subject to $x_1 - x_2 \ge 0$
$2x_1 + 3x_2 \le -6$
and x_1, x_2 unrestricted.

5. Solve the following LP problems to show that these have alternative optimal solutions.

(i) Max $Z = 6x_1 + 3x_2$
subject to $2x_1 + x_2 \le 8$
$3x_1 + 3x_2 \le 18$
$x_2 \le 3$
and $x_1, x_2 \ge 0$

(ii) Min $Z = 2x_1 + 8x_2$
subject to $5x_1 + x_2 \ge 10$
$2x_1 + 2x_2 \ge 14$
$x_1 + 4x_2 \ge 12$
and $x_1, x_2 \ge 0$

(iii) Max $Z = x_1 + 2x_2 + 3x_3 - x_4$
subject to $x_1 + 2x_2 + 3x_3 = 15$
$2x_1 + x_2 + 5x_3 \ge 20$
$x_1 + x_2 + x_3 + x_4 \ge 10$
and $x_1, x_2, x_3, x_4 \ge 0$

6. Solve the following LP problems to show that these have an unbounded solution.

(i) Max $Z = -2x_1 + 3x_2$
subject to $x_1 \le 5$
$2x_1 - 3x_2 \le 6$
and $x_1, x_2 \ge 0$

(ii) Max $Z = 3x_1 + 6x_2$
subject to $3x_1 + 4x_2 \ge 12$
$-2x_1 + x_2 \le 4$
and $x_1, x_2 \ge 0$

(iii) Max $Z = 107x_1 + x_2 + 2x_3$
subject to $14x_1 + x_2 - 6x_3 + 3x_4 = 7$
$16x_1 + 0.5x_2 - 6x_3 \le 5$
$3x_1 - x_2 - x_3 \le 0$
and $x_1, x_2 \ge 0$

(iv) Max $Z = 6x_1 - 2x_2$
subject to $2x_1 - x_2 \le 2$
$x_1 \le 4$
and $x_1, x_2, x_3, x_4 \ge 0$

7. Solve the following LP problems to show that these have no feasible solution.

(i) Max $Z = 2x_1 + 3x_2$
subject to $x_1 - x_2 \ge 4$
$x_1 + x_2 \le 6$
$x_1 \le 2$
and $x_1, x_2 \ge 0$

(ii) Max $Z = 4x_1 + x_2 + 4x_3 + 5x_4$
subject to $4x_1 + 6x_2 - 5x_3 + 4x_4 \ge -20$
$3x_1 - 2x_2 + 4x_3 + x_4 \le 10$
$3x_1 - 2x_2 + 4x_3 + x_4 \le 10$
$8x_1 - 3x_2 - 3x_3 + 2x_4 \le 20$
and $x_1, x_2, x_3, x_4 \ge 0$

(iii) Max $Z = x_1 + 3x_2$
subject to $x_1 - x_2 \ge 1$
$3x_1 - x_2 \le -3$
and $x_1, x_2 \ge 0$

(iv) Max $Z = 3x_1 + 2x_2$
subject to $2x_1 + x_2 \le 2$
$3x_1 + 4x_2 \ge 12$
and $x_1, x_2 \ge 0$

[*Meerut, MSc* (*Maths*), *2006*]

HINTS AND ANSWERS

1. (i) $x_1 = 3$, $x_2 = 1$ and Max $Z = 11$
(ii) $x_1 = 2$, $x_2 = 0$ and Max $Z = 10$
(iii) $x_1 = 0$, $x_2 = 100$, $x_3 = 230$ and Max $Z = 1{,}350$
(iv) Max $Z^* = -x_1 + 3x_2 - 2x_3$ where $Z^* = -Z$
$x_1 = 4, x_2 = 5, x_3 = 0$ and $Z^* = 11$.
(v) $x_1 = 50/7, x_2 = 0, x_3 = 55/7$ and Max $Z = 695/7$
(vi) $x_1 = 0$, $x_2 = 0$, $x_3 = 5$ and Max $Z = 5$
(vii) $x_1 = 0$, $x_2 = 0$, $x_3 = 1$ and Max $Z = 3$
(viii) $x_1 = 1$, $x_2 = 1$, $x_3 = 1/2$ and Max $Z = 13/2$
(ix) Divide the first equation by 3 (coefficient of x_4)
(x) $x_1 = 0$, $x_2 = 0$ and Max $Z = 200$
(xi) $x_1 = 0$, $x_2 = 6$, $x_3 = 4$ and Max $Z = 6$

2. (i) $x_1 = 1$, $x_2 = 0$, and Max $Z = 6$
(ii) All $c_j - z_j \le 0$ but $Z = -5/4$ (< 0) and artificial variable $A_1 = 5/4$ appears in the basis with positive value. Thus the given LP problem has no feasible solution.
(iii) $x_1 = 5/4$, $x_2 = 0$, $x_3 = 0$ and Min $Z = 75/8$
(iv) $x_1 = 2$, $x_2 = 0$ and Max $Z = 6$
(v) $x_1 = 0$, $x_2 = 5$ and Max $Z = 40$
(vi) All $c_j - z_j \ge 0$, but $Z = -4$ (< 0) and artificial variable $A_1 = 4$ appears in the basis with a positive value. Thus the given LP problem has no feasible solution.

3. (i) $x_1 = 3$, $x_2 = 0$ and Max $Z = 9$
(ii) $x_1 = 3/5$, $x_2 = 6/5$ and Min $Z = 12/5$
(iii) All $c_j - z_j \ge 0$ artificial variable $A_1 = 0$ appears in the basis with zero value. Thus an optimal solution to the given LP problem exists.
(iv) Introduce artificial variable only in the third constraint.
$x_1 = 0, x_2 = 0, x_3 = 0, x_5 = 0$ and Max $Z = 4$.
(v) $x_1 = 4, x_2 = 5$ and Min $Z = -11$
(vi) $x_1 = 0, x_2 = 10, x_3 = 0$ and Min $Z = 20$

4. (i) $x_1 = 0$, $x_2 = 1$, $x_3 = 0$ and Max $Z = 3$
(ii) Degeneracy occurs at the initial stage. One of the variable eligible to leave the basis is artificial variable, therefore, there is no need of resolving degeneracy. Remove the artificial variable from the basis.
$x_1 = 23/3$, $x_2 = 5$, $x_3 = 0$ and Max $Z = 85/3$
(iii) $x_1 = 2$, $x_2 = 0$ and Max $Z = 10$
(iv) $x_1 = 100/3$, $x_2 = 50/3$, $x_3 = 50/3$ and Max $Z = 1{,}650$
(v) $x_1'' = 6/5$ or $x_1 = -6/5$ or $x_2'' = 6/5$ or $x_2 = -6/5$ and Max $Z = -48/5$.

5. (i) (a) $x_1 = 4$, $x_2 = 0$ and Max $Z = 24$
(b) $x_1 = 5/2$, $x_2 = 3$, and Max $Z = 24$
(ii) (a) $x_1 = 32/6$, $x_2 = 10/6$ and Min $Z = 24$
(b) $x_1 = 12$, $x_2 = 0$ and Min $Z = 24$

6. (i) At the current solution: $x_1 = 5$, $x_2 = 9$ and Max $Z = 15$, it may be observed that $c_2 - z_2 = 3/2$ but all elements in the second column are negative. Solution is unbounded.

(ii) At the current solution: $x_2 = 4$, $s_1 = 4$ and Max $Z = 24$, it may be observed that $c_2 - z_2 = 15$ but all elements in the second column are negative. Solution is unbounded.

(iii) At second best solution, $c_3 - z_3 = 113/3$ but all elements in the third column are negative. Solution is unbounded; $x_1 = 0$, $x_4 = 7/3$, $s_1 = 5$ and Max $Z = 0$.

(iv) Optimal solution: $x_1 = 4$, $x_2 = 6$ and Max $Z = 12$. Since in the initial simplex table all the elements are negative in the second column, the feasible solution is unbounded but the optimal solution is bounded.

7. (i) $x_1 = 2, x_2 = 0, A_1 = 2$ and M Max $Z = 9 - 4M$; Infeasible solution.

(iv) $x_1 = 0, x_2 = 2, A_1 = 2$ and Max $Z = 4 - 4M$; Infeasible solution.

CHAPTER SUMMARY

The simplex method is an interactive procedure for reaching the optimal solution to any LP problem. It consists of a series of rules that, in effect, algebraically examine corner (extreme) points of the solution space in a systematic way. Each step moves towards the optimal solution by increasing profit or decreasing cost, while maintaining feasibility. The simplex method consists of five steps: (i) identifying the pivot column, (ii) identifying the pivot row and key element, (iii) replacing the pivot row, (iv) computing new values for each remaining row; and (v) computing the z_j and $c_j - z_j$ values and examining for optimality. Each step of this iterative procedure is displayed and explained in a simplex table both for maximization and minimization LP problems.

CHAPTER CONCEPTS QUIZ

True or False

1. The major difference between slack and artificial variables is that an artificial can never be zero.
2. If an optimal solution is degenerate, then there are alternate optimal solutions of the LP problem.
3. An infeasible solution is characterized as one where one constraint is violated.
4. If the objective function coefficient in the c_j row above an artificial variable is – M, then the problem is a minimization problem.
5. In the simplex method, the initial solution contains only slack variable in the product mix.
6. If there is a tie between decision variable and a slack (or surplus) variable, then select the decision variable to enter into the first basis.
7. An optimal solution to the maximized LP problem is reached if all $c_j - z_j \geq 0$.
8. Variables which can assume negative, positive or zero value are called unrestricted variables.
9. All the rules and procedures of the simplex method are identical whether solving a maximization or minimization LP problem.
10. Artificial variables are added to a linear programming problem to aid in the finding an optimal solution.

Fill in the Blanks

11. In the simplex method, the ———— column contains the variables which are currently in the solution; the values of these variables can be read from the ———— column.
12. A ———— variable represents amounts by which solution values exceed a resource.
13. The simplex method examines the ———— points in a systematic manner, repeating the same set of steps of the algorithm until an ———— solution is reached.
14. ———— occurs when there is no solution that satisfies all of the constraints in the linear programming problem.
15. The value for the replacing row must be ———— before computing the values for the ———— rows.
16. Entries in the $c_j - z_j$ rows are known as ———— costs.
17. Optimality is indicated for a maximization problem when all elements in the $c_j - z_j$ rows are ————, while for a minimization problem all elements must be ————.
18. In Big-M method, ———— basic feasible solution is obtained by assigning ———— value to the original value.
19. In the two-phase method, an ———— variable is never considered for re-entry into the basis.
20. ———— occurs when there is no finite solution in the LP problem.

Multiple Choice

21. The role of artificial variables in the simplex method is
    (a) to aid in finding an initial solution
    (b) to find optimal dual prices in the final simplex table
    (c) to start phases of simplex method
    (d) all of the above
22. For a maximization problem, the objective function coefficient for an artificial variable is
    (a) + M (b) – M
    (c) Zero (d) none of the above
23. If a negative value appears in the solution values ($\mathbf{x_B}$) column of the simplex table, then
    (a) the solution is optimal
    (b) the solution is infeasible
    (c) the solution is unbounded
    (d) all of the above
24. At every iteration of the simplex method, for minimization problem, a variable in the current basis is replaced with another variable that has
    (a) a positive $c_j - z_j$ value
    (b) a negative $c_j - z_j$ value
    (c) $c_j - z_j = 0$
    (d) none of the above

25. In the optimal simplex table, $c_j - z_j = 0$ value indicates
    (a) unbounded solution (b) cycling
    (c) alternative solution (d) infeasible solution
26. For maximization LP model, the simplex method is terminated when all values
    (a) $c_j - z_j \leq 0$ (b) $c_j - z_j \geq 0$
    (c) $c_j - z_j = 0$ (d) $z_j \leq 0$
27. A variable which does not appear in the basic variable (***B***) column of simplex table is
    (a) never equal to zero (b) always equal to zero
    (c) called a basic variable (d) none of the above
28. If for a given solution a slack variable is equal to zero then
    (a) the solution is optimal (b) the solution is infeasible
    (c) the entire amount of resource with the constraint in which the slack variable appears has been consumed.
    (d) all of the above
29. If an optimal solution is degenerate, then
    (a) there are alternative optimal solutions
    (b) the solution is infeasible
    (c) the solution is of no use to the decision-maker
    (d) none of the above
30. To formulate a problem for solution by the simplex method, we must add artificial variable to
    (a) only equality constraints
    (b) only 'greater than' constraints
    (c) both (a) and (b)
    (d) none of the above
31. If any value in x_B – Column of final simplex table is negative, then the solution is
    (a) unbounded (b) infeasible
    (c) optimal (d) none of the above
32. If all a_{ij} values in the incoming variable column of the simplex table are negative, then
    (a) solution is unbounded (b) there are multiple solutions
    (c) there exist no solution (c) the solution is degenerate
33. If an artificial variable is present in the 'basic variable' column of optimal simplex table, then the solution is
    (a) infeasible (b) unbounded
    (c) degenerate (d) none of the above
34. The per unit improvement in the solution of the minimization LP problem is indicated by the negative value of
    (a) $c_j - z_j$ in the decision-variable column
    (b) $c_j - z_j$ in the slack variable column
    (c) $c_j - z_j$ in the surplus variable column
    (d) none of the above
35. To convert ≥ inequality constraints into equality constraints, we must
    (a) add a surplus variable
    (b) subtract an artificial variable
    (c) substract a surplus variable and an artificial variable
    (d) add a surplus variable and substract an artificial variable

Answers to Quiz

1. F	2. F	3. T	4. F	5. F	6. T	7. T	8. T	9. F	10. T
11. product mix; quantity		12. Suplus		13. Extreme; optimal		14. Infeasibility		15. Computed; remaining	
16. reduced		17. non positive, non negative				18. Initial; zero		19. Artificial	20. Unboundedness
21. (d)	22. (b)	23. (b)	24. (b)	25. (c)	26. (a)	27. (b)	28. (c)	29. (d)	30. (c)
31. (b)	32. (a)	33. (a)	34. (d)	35. (c)					

CASE STUDY

Case 4.1: New Bharat Bank

New Bharat Bank is planning its operations for the next year. The bank makes five types of loans. These are listed below, together with the annual return (in per cent);

Type of Loan	*Annual Return (per cent)*
Education	15
Furniture	12
Automobile	9
Home Construction	10
Home repair	7

Legal requirements and bank policy place the following limits on the amounts of the various types of loans: Education loans cannot exceed 10 per cent of the total amount of loans. The amount of education and furniture loans together cannot exceed 20 per cent of the total amount of loans. Home construction loan must be at least 40 per cent of the total amount of home loans and at least 20 per cent of the total amount of loans. Home construction loan must not exceed 25 per cent of the total amount of loans.

The bank wishes to maximize its revenue from loan interest, subject to the above restrictions. The bank can lend a maximum of Rs 75 crore. You are expected to suggest to the management of the bank for maximizing revenue keeping in view operational restrictions.

Case 4.2: Accurate Transformers

Accurate Transformers produces electric transformers. The company has orders for transformer for the next six months. The cost of manufacturing a transformer is expected to somewhat vary over the next few months due to expected changes in materials costs and in labor rates. The company can produce up to 50 units per month in regular time and up to an additional 20 units per month during overtime. The costs for both regular and overtime production are shown in the table below:

	Month					
	Jan.	*Feb.*	*Mar.*	*April*	*May*	*June*
Orders (units)	58	36	34	69	72	43
Cost per unit at regular time (in '000 Rs)	180	170	170	185	190	190
Cost per unit at overtime (in '000 Rs)	200	190	190	210	220	220

The cost of carrying an unsold transformer in stock is Rs 25,000 per month. As on January 1 the company has 15 transformers in stock on and it wishes to have no less than 5 in stock on June 30. Suggest to the management of the company to determine the optimal production schedule.

Case 4.3: Nationwide Air lines*

Nationwide Airlines, faced with a sharply escalating cost of jet fuel, is interested in optimizing its purchase of jet fuel at its various locations around the country. Typically, there is some choice concerning the amount of fuel that can be placed on board any aircraft for any flight segment, as long as minimum and maximum limits are not violated. The flight schedule is considered as a chain of flight segments, or legs, that each aircraft follows. The schedule ultimately returns the aircraft to its starting point, resulting in a 'rotation'. Consider the following rotation: Delhi – Hyderabad – Cochin – Chennai – Delhi

Fuel Requirements and Limits (1,000 gallons unless specified otherwise)

City	*Flight Sequence*	*Minimum Fuel Required*	*Maximum Fuel Allowed*	*Regular Fuel Consumption if Minimum Fuel Boarded*	*Additional Fuel Burned per Gallon of Tankered Fuel (i.e. fuel above minimum – in gallons)*	*Price per Gallons (Rs)*
1.	Delhi to Hyderabad	23	33	12.1	0.040	8,200
2.	Hyderabad to Cochin	8	19	2.0	0.005	7,500
3.	Cochin to Chennai	19	33	9.5	0.025	7,700
4.	Chennai to Delhi	25	33	13.0	0.045	8,900

The fuel for any one of these flight segments may be bought at its departure city, or it may be purchased at a previous city in the sequence and 'tankered' for the flight. Of course, it takes fuel to carry fuel, and thus an economic trade-off between purchasing fuel at the lowest-cost location and tankering it all around the country must be made.

In the table above the column 'Regular Fuel Consumption' takes into account fuel consumption if the minimum amount of fuel is on board, and the column 'Additional Fuel Burned' indicates the additional fuel burned in each flight segment per gallon of 'tankered' fuel carried; tankered fuel refers to fuel above the minimum amount.

The fuel originally carried into Delhi should equal fuel carried into Delhi on the next rotation, in order for the system to be in equilibrium.

If l_1= Leftover fuel inventory coming into city i (1,000 gallons) and x_i = Amount of fuel purchased at city i (1,000 gallons) then, $(l_i + x_i)$ is the amount of fuel on board the aircraft when it departs city i.

Suggest the optimal fuel quantity to be purchased by the Airlines.

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* This case is based on 'D. Wayne Darnell and Carln Loflin, *National Airlines Fuel Management and Allocation Model*, Interfaces' Feburary, 1977.

Chapter 5

Duality in Linear Programming

"A boat can't have two captains."

– Akira Mori

PREVIEW

The chapter deals with how to find the marginal value (also known as shadow price) of each resource. This value reflects an additional cost to be paid to obtain one additional unit of the resource to get the optimal value of objective function under resource constraints.

LEARNING OBJECTIVES

After studying this chapter, you should be able to

- appreciate the significance of the duality concept.
- formulate the dual LP problem and understand the relationship between primal and dual LP problems.
- understand the concept of shadow prices.

CHAPTER OUTLINE

5.1 Introduction

5.2 Formulation of Dual Linear Programming Problem

- Self Practice Problems A
- Hints and Answers

5.3 Standard Results on Duality

5.4 Managerial Significance of Duality

5.5 Advantages of Duality

- Conceptual Questions
- Self Practice Problems B
- Hints and Answers

❑ Chapter Summary

❑ Chapter Concepts Quiz

❑ Case Study

❑ Appendix: Theorems of Duality

5.1 INTRODUCTION

The term 'dual', in general, implies two or double. The concept of duality is very useful in mathematics, physics, statistics, engineering and managerial decision-making. For example, in a two-person game theory, one competitor's problem is the dual of the opponent's problem.

In linear programming, duality implies that each linear programming problem can be analyzed in two different ways but would have equivalent solutions. Any LP problem (either maximization and minimization) can be stated in another equivalent form based on the same data. The new LP problem is called *dual linear programming problem or in short dual*. In general, it is immaterial which of the two problems is called primal or dual, since the dual of the dual is primal.

Dual LP problem provides useful economic information about worth of resources to be used.

For example, consider the problem of production planning. The production manager attempts to determine quantities for each product to be produced with an objective to optimize the use of available resources so that profit is maximum. But through a dual LP problem approach, he may develop a production plan that optimizes resource utilization so that *marginal opportunity cost of each unit of a resource is equal to its marginal return* (*also called shadow price*). The shadow price indicates an additional price to be paid to obtain one additional unit of the resources in order to maximize profit under the resource constraints. If a resource is not completely used, i.e. there is slack, then its marginal return is zero.

The shadow price is also defined as the rate of change in the optimal objective function value with respect to the unit change in the availability of a resource. To be more precise for any constraint, we have

$$\text{Shadow price} = \frac{\text{Change in optimal objective function value}}{\text{Unit change in the availability of resource}}$$

The interpretation of rate of change (increase or decrease) in the value of objective function depends on whether LP problem is of maximization or minimization type. The shadow price for a less than or equal to (≤) type constraint will always be positive. This is because increasing the right-hand side resource value cannot decrease the value of objective function. Similarly, the shadow price for a greater than or equal to (≥) type constraint will always be negative because increasing the right-hand side resonance value cannot increase the value of the objective function.

There is no need to solve both LP problems separately. Solving one LP problem is equivalent to solving the other simultaneously. Thus, if the optimal solution to one is known, the optimal solution of the other can also be read from the $c_j - z_j$ row. In some cases, considerable computing time can be reduced by solving the dual LP problem.

5.2 FORMULATION OF DUAL LINEAR PROGRAMMING PROBLEM

There are two important forms of primal and dual LP problems, namely the *symmetrical (canonical) form* and the *standard form.*

Shadow price represents increase in the objective function value due to one-unit increase in the right hand side (resource) of any constraint.

5.2.1 Symmetrical Form

Suppose the *primal LP problem* is given in the form

Maximize $Z_x = c_1 x_1 + c_2 x_2 + \ldots + c_n x_n$

subject to the constraints

$$\begin{aligned} a_{11} x_1 + a_{12} x_2 + \ldots + a_{1n} x_n &\le b_1 \\ a_{21} x_1 + a_{22} x_2 + \ldots + a_{2n} x_n &\le b_2 \\ \vdots \qquad\qquad\qquad\qquad &\quad \vdots \\ a_{m1} x_1 + a_{m2} x_2 + \ldots + a_{mn} x_n &\le b_m \end{aligned}$$

and $$x_1, x_2 \ldots, x_n \ge 0$$

Then the corresponding *dual LP problem* is written as:

Minimize $Z_y = b_1 y_1 + b_2 y_2 + \ldots + b_m y_m$

subject to the constraints

$$\begin{aligned} a_{11} y_1 + a_{21} y_2 + \ldots + a_{m1} y_m &\le c_1 \\ a_{12} y_1 + a_{22} y_2 + \ldots + a_{m2} y_m &\le c_2 \\ \vdots \qquad\qquad\qquad\qquad\qquad &\quad \vdots \\ a_{1n} y_1 + a_{2n} y_2 + \ldots + a_{mn} y_m &\le c_n \end{aligned}$$

and $y_1, y_2, \ldots, y_m \ge 0$

In general, the primal-dual relationship between a pair of LP problems can be expressed as follows:

Dual variables represent the potential value of resources

Primal	*Dual*
Max $Z_x = \sum_{j=1}^{n} c_j x_j$	Min $Z_y = \sum_{i=1}^{m} b_i y_i$
subject to the constraints	subject to the constraints
$\sum_{j=1}^{n} a_{ij} x_j \le b_i ;\ i = 1, 2, \ldots, m$	$\sum_{i=1}^{m} a_{ji} y_i \ge c_j ;\ j = 1, 2, \ldots, n$
$a_{ij} = a_{ji}$	
and $x_j \ge 0;\quad j = 1, 2, \ldots, n$	and $y_i \ge 0;\quad i = 1, 2, \ldots, m$

A summary of the general relationships between primal and dual LP problems is given in Table 5.1.

If Primal	*Then Dual*
(i) Objective is to maximize	(i) Objective is to minimize
(ii) jth primal variable, x_j	(ii) jth dual constraint
(iii) ith primal constraint	(iii) ith dual variable, y_i
(iv) Primal variable x_j unrestricted in sign	(iv) Dual constraint j is = type
(v) Primal constraint i is = type	(v) Dual variable y_i is unrestricted in sign
(vi) Primal constraints ≤ type	(vi) Dual constraints ≥ type

Table 5.1 Primal-Dual Relationship

5.2.2 Economic Interpretation of Dual Variables

In the maximization primal LP model, we may define each parameter as follows:

Z = return
x_j = number units of variable j
b_i = maximum units of resource, i available
c_j = profit (or return) per unit of variable (activity) x_j
a_{ij} = units of resource, i consumed (required) per unit of variable j

The new variables introduced in the dual problem are Z_y and y_i (dual variables). When both the primal and the dual solutions are optimal, the value of objective function satisfies the strict equality, i.e. $Z_x = Z_y$. The interpretation associated with the dual variables y_i (i = 1, 2, . . ., m) is discussed below. Rewrite the primal LP problem as follows:

Primal LP Problem

Maximize (return) $Z_x = \sum_{j=1}^{n} c_j x_j = \sum_{j=1}^{n}$ (Profit per unit of variable x_j) (Units of variable x_j)

subject to the constraints

$$\sum_{j=1}^{n} a_{ij} x_j \le b_i$$

or $\sum_{j=1}^{n}$ (Units of resource i, consumed per unit of variable, x_j) (Units of variable x_j) ≤ Units of resource, i available

and $x_j \ge 0$, for all j

Dual LP Problem

$$\text{Minimize (cost) } Z_y = \sum_{i=1}^{m} b_i y_i = \sum_{i=1}^{m} (\text{Units of resource, } i)\ (\text{Cost per unit of resource, } i)$$

subject to the constraints

$$\sum_{i=1}^{m} a_{ji}\, y_i \geq c_j$$

or $\sum_{i=1}^{m}$ (Units of a resource, i consumed per unit of variable y_i) (Cost per unit of resource, i) $\geq$ Profit per unit for each variable, x_j

and $y_i \geq 0$, for all i

From these expressions of parameters of both primal and dual problems, it is clear that for the unit of measurement to be consistent, *the dual variable* (y_i) *must be expressed in terms of return (or worth) per unit of resource i*. This is called *dual price* (*simplex multiplier* or *shadow price*) of resource *i.* In other words, optimal value of a dual variable associated with a particular primal constraint indicates the *marginal change (increase, if positive or decrease, if negative) in the optimal value of the primal objective function*. For example, if $y_2 = 5$, then this indicates that for every additional unit (up to a certain limit) of resource 2 (resource associated with constraint 2 in the primal), the objective function value will increase by 5 units. The value $y_2 = 5$ is also called the marginal (or shadow or implicit) price of resource 2.

Similarly, for *feasible* solutions of both primal and dual LP problems, objective function value satisfy the inequality $Z_x \leq Z_y$. This inequality is interpreted as: *Profit* $\leq$ *Worth of resources*. Thus, so long as the total profit (return) from all activities is less than the worth of the resources, the feasible solution of both primal and dual is not optimal. The optimality (maximum profit or return) is reached only when the resources have been completely utilized. This is only possible if the worth of the resources (i.e. input) is equal to profit (i.e. output).

5.2.3 Economic Interpretation of Dual Constraints

As stated earlier, the dual constraints are expressed as:

$$\sum_{i=1}^{m} a_{ji} y_i - c_j \geq 0$$

Since coefficients a_{ji} represents the amount of resource b_i consumed by per unit of activity x_j, and the dual variable y_i represents shadow price per unit of resource b_i, the quantity $\Sigma\, a_{ji} y_i$ $(= z_j)$ should be total shadow price of all resources required to produce one unit of activity x_j.

For maximization LP problem, if $c_j - z_j > 0$ value corresponds to any non-basic variable, then the value of objective function can be increased. This implies that the value of variable, x_j can be increased from zero to a positive level provided its unit profit (c_j) is more than its shadow price, i.e.

$$c_j \geq \sum_{i=1}^{m} a_{ji} y_i$$

Profit per unit of activity $x_j \geq$ Shadow price of resources used per unit of activity, x_j.

5.2.4 Rules for Constructing the Dual from Primal

The rules for constructing the dual from the primal and vice-versa using the symmetrical form of LP problem are:

1. A dual variable is defined corresponds to each constraint in the primal LP problem and vice versa. Thus, for a primal LP problem with m constraints and n variables, there exists a dual LP problem with m variables and n constraints and vice-versa.
2. The right-hand side constants $b_1, b_2, \ldots, b_m$ of the primal LP problem becomes the coefficients of the dual variables $y_1, y_2, \ldots, y_m$ in the dual objective function Z_y. Also the coefficients $c_1, c_2, \ldots, c_n$ of the primal variables $x_1, x_2, \ldots, x_n$ in the objective function become the right-hand side constants in the dual LP problem.

3. For a maximization primal LP problem with all ≤ (less than or equal to) type constraints, there exists a minimization dual LP problem with all ≥ (greater than or equal to) type constraints and vice versa. Thus, the inequality sign is reversed in all the constraints except the non-negativity conditions.
4. The matrix of the coefficients of variables in the constraints of dual is the transpose of the matrix of coefficients of variables in the constraints of primal and vice versa.
5. If the objective function of a primal LP problem is to be maximized, the objective function of the dual is to be minimized and vice versa.
6. If the *i*th primal constraint is = (equality) type, then the *i*th dual variables is unrestricted in sign and vice versa.

The primal-dual relationships may also be memorized by using the following table:

Dual Variables	*Primal Variables*						*Maximize* Z_x
	x_1	x_2	. . .	x_j	. . .	x_n	
y_1	a_{11}	a_{12}	. . .	a_{1j}	. . .	a_{1n}	$\le b_1$
y_2	a_{21}	a_{22}	. . .	a_{2j}	. . .	a_{2n}	$\le b_2$
⋮						⋮	⋮
y_m	a_{m1}	a_{m2}	. . .	a_{mj}	. . .	a_{mn}	$\le b_m$
Minimize Z_y	$\ge c_1$	$\ge c_2$	. . .	$\ge c_j$	. . .	$\ge c_n$	↑ Dual objective function coefficients

↑ *j*th dual constraint

The primal constraints should be read across the rows, and the dual constraints should be read across the columns.

Example 5.1 Write the dual to the following LP problem.

Maximize $Z = x_1 - x_2 + 3x_3$

subject to the constraints

(i) $x_1 + x_2 + x_3 \le 10$, (ii) $2x_1 - x_2 - x_3 \le 2$, (iii) $2x_1 - 2x_2 - 3x_3 \le 6$

and $x_1, x_2, x_3 \ge 0$

Solution In the given LP problem there are $m = 3$ constraints and $n = 3$ variables. Thus, there must be $m = 3$ dual variables and $n = 3$ constraints. Further, the coefficients of the primal variables, $c_1 = 1$, $c_2 = -1$, $c_3 = 3$ become right-hand side constants of the dual. The right-hand side constants $b_1 = 10$, $b_2 = 2$, $b_3 = 6$ become the coefficients in the dual objective function. Finally, the dual must have a minimizing objective function with all ≥ type constraints. If y_1, y_2 and y_3 are dual variables corresponding to three primal constraints in the given order, the resultant dual is

Minimize $Z_y = 10y_1 + 2y_2 + 6y_3$

subject to the constraints

(i) $y_1 + 2y_2 + 2y_3 \ge 1$, (iii) $y_1 - y_2 - 2y_3 \ge -1$, (iii) $y_1 - y_2 - 3y_3 \ge 3$

and $y_1, y_2, y_3 \ge 0$

Example 5.2 Write the dual of the following LP problem.

Minimize $Z_x = 3x_1 - 2x_2 + 4x_3$

subject to the constraints

(i) $3x_1 + 5x_2 + 4x_3 \ge 7$, (ii) $6x_1 + x_2 + 3x_3 \ge 4$, (iii) $7x_1 - 2x_2 - x_3 \le 10$

(iv) $x_1 - 2x_2 + 5x_3 \ge 3$, (v) $4x_1 + 7x_2 - 2x_3 \ge 2$

and $x_1, x_2, x_3 \ge 0$

Solution Since the objective function of the given LP problem is of minimization, the direction of each inequality has to be changed to ≥ type by multiplying both sides by – 1. The standard primal LP problem so obtained is:

Minimize $Z_x = 3x_1 - 2x_2 + 4x_3$

subject to the constraints

(i) $3x_1 + 5x_2 + 4x_3 \geq 7$, (ii) $6x_1 + x_2 + 3x_3 \geq 4$, (iii) $-7x_1 + 2x_2 + x_3 \geq -10$

(iv) $x_1 - 2x_2 + 5x_3 \geq 3$, (v) $4x_1 + 7x_2 - 2x_3 \geq 2$

and $x_1, x_2, x_3 \geq 0$

If y_1, y_2, y_3, y_4 and y_5 are dual variables corresponding to the five primal constraints in the given order, the dual of this primal LP problem is stated as:

Maximize $Z_y = 7y_1 + 4y_2 - 10y_3 + 3y_4 + 2y_5$

subject to the constraints

(i) $3y_1 + 6y_2 - 7y_3 + y_4 + 4y_5 \leq 3$, (ii) $5y_1 + y_2 + 2y_3 - 2y_4 + 7y_5 \leq -2$

(iii) $4y_1 + 3y_2 + y_3 + 5y_4 - 2y_5 \leq 4$

and $y_1, y_2, y_3, y_4, y_5 \geq 0$

Example 5.3 Obtain the dual LP problem of the following primal LP problem:

Minimize $Z = x_1 + 2x_2$

subject to the constraints

(i) $2x_1 + 4x_2 \leq 160$, (ii) $x_1 - x_2 = 30$, (iii) $x_1 \geq 10$

and $x_1, x_2 \geq 0$

Solution Since the objective function of the primal LP problem is of minimization, change all ≤ type constraints to ≥ type constraints by multiplying the constraint on both sides by –1. Also write = type constraint equivalent to two constraints of the type ≥ and ≤. Then the given primal LP problem can be rewritten as:

Minimize $Z_x = x_1 + 2x_2$

subject to the constraint

(i) $-2x_1 - 4x_2 \geq -160$, (ii) $x_1 - x_2 \geq 30$

(iii) $x_1 - x_2 \leq 30$ or $-x_1 + x_2 \geq -30$, (iv) $x_1 \geq 10$

and $x_1, x_2 \geq 0$

Let y_1, y_2, y_3 and y_4 be the dual variables corresponding to the four constraints in the given order. The dual of the given primal LP problem can then be formulated as follows:

Maximize $Z_y = -160y_1 + 30y_2 - 30y_3 + 10y_4$

subject to the constraints

(i) $-2y_1 + y_2 - y_3 + y_4 \leq 1$, (ii) $-4y_1 - y_2 + y_3 \leq 2$

and $y_1, y_2, y_3, y_4 \geq 0$

Let $y = y_2 - y_3$ ($y_2, y_3 \geq 0$). The above dual problem then reduces to the form

Maximize $Z_y = -160y_1 + 30y + 10y_4$

subject to the constraints

(i) $-2y_1 + y + y_4 \leq 1$, (ii) $-4y_1 - y \leq 2$

and $y_1, y_4 \geq 0$; y unrestricted in sign

Remark Since second constraint in the primal LP problem is equality, therefore as per rule 6 corresponding second dual variable y ($= y_2 - y_3$) should be unrestricted in sign.

Example 5.4 Obtain the dual LP problem of the following primal LP problem:

Minimize $Z_x = x_1 - 3x_2 - 2x_3$

subject to the constraints

(i) $3x_1 - x_2 + 2x_3 \leq 7$, (ii) $2x_1 - 4x_2 \geq 12$, (iii) $-4x_1 + 3x_2 + 8x_3 = 10$

and $x_1, x_2 \geq 0$; x_3 unrestricted in sign.

Solution Let y_1, y_2 and y_3 be the dual variables corresponding to three primal constraints in the given order. As the given LP problem is of minimization, all constraints can be converted to ≥ type by multiplying both sides by –1, i.e., $-3x_1 + x_2 - 2x_3 \geq -7$. Since the third constraint of the primal is an equation, the third dual variable y_3 will be unrestricted in sign. The dual of the given LP primal can be formulated as follows:

Maximize $Z_y = -7y_1 + 12y_2 + 10y_3$

subject to the constraints

(i) $-3y_1 + 2y_2 - 4y_3 \leq 1$, (ii) $y_1 - 4y_2 + 3y_3 \leq -3$, (iii) $-2y_1 + 8y_3 \leq -2$

and $y_1, y_2 \geq 0$; y_3 unrestricted in sign.

Example 5.5 Obtain the dual of the following primal LP problem

Maximize $Z_x = x_1 - 2x_2 + 3x_3$

subject to the constraints

(i) $-2x_1 + x_2 + 3x_3 = 2$, (ii) $2x_1 + 3x_2 + 4x_3 = 1$

and $x_1, x_2, x_3 \geq 0$

Solution Since both the primal constraints are of the equality type, the corresponding dual variables y_1 and y_2, will be unrestricted in sign. Following the rules of duality formulation, the dual of the given primal LP problem is

Minimize $Z_y = 2y_1 + y_2$

subject to the constraints

(i) $-2y_1 + 2y_2 \geq 1$, (ii) $y_1 + 3y_2 \geq -2$, (iii) $3y_1 + 4y_2 \geq 3$

and y_1, y_2 unrestricted in sign.

Example 5.6 Write the dual of the following primal LP problem

Maximize $Z = 3x_1 + x_2 + 2x_3 - x_4$

subject to the constraints

(i) $2x_1 - x_2 + 3x_3 + x_4 = 1$, (ii) $x_1 + x_2 - x_3 + x_4 = 3$

and $x_1, x_2 \geq 0$ and x_3, x_4 unrestricted in sign.

Solution Here we may apply the following rules of forming a dual of given primal LP problem.

(i) The x_3 and x_4 variables in the primal are unrestricted in sign, third and fourth constraints in the dual shall be equalities.

(ii) The given primal LP problem is of maximization; the first two constraints in the dual LP problem will therefore be ≥ type constraints.

(iii) Since both the constraints in the primal are equalities, the corresponding dual variables y_1 and y_2 will be unrestricted in sign.

If y_1 and y_2 are dual variables corresponding to the two primal constraints in the given order, the dual of the given primal can be written as:

Minimize $Z_y = y_1 + 3y_2$

subject to the constraints

(i) $2y_1 + y_2 \geq 3$, (ii) $-y_1 + y_2 \geq 1$, (iii) $3y_1 - y_2 = 2$

(iv) $y_1 + y_2 = -1$

and y_1, y_2 unrestricted in sign.

SELF PRACTICE PROBLEMS A

Write the dual of the following primal LP problems

1. Max $Z_x = 2x_1 + 5x_2 + 6x_3$
   subject to (i) $5x_1 + 6x_2 - x_3 \le 3$
   (ii) $-2x_1 + x_2 + 4x_3 \le 4$
   (iii) $x_1 - 5x_2 + 3x_3 \le 1$
   (iv) $-3x_1 - 3x_2 + 7x_3 \le 6$
   and $x_1, x_2, x_3 \ge 0$
   [*Sambalpur MSc* (*Maths*), *1996*]
2. Min $Z_x = 7x_1 + 3x_2 + 8x_3$
   subject to (i) $8x_1 + 2x_2 + x_3 \ge 3$
   (ii) $3x_1 + 6x_2 + 4x_3 \ge 4$
   (iii) $4x_1 + x_2 + 5x_3 \ge 1$
   (iv) $x_1 + 5x_2 + 2x_3 \ge 7$
   and $x_1, x_2, x_3 \ge 0$
3. Max $Z_x = 2x_1 + 3x_2 + x_3$
   subject to (i) $4x_1 + 3x_2 + x_3 = 6$, (ii) $x_1 + 2x_2 + 5x_3 = 4$
   and $x_1, x_2, x_3 \ge 0$
4. Max $Z_x = 3x_1 + x_2 + 3x_3 - x_4$
   subject to (i) $2x_1 - x_2 + 3x_3 + x_4 = 1$
   (ii) $x_1 + x_2 - x_3 + x_4 = 3$
   and $x_1, x_2, x_3, x_4 \ge 0$
5. Min $Z_x = 2x_1 + 3x_2 + 4x_3$
   subject to (i) $2x_1 + 3x_2 + 5x_3 \ge 2$
   (ii) $3x_1 + x_2 + 7x_3 = 3$
   (iii) $x_1 + 4x_2 + 6x_3 \le 5$
   and $x_1, x_2, \ge 0, x_3$ is unrestricted
6. Min $Z_x = x_1 + x_2 + x_3$
   subject to (i) $x_1 - 3x_2 + 4x_3 = 5$, (ii) $x_1 - 2x_2 \le 3$
   (ii) $2x_2 - x_3 \ge 4$
   and $x_1, x_2 \ge 0, x_3$ is unrestricted.
   [*Meerut, MSc* (*Maths*), *2005*]
7. Min $Z_x = 8x_1 + 3x_2$
   subject to (i) $x_1 - 6x_2 \ge 2$, (ii) $5x_1 + 7x_2 = -4$
   and $x_1, x_2 \ge 0$.
8. Max $Z_x = 8x_1 + 8x_2 + 8x_3 + 12x_4$
   subject to (i) $30x_1 + 20x_2 + 25x_3 + 40x_4 \le 800$
   (ii) $25x_1 + 10x_2 + 7x_3 + 15x_4 \le 250$
   (iii) $4x_1 - x_2 = 0$
   (iv) $x_3 \ge 5$
   and $x_1, x_2, x_3, x_4 \ge 0$
9. Min $Z_x = 18x_1 + 10x_2 + 11x_3$
   subject to (i) $4x_1 + 6x_2 + 5x_3 \ge 480$
   (ii) $12x_1 + 10x_2 + 10x_3 \ge 1{,}200$
   (iii) $10x_1 + 15x_2 + 7x_3 \le 1{,}500$
   (iv) $x_3 \ge 50$
   (v) $x_1 - x_2 \le 0$
   and $x_1, x_2, x_3 \ge 0$
10. Min $Z_x = 2x_1 - x_2 + 3x_3$
    subject to (i) $x_1 + 2x_2 + x_3 \ge 12$
    (ii) $x_2 - 2x_3 \ge -6$
    (iii) $6 \le x_1 + 2x_2 + 4x_3 \ge 24$
    and $x_1, x_2 \ge 0, x_3$ unrestricted.

HINTS AND ANSWERS

1. Min $Z_y = 3y_1 + 4y_2 + y_3 + 6y_4$
   subject to (i) $5y_1 - 2y_2 + y_3 - 3y_4 \ge 2$
   (ii) $6y_1 + y_2 - 5y_3 - 3y_4 \ge 5$
   (iii) $-y_1 + 4y_2 + 3y_3 + 7y_4 \ge 6$
   and $y_1, y_2, y_3, y_4 \ge 0$
2. Max $Z_y = 3y_1 + 4y_2 + y_3 + 7y_4$
   subject to (i) $8y_1 + 3y_2 + 4y_3 + y_4 \le 7$
   (ii) $2y_1 + 6y_2 + y_3 + 5y_4 \le 3$
   (iii) $y_1 + 4y_2 + 5y_3 + 2y_4 \le 8$
   and $y_1, y_2, y_3, y_4 \ge 0$
3. Min $Z_y = 6y_1 + 4y_2$
   subject to (i) $4y_1 + y_2 \ge 2$, (ii) $3y_1 + 2y_2 \ge 3$
   (iii) $y_1 + 5y_2 \ge 1$
   and y_1, y_2 unrestricted in sign.
4. Min $Z_y = y_1 + 3y_2$
   subject to (i) $2y_1 + y_2 \ge 3$
   (ii) $-y_1 + y_2 \ge 1$
   (iii) $3y_1 - y_2 \ge 3$
   (iv) $y_1 + y_2 \ge -1$
   and y_1, y_2 unrestricted in sign.
5. Max $Z_y = 2y_1 + 3y_2 - 5y_3$
   subject to $2y_1 + 3y_2 - y_3 \le 2$
   $3y_1 + y_2 - 4y_3 \le 3$
   $5y_1 + 7y_2 - 6y_3 = 4$
   and $y_1, y_3 \ge 0$ and y_2 unrestricted.
6. Max $Z_y = -5y_1 - 3y_2 + 4y_3$
   subject to (i) $-y_1 - y_2 \le 1$, (ii) $-3y_1 + 2y_2 + 2y_3 \le 1$
   $4y_1 - y_3 \le 1$
   and $y_2, y_3 \ge 0$ and y_1 is unrestricted.
7. Max $Z_y = 2y_1 - 4y_2$
   subject to (i) $y_1 + 5y_2 \le 8$, (ii) $-6y_1 + 7y_2 \le 3$
   and $y_1 \ge 0$ and y_2 is unrestricted.
8. Min $Z_y = 800y_1 + 250y_2 + y_3 + 5y_4$
   subject to $30y_1 + 25y_2 + 4y_3 \ge 8$
   $20y_1 + 10y_2 - y_3 \ge 8$
   $25y_1 + 7y_2 + y_4 \ge 8$
   $40y_1 + 15y_2 \ge 12$
   $y_1, y_2 \ge 0$ and y_3, y_4 are unrestricted.

5.3 STANDARD RESULTS ON DUALITY

See appendix for detail proof of the following standard results:

1. The dual of the dual LP problem is again the primal problem.
2. If either the primal or the dual problem has an unbounded objective function value, the other problem has no feasible solution.
3. If either the primal or dual problem has a finite optimal solution, the other one also possesses the same, and the optimal value of the objective functions of the two problems are equal, i.e. Max Z_x = Min Z_y. This analytical result is known as the *fundamental primal-dual relationship*. These results are summarized as follows.

Dual Problem (Max)	*Primal Problem (Min)*	
	Feasible	*Infeasible*
Feasible	Max Z_y = Min Z_x	Max $Z_y \rightarrow + \infty$
Infeasible	Min $Z_x \rightarrow - \infty$	Unbounded or infeasible

4. *Complementary slackness* property of primal-dual relationship states that for a positive basic variable in the primal, the corresponding dual variable will be equal to zero. Alternatively, for a non-basic variable in the primal (which is zero), the corresponding dual variable will be basic and positive.

5.3.1 Principle of Complementary Slackness

The principle of complementary slackness establishes the relationship between the optimal value of the main variables in one problem with their counterpart slack or surplus variables in other problem. Thus, this principle can also be helpful in obtaining the primal LP problem solution when only the dual solution is known. To illustrate this concept, let us consider the following example.

1. If at the optimal solution of a primal problem, a primal constraint has a positive value of a slack variable, the corresponding resource is not completely used and must have zero opportunity cost (shadow price). This means that having more of this resource will not improve the value of the objective function. But if the value of slack variable is zero in that constraint, the entire resource is being used and must have a positive opportunity cost, i.e. additional resource will improve the value of objective function by allowing more production.

   Since resources are represented by slack variables in the primal and by main variables in the dual, therefore, the principle of complementary slackness states that for every resource, the following condition must hold:

   Primal slack variable × Dual main variable = 0

2. In cases where the resources are not completely used, the opportunity cost of such resources exceeds the unit profit from that variable. But, if few units of the variable are obtained, then the opportunity cost of resources used must equal the profit from each unit of the variable, so that the surplus cost is zero. A primal variable in primal LP problem represents the quantity to be obtained, and the surplus variables in dual represent surplus costs. In this case the principle of complementary slackness states that for every unit of the variable, the following condition must hold.

   Primal main variable × Dual surplus variable = 0

5.4 MANAGERIAL SIGNIFICANCE OF DUALITY

The study of dual LP problem provides information about the value of the resources. Thus, a thorough economic analysis is required to decide whether additional units of any resource are needed. If yes, then how much cost per unit is required to be paid for any resource.

The significance of the study of dual is summarized as follows:

(i) The right-hand side of the constraint in primal LP problem represents the amount of a resource available and the associated dual variable value is interpretated as the maximum amount likely to be paid for an additional unit of this resource.

(ii) The maximum amount that should be paid for one additional unit of a resource is called its *shadow price* (also called *simplex multiplier*).
(iii) The total shadow price (or marginal value) of the resources equals the optimal value of objective function.
(iv) The value of the *i*th dual variable represents the rate at which the primal objective function value will increase by increasing the right-hand side (resource value) of constraint *i*, assuming that all other data remain unchanged.

Remark When the cost of a resource is *sunk* (cost that is incurred no matter irrespective to the value the dual variables). When the cost of a resource vary with the value of decision variables the shadow price represents the extra cost likely to be paid for one unit of the resource.

Example 5.7 The optimal solution simplex table for the primal LP problem

Maximize $Z = 3x_1 + 4x_2 + x_3$

subject to the constraints

(i) $x_1 + 2x_2 + 3x_3 \le 90$ (time for operation 1), (ii) $2x_1 + x_2 + x_3 \le 60$ (time for operation 2)
(iii) $3x_1 + x_2 + 2x_3 \le 80$ (time for operation 3)

and $x_1, x_2, x_3 \ge 0$

is given below:

		$c_j \rightarrow$	3	4	1	0	0	0
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value x_B (= b)	x_1	x_2	x_3	s_1	s_2	s_3
4	x_2	40	0	1	10/6	4/6	– 1/3	0
3	x_1	10	1	0	– 1/3	– 1/3	2/3	0
0	s_3	10	0	0	8/6	8/6	– 10/6	1
Z = 190		$c_j - z_j$	0	0	– 28/6	– 10/6	– 2/3	0

(a) Find the solution, maximum profit, idle capacity and the loss of the total contribution of every unit reduced from the right-hand side of the constraints.
(b) Write the dual of the given problem and give the initial simplex table.

Solution Considering this LP problem in the context of profit maximization production problem with constraints on time input resource.

The primal LP problem is concerned with maximizing the profit contribution from three products say A, B and C, while the dual will be concerned with evaluating the time used in the three operations for producing the three products.

The production capacity of the three operations is a resource to the firm. This LP problem can be solved along the following lines.

(a) The optimal solution of the primal LP problem can be read from the given table as:

$x_1 = 10, x_2 = 40, x_3 = 0$ and Max (profit) Z_x = Rs 190

The value of slack variable, $s_3 = 10$ units in the optimal solution is representing idle capacity in operation 3.

Interpretation of dual variables The absolute values of $c_j - z_j$ in the optimal solution simplex table, under the slack variables columns represent the values of dual variables, i.e. marginal value (shadow price) or worth of one hour in each operation 1 and 2. The optimal solution of dual LP problem is:

$$y_1 = 10/6, y_2 = 2/3, y_3 = 0 \text{ and Mix } Z_y = 190$$

Thus optimal value of the dual variable y_1 = Rs 10/6 or Rs 1.66 represents the per unit price (worth or marginal price) of the first resource (i.e. time available in operation 1). This means that marginal contribution of operation 1 to the total profit is Rs 1.66. In other words, if one productive unit is removed from the right-hand side of the first constraint, the total contribution would reduce by Rs 10/6, or Rs 1.66.

However, if one productive unit is removed up to the extent of idle capacity from the right-hand side of the third constraint (time available in operation 1), the total contribution would not be affected.

(b) Let y_1, y_2 and y_3 be the dual variables representing per unit price (worth) for time available in the operations 1, 2 and 3, respectively. Then the dual of the given problem is stated below:

Minimize $Z = 90y_1 + 60y_2 + 80y_3$

subject to the constraints

(i) $y_1 + 2y_2 + 3y_3 \geq 3$, (ii) $2y_1 + y_2 + y_3 \geq 4$, (iii) $3y_1 + y_2 + 2y_3 \geq 1$

and $y_1, y_2, y_3 \geq 0$

Interpretation of dual constraints Suppose company wants to continue production of product *C*. To see economic worth of *C*, examine $c_3 - z_3$ value under column-x_3. Producing *C* will be economical only if $c_3 > z_3$. This is possible either by increasing the profit per unit c_3 or decreasing the cost of the used resources z_3 ($= 3y_1 + y_2 + 2y_3$). An increase in the unit profit may not be possible because of competitive market. A decrease in z_3 is possible by reducing the unit usage of time in operation 3. Let t_1, t_2 and t_3 be the unit times of three operations to be reduced. Then to determine t_1, t_2 and t_3 such that the new cost, z_3 of three operations falls below the unit profit, i.e.

The shadow price is represented by the $c_j - z_j$ value under slack variable columns in the optimal simplex table.

$$3(1-t_1)y_1 + (1-t_2)y_2 + 2(1-t_3)y_3 < 1$$

For given values of $y_1 = 5/3$, $y_2 = 2/3$ and $y_3 = 0$, we get

$$3(1-t_1)(5/3) + (1-t_2)(2/3) < 1 \quad \text{or} \quad 5t_1 + 2t_2 > 16$$

Thus, any values of t_1 and t_2 that satisfy $5t_1 + 2t_2 > 16$ should make the product profitable. However, this may not be possible because this requires a reduction in the times of operation 1 and 2.

Example 5.8 XYZ Company has three departments – Assembly, Painting and Packing. The company can make three types of almirahs. An almirah of type I requires one hour of assembly, 40 minutes of painting and 20 minutes of packing time, respectively. Similarly, an almirah of type II needs 80 minutes, 20 minutes and one hour, respectively. The almirah of type III requires 40 minutes each of assembly, painting and packing time. The total time available at assembly, painting and packing departments is 600 hours, 400 hours and 800 hours, respectively. Determine the number of each type of almirahs that should be produced in order to maximize the profit. The unit profit for types I, II and III is Rs 40, 80 and 60, respectively.

Suppose that the manager of this XYZ company is thinking of renting the production capacities of the three departments to another almirah manufacturer – ABC company. ABC company is interested in minimizing the rental charges. On the other hand, the XYZ company would like to know the worth of production hours to them, in each of the departments, in order to determine the rental rates. (a) Formulate this problem as an LP problem and solve it to determine the number of each type of almirahs that should be produced by the XYZ company in order to maximize its profit. (b) Formulate the dual of the primal LP problem and interpret your results. [*Delhi Univ., MBA, 2002, 2005*]

Mathematical formulation The production data given in the problem may be summarized as below:

Types of Almirah	*Number of Hours Required (per Unit)*			*Profit per Unit*
	Assembly	*Painting*	*Packing*	*of Almirah (Rs)*
I	1	2/3	1/3	40
II	4/3	1/3	1	80
III	2/3	2/3	2/3	60
Total availability (hrs)	600	400	800	

Let x_1, x_2 and x_3 be the number of units of the three types of almirahs to be produced, respectively. The given problem can then be represented as an LP model:

Maximize $Z = 40x_1 + 80x_2 + 60x_3$

subject to the constraints

(i) $x_1 + (4/3)x_2 + (2/3)x_3 \le 600$, (ii) $(2/3)\,x_1 + (1/3)\,x_2 + (2/3)\,x_3 \le 400$

(iii) $(1/3)\,x_1 + x_2 + (2/3)\,x_3 \le 800$

and $x_1, x_2, x_3 \ge 0$

Let y_1, y_2 and y_3 be the per hour rental price for assembly, painting and packing departments, respectively to be paid by ABC company. Since company ABC desires to minimize total rental changes, then objective function becomes:

$$\text{Min (total rental charges) } Z_y = 600y_1 + 400y_2 + 800y_3$$

The given data show that an almirah of type I requires 1 assembly hour, 2/3 painting hour and 1/3 packing hour. The cost (i.e. rent) of time used for making almirah I is: $y_1 + (2/3)\,y_2 + (1/3)\,y_3$. If ABC Company used that time to make almirah I, it would earn Rs $c_1 = 40$ in contribution to profit, and so it will not rent out the time unless: $y_1 + (2/3)\,y_2 + (1/3)\,y_3 \ge 40$. Similarly, the company will work for almirahs II and III.

Since the total rent of all the departments should be greater than or equal to the profit from one unit of the almirah, the dual objective function along with the constraints that determines for ABC, the value of the productive resources, can be written as:

Minimize (total rent) $Z_y = 600y_1 + 400y_2 + 800y_3$

subject to the constraints

(i) $y_1 + (2/3)\,y_2 + (1/3)\,y_3 \ge 40$, (ii) $(4/3)\,y_1 + (1/3)\,y_2 + y_3 \ge 80$

(iii) $(2/3)\,y_1 + (2/3)\,y_2 + (2/3)\,y_3 \ge 60$

and $y_1, y_2, y_3 \ge 0$

This dual objective function (minimum total rent acceptable by the ABC company) is equal to the primal objective function (maximum profit that could be earned by XYZ company) from its resources.

Example 5.9 A firm manufactures two products A and B on machines I and II as shown below:

Machine	*Product*		*Available Hours*
	A	*B*	
I	30	20	300
II	5	10	110
Profit per unit (Rs)	6	8	

The total time available is 300 hours and 110 hours on machines I and II, respectively. Products A and B contribute a profit of Rs 6 and Rs 8 per unit, respectively. Determine the optimum product mix. Write the dual of this LP problem and give its economic interpretation.

Mathematical formulation The primal and the dual LP problems of the given problem are

Primal problem

x_1 and x_2 = number of units of A and B to be produced, respectively

$\text{Max } Z_x = 6x_1 + 8x_2$

subject to the constraints

$30x_1 + 20x_2 \le 300$

$5x_1 + 10x_2 \le 110$

and $x_1, x_2 \ge 0$

Dual problem

y_1 and y_2 = cost of one hour on machines I and II, respectively

$\text{Min } Z_y = 300y_1 + 110y_2$

subject to the constraints

$30y_1 + 5y_2 \ge 6$

$20y_1 + 10y_2 \ge 8$

and $y_1, y_2 \ge 0$

Solution of the primal problem The optimal solution of the primal problem is given in Table 5.2.

		$c_j \rightarrow$	6	8	0	0
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value x_B (= b)	x_1	x_2	s_{1p}	s_{2p}
6	x_1	4	1	0	1/20	– 1/10
8	x_2	9	0	1	– 1/10	3/20
Z = 96		$c_j - z_j$	0	0	– 1/10	– 6/10

Table 5.2 Optimal Solution of Primal Problem

Table 5.2 indicates that the optimal solution is to produce: x_1 = 4 units of product A; x_2 = 9 units of product B and Z_x = total maximum profit, Rs 96.

Solution of the dual problem The optimal solution of the dual problem can be obtained by applying the Big-M method. The optimal solution is shown in Table 5.3.

		$b_i \rightarrow$	300	110	0	0
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value y_B	y_1	y_2	s_{1d}	s_{2d}
300	y_1	1/10	1	0	– 1/20	1/40
110	y_2	6/10	0	1	1/10	– 3/20
Z = 96		$b_i - z_j$	0	0	4	9

Table 5.3 Optimal Solution of Dual Problem

The optimal solution as given in Table 5.3 is: y_1 = Rs 1/10 per hour on machine I; y_2 = Rs 6/10 per hour on machine II and Z_y = total minimum cost, Rs 96.

The values of y_1 = Rs 1/10 and y_2 = Rs 3/5, indicate the worth of one hour of machine time of machines I and II, respectively. However, since the machines hours can be increased beyond a specific limit, this result holds true only for a specific range of hours available on machines I and II.

Comparison of the solutions For interpreting the optimal solution of the primal (or dual), its solution values can be read directly from the optimal simplex table of the dual (or primal). The method can be summarized in the following steps.

1. The slack variables in the primal, correspond to the dual basic variables in the optimal solution and vice versa. For example, s_{1p} is the slack variable of the first primal constraint. It corresponds to the first dual variable y_1. Likewise s_{2p} corresponds to the second dual variable y_2. Moreover, s_{1d} and s_{2d} are the dual surplus variables and they correspond to x_1 and x_2, respectively of the primal problem. The correspondence between primal and dual variables is summarized in Table 5.4.

Primal		Dual	
Main variables	x_1, x_2	s_{1d}, s_{2d}	Surplus variables
Slack variables	s_{1p}, s_{2p}	y_1, y_2	Main variables

Table 5.4

2. The value in the $c_j - z_j$ row under columns of the slack/surplus variables, ignoring the negative sign, directly gives the optimal values of the dual/primal basic variables. For example, $c_3 - z_3$ = 1/10 and $c_4 - z_4$ = 6/10 values in Table 5.2 under primal slack s_{1p} and s_{2p} correspond to solution values of dual variables y_1 and y_2 in Table 5.3 and vice versa. The primal-dual relationship for these problems which are feasible can be summarized in Table 5.5.

Primal	*Dual*
• Values of the basic variables	• $(c_j - z_j)$ values in the non-basic surplus variable columns
• $(c_j - z_j)$ values in the non-basic slack variable columns	• Value of the basic variables

3. The optimal value of the objective function is the same for primal and dual LP problems.
4. The dual variable y_i represents the worth (dual price or shadow price) of one unit of resource *i*. For example, for y_1 the worth of time on machine I, equals Rs 1/10 and for y_2, the worth of time on machine II, equals Rs 6/10, as shown in Table 5.3.

Example 5.10 A company wishes to get at least 160 million 'audience exposures' the number of times one of the advertisements is seen or heard by a person. Because of the nature of the product the company wants at least 60 million month and at least 80 million of the exposures to involve persons between 18 and 40 years of age. The relevant information pertaining to the two advertising media under consideration—magazine and television is given below:

	Magazine	*Television*
• Cost per advertisement (Rs. thousand)	40	200
• Audience per advertisement (million)	4	40
• Audience per advertisement with monthly income over Rs. 10,000 (million)	3	10
• Audience (per advertisement) in the age group 18–40 (million)	8	10

The company wishes to determine the number of advertisements to be released each in magazine and television so as to keep the advertisement expenditure to the minimum. Formulate this problem as a LP problem. What will be the minimum expenditure and its allocation among the two media? Write 'dual' of this problem. Solve the 'dual' problem to find answer to the problem.

Solution The primal and dual LP problems of the given problem are:

Primal Problem

x_1, x_2 = number of advertisements in magazine and television, respectively.

$$\text{Minimize } Z_x = 40x_1 + 200x_2$$
$$\text{subject to } 4x_1 + 40x_2 \geq 160$$
$$3x_1 + 10x_2 \geq 60$$
$$8x_1 + 10x_2 \geq 80$$
$$\text{and } x_1, x_2 \geq 0.$$

Dual Problem

y_1, y_2, y_3 = shadow price (or worth) of one unit of advertisement over audience characteristics, respectively.

$$\text{Maximize } Z_y = 160y_1 + 60y_2 + 80y_3$$
$$\text{subject to } 4y_1 + 3y_2 + 8y_3 \leq 40$$
$$40y_1 + 10y_2 + 10y_3 \leq 200$$
$$\text{and } y_1, y_2, y_3 \geq 0.$$

Example 5.11 XYZ manufacturing company operates a three-shift system at one of its plants. In a certain section of the plant, the number of operators required on each of the three shifts is as follows:

Shift	*Number of Operators*
Day (6 a.m. to 2 p.m.)	50
Afternoon (2 p.m. to 10 p.m.)	24
Night (10 p.m. to 6 p.m.)	10

The company pays its operators at the basic rate of Rs. 10 per hour for those working on the day shift. For the afternoon and night shifts, the rates are one and a half times the basic rate and twice the basic rate, respectively. In agreement with each operator at the commencement of his employment, he is allocated to one of three schemes *A*, *B* or *C*. These are as follows:

A : Work (on average) one night shift, one afternoon shift, and two day shifts in every four shifts.

B : Work (one average) equal number of day and afternoon shifts.

C : Work day shifts only.

In schemes A and B, it is necessary to work strictly alternating sequences of specified shifts, as long as the correct proportion of shifts is worked in the long run.

(a) Formulate a linear programming model to obtain the required number of operators at minimum cost.

(b) By solving the dual of the problem, determine how many operators must be employed under each of the three schemes. Does this result in over-provision of operators on any one of the three shifts?

Solution The primal and dual LP problems of the given problem are:

Primal Problem

x_1, x_2 and x_3 = number of operators employed under scheme *A*, *B* and *C,* respectively.

$$\text{Minimize } Z_x = 20 \times \frac{1}{4}x_1 + 15\left(\frac{1}{4}x_1 + \frac{1}{2}x_2\right) + 10\left(\frac{1}{2}x_1 + \frac{1}{2}x_2 + x_3\right)$$

$$= \frac{55}{4}x_1 + \frac{25}{2}x_2 + 10x_3$$

subject to the constraints

$$(1/4)x_1 \geq 10,$$
$$(1/4)x_1 + (1/2)x_2 \geq 24$$
$$(1/2)x_1 + (1/2)x_2 + x_3 \geq 50$$

and $x_1, x_2, x_3 \geq 0.$

Dual Problem

y_1, y_2 and y_3 = shadow price (or worth) per unit of resources—operators in three shifts, respectively.

$$\text{Maximize } Z_y = 10y_1 + 24x_2 + 50y_3$$

subject to the constraints

$$(1/4)y_1 + (1/4)y_2 + (1/2)y_3 \leq 55/4$$
$$(1/2)y_2 + (1/2)y_3 \leq 25/2$$
$$y_3 \leq 10$$

and $y_1, y_2, y_3 \geq 0.$

5.5 ADVANTAGES OF DUALITY

1. It is advantageous to solve the dual of a primal that has a less number of constraints because the number of constraints usually equals the number of iterations required to solve the problem.
2. This avoids the necessity for adding surplus or artificial variables and solves the problem quickly (the technique is known as the *primal-dual method*). In economics, duality is useful in the formulation of the input and output systems. It is also useful in physics, engineering, mathematics, etc.
3. The dual variables provide an important economic interpretation of the final solution of an LP problem.
4. It is quite useful when investigating changes in the parameters of an LP problem (the technique is known as the *sensitivity analysis*).
5. Duality is used to solve an LP problem by the simplex method in which the initial solution is infeasible (the technique is known as the *dual simplex method*).

CONCEPTUAL QUESTIONS

1. Define the dual of a linear programming problem. State the functional properties of duality.
2. Explain the primal-dual relationship.
3. Briefly discuss 'duality' in linear programming.
4. What is the principle of duality in linear programming? Explain its advantages.
5. What is duality? What is the significance of dual variables in an LP model?
6. State the general rules for formulating a dual LP problem from its primal.
7. What is a shadow price? How does the concept relate to the dual of an LP problem?
8. How can the concept of duality be useful in managerial decision-making?
9. State and prove the relationship between the feasible solutions of an LP problem and its dual.
10. Prove that the necessary and sufficient condition for any LP problem and its dual, in order to have optimal solutions is that both have feasible solutions.

SELF PRACTICE PROBLEMS B

Write the dual of the following primal LP problems

1. Max $Z_x = 2x_1 + 5x_2 + 6x_3$
   subject to (i) $5x_1 + 6x_2 - x_3 \leq 3$ (ii) $-2x_1 + x_2 + 4x_3 \leq 4$
   (iii) $x_1 - 5x_2 + 3x_3 \leq 1$ (iv) $-3x_1 - 3x_2 + 7x_3 \leq 6$
   and $x_1, x_2, x_3 \geq 0.$

2. Min $Z_x = 7x_1 + 3x_2 + 8x_3$
   subject to (i) $8x_1 + 2x_2 + x_3 \geq 3$ (ii) $3x_1 + 6x_2 + 4x_3 \geq 4$
   (iii) $4x_1 + x_2 + 5x_3 \geq 1$ (iv) $x_1 + 5x_2 + 2x_3 \geq 7$
   and $x_1, x_2, x_3 \geq 0.$

3. Max $Z_x = 2x_1 + 3x_2 + x_3$
subject to (i) $4x_1 + 3x_2 + x_3 = 6$ (ii) $x_1 + 2x_2 + 5x_3 = 4$
and $x_1, x_2, x_3 \geq 0$.

4. Max $Z_x = 3x_1 + x_2 + 3x_3 - x_4$
subject to (i) $2x_1 - x_2 + 3x_3 + x_4 = 1$ (ii) $x_1 + x_2 - x_3 + x_4 = 3$
and $x_1, x_2, x_3, x_4 \geq 0$.

5. Min $Z_x = 2x_1 + 3x_2 + 4x_3$
subject to (i) $2x_1 + 3x_2 + 5x_3 \geq 2$ (ii) $3x_1 + x_2 + 7x_3 = 3$
(iii) $x_1 + 4x_2 + 6x_3 \leq 5$
and $x_1, x_2 \geq 0$, x_3 is unrestricted.

6. Min $Z_x = x_1 + x_2 + x_3$
subject to (i) $x_1 - 3x_2 + 4x_3 = 5$ (ii) $x_1 - 2x_2 \leq 3$
(iii) $2x_2 - x_3 \geq 4$
and $x_1, x_2 \geq 0$, x_3 is unrestricted.
[*Meerut Univ., MSc (Maths), 2004*]

7. One unit of product A contributes Rs. 7 and requires 3 units of raw material and 2 hours of labour. One unit of product B contributes Rs. 5 and requires one unit of raw material and one hour of labour. Availability of raw material at present is 48 units and there are 40 hours of labour.
   (a) Formulate this problem as a linear programming problem.
   (b) Write its dual.
   (c) Solve the dual by the simplex method and find the optimal product mix and the shadow prices of the raw material and labour.

8. A company makes three products: X, Y and Z out of three raw materials A, B and C. The raw material requirements are given below:

Raw Materials	*Number of Units of Raw Material Required to Produce One Unit of Product*		
	X	*Y*	*Z*
A	1	2	1
B	2	1	4
C	2	5	1

The unit profit contribution of the products: X, Y and Z is Rs. 40, 25 and 50, respectively. The number of units of raw material available are 36, 60 and 45, respectively.
   (a) Determine the product mix that will maximize the total profit.
   (b) Using the final simplex table, write the solution to the dual problem and give its economic interpretation.

9. Three food products are available at costs of Rs. 10, Rs. 36 and Rs. 24 per unit, respectively. They contain 1,000, 4,000 and 2,000 calories per unit, respectively and 200, 900 and 500 protein units per unit, respectively. It is required to find the minimum-cost diet containing at least 20,000 calories and 3,000 units of protein. Formulate and solve the given problem as an LP problem. Write the dual and use it to check the optimal solution of the given problem.

10. A company produces three products: P, Q and R from three raw materials A, B and C. One unit of product P requires 2 units of A and 3 units of B. A unit of product Q requires 2 units of B and 5 units of C and one unit of product R requires 3 units of A, 2 units of B and 4 units of C. The company has 8 units of material A, 10 units of material B and 15 units of material C available to it. Profits per unit of products P, Q and R are Rs. 3, Rs. 5 and Rs. 4, respectively.
   (a) Formulate this problem as an LP problem.
   (b) How many units of each product should be produced to maximize profit?
   (c) Write the dual of this problem.

11. A diet conscious housewife wishes to ensure certain minimum intake of vitamins A, B and C for the family. The minimum daily (quantity) needs of the vitamins A, B and C for the family are respectively 30, 20 and 16 units. For the supply of these minimum vitamin requirements, the housewife relies on two fresh food I and II. The first one provides 7, 5 and 2 units of the three vitamins per gram, respectively and the second one provides 2, 4 and 8 units of the same three vitamins per gram of the foodstuff, respectively. The first foodstuff costs Rs. 3 per gram and the second Rs. 2 per gram. The problem is how many grams of each foodstuff should the housewife buy everyday to keep her food bill as low as possible?
   (a) Formulate this problem as an LP model.
   (b) Write and then solve the dual problem.

12. A large distributing company buys coffee seeds from four different plantations. On these plantations, the seeds are available only in a blend of two types A and B. The company wants to market a blend consisting of 30 per cent of type A and 70 per cent of type B. The percentage of each type used by each plantation and the selling prices per 10 kg of the blends of each plantation are as follows:

Type	*Plantation*				*Desired*
	1	*2*	*3*	*4*	
A	40%	20%	60%	80%	30%
B	60%	80%	40%	20%	70%
Selling price per 5 kg	Rs. 3	Rs. 2	Rs. 1.20	Rs. 1.50	

What quantity of coffee seeds should the company buy from each plantation so that the total mixture will contain the desired percentages of A and B and at the same time keep the purchasing cost at a minimum? Also write the dual of the problem.

13. The XYZ Plastic Company has just received a government contract to produce three different plastic valves. These valves must be highly heat and pressure resistant and the company has developed a three-stage production process that will provide the valves with the necessary properties involving work in three different chambers. Chamber 1 provides the necessary pressure resistance and can process valves for 1,200 minutes each week. Chamber 2 provides heat resistance and can process valves for 900 minutes per week. Chamber 3 tests the valves and can work 1,300 minutes per week. The three valve types and the time in minutes required in each chamber are:

Valve Type	*Time Required in*		
	Chamber 1	*Chamber 2*	*Chamber 3*
A	5	7	4
B	3	2	10
C	2	4	5

The government will buy all the valves that can be produced and the company will receive the following profit margins on each valve: A, Rs. 15; B, Rs. 13.50; and C, Rs. 10.

How many valves of each type should the company produce each week in order to maximize profits? Write the dual of the given LP problem and give its economic interpretation?

14. A medical scientist claims to have found a cure for the common cold that consists of three drugs called K, S and H. His results indicate that the minimum daily adult dosage for effective treatment is 10 mg of drug K, 6 mg of drug S and 8 mg of drug H. Two substances are readily available for preparing pills for distribution to cold sufferers. Both substances contain all three of the required drugs. Each unit of substance A contains 6 mg, 1 mg and 2 mg of drugs K, S and H, respectively and each unit of substance B contains 2 mg, 3 mg and 2 mg of the same drugs. Substance A costs Rs. 3 per unit and substance B costs Rs. 5 per unit.
   (a) Find the least-cost combination of the two substances that will yield a pill designed to contain the minimum daily recommended adult dosage.
   (b) Suppose that the costs of the two substances are interchanged so that substance A costs Rs. 5 per unit and substance B costs Rs. 3 per unit. Find the new optimal solution.
[*Delhi Univ., MBA (HCA), 2008*]

15. The XYZ company has the option of producing two products during the period of slack activity. For the next period, production has been scheduled so that the milling machine is free for 10 hours and skilled labour will have 8 hours of time available.

Product	*Machine Time per Unit*	*Skilled Labour Time per Unit*	*Profit Contribution per Unit (Rs.)*
A	4	2	5
B	2	2	3

Solve the primal and dual LP problems and bring out the fact that the optimum solution of one can be obtained from the other. Also explain in the context of the example, what you understand by shadow prices (or dual prices or marginal value) of resource. [*Jammu Univ. MBA 2006*]

16. A company produces three products *P*, *Q* and *R* whose prices per unit are 3, 5 and 4 respectively. On unit of product *P* requires 2 units of m_1 and 3 units of m_2. A unit of product *Q* requires 2 units of m_2 and 5 units of m_3 and one unit of product *R* requires 3 units of m_1, 2 units of m_2 and 4 units of m_3. The company has 8 units of material m_1, 10 units of material m_2 and 15 units of material m_3 available to it.
    (a) Formulate the problem as on LP model.
    (b) How many units of each product should be produced to maximize revenue?
    (c) Write the dual problem.

17. A person consumes two types of food A and B everyday to obtain 8 units of proteins, 12 units of carbohydrates and 9 units of fats which is his daily minimum requirements. 1 kg of food A contains 2, 6 and 1 units of protein, carbohydrates and fats, respectively. 1 kg of food B contains 1, 1 and 3 units of proteins, carbohydrates and fats respectively. Food A costs Rs. 8.50 per kg, while B costs Rs. 4 per kg. Determine how many kg of each food should he buy daily to minimize his cost of food and still meet the minimum requirements.

    Formulate on LP problem mathematically. Write its dual and solve the dual by the simplex method. *[Gujarat Univ. MBA, 2006]*

18. A firm produces three articles *X*, *Y*, *Z* at a total cost of Rs. 4 , Rs. 3, and Rs. 6 per item respectively. Total number of *X* and *Z* item produced should be at least 2 and number of *Y* and *Z* together be at least 5. The firm wants to minimize the cost. Formulate this problem as an LP problem. Write its dual. Solve the dual by the simplex method. Can you point out the solution of the primal problem? If yes, what is it?

19. A firm produces three types of biscuits *A*, *B* and *C*. It packs them in assortments of two sizes I and II. The size I contains 20 biscuits of type *A*, 50 of type *B* and 10 of type *C*. The size II contains 10 biscuits of type *A*, 80 of type *B* and 60 of type *C*. A buyer intends to buy at least 120 biscuits of type *A*, 740 of type *B* and 240 of type *C*. Determine the least number of packets he should buy. Write the dual LP problem and interpret your answer.

20. Consider the following product mix problem: Let x_1 denote number of units of Product 1 to be produced daily and x_2 the number of units of Product 2 to be produced daily. The production of Product 1 requires one hour of processing time in department D_1. Production of 1 unit of Product 2 requires 2 hours of processing time in department D_1 and one hour in department D_2. The number of hours available in department D_1 are 32 hours and in department D_2, 8 hours. The contribution of one unit of Product 1 is Rs. 200 and of Product 2 is Rs. 300.

    The solution to this LP model is given below:

		$c_j \rightarrow$	*200*	*300*	*0*	*0*
c_B	*Basic Variables*	*Solution Values* x_B	x_1	x_2	s_1	s_2
200	x_1	32	1	2	1	0
0	s_2	8	0	1	0	1
Z = 6400		$c_j - z_j$	–100	–200	0	6400

Given the dual of the primal model. Obtain the optimum solution to the dual LP model from the above table. Interpret the dual variables.

HINTS AND ANSWERS

1. Min $Z_y = 3y_1 + 4y_2 + y_3 + 6y_4$
   subject to (i) $5y_1 - 2y_2 + 5y_3 - 3y_4 \geq 2$
   (ii) $6y_1 + y_2 - 5y_3 - 3y_4 \geq 5$
   (iii) $-y_1 + 4y_2 + 3y_3 + 7y_4 \geq 6$
   and $y_1, y_2, y_3, y_4 \geq 0$.

2. Max $Z_y = 3y_1 + 4y_2 + y_3 + 7y_4$
   subject to (i) $8y_1 + 3y_2 + 4y_3 + y_4 \leq 7$
   (ii) $2y_1 + 6y_2 + y_3 + 5y_4 \leq 3$
   (iii) $y_1 + 4y_2 + 5y_3 + 2y_4 \leq 8$
   and $y_1, y_2, y_3, y_4 \geq 0$.

3. Min $Z_y = 6y_1 + 4y_2$
   subject to (i) $4y_1 + y_2 \geq 2$ (ii) $3y_1 + 2y_2 \geq 3$
   (iii) $y_1 + 5y_2 \geq 1$
   and y_1, y_2 unrestricted in sign.

4. Min $Z_y = y_1 + 3y_2$
   subject to (i) $2y_1 + y_2 \geq 3$ (ii) $-y_1 + y_2 \geq 1$
   (iii) $3y_1 - y_2 \geq 3$ (iv) $y_1 + y_2 \geq -1$
   and y_1, y_2, unrestricted in sign.

5. Max $Z_y = 2y_1 + 3y_2 - 5y_3$
   subject to (i) $2y_1 + 3y_2 - y_3 \leq 2$ (ii) $3y_1 + y_2 - 4y_3 \leq 3$
   (iii) $5y_1 + 7y_2 - 6y_3 = 4$
   and $y_1, y_3 \geq 0$ and y_2 unrestricted.

6. Max $Z_y = -5y_1 - 3y_2 + 4y_3$
   subject to (i) $-y_1 - y_2 \leq 1$ (ii) $3y_1 + 2y_2 + 2y_3 \leq 1$
   (iii) $-4y_1 - y_3 \leq 1$
   and $y_2, y_3 \geq 0$ and y_1 is unrestricted.

7. *Primal*
   x_1 and x_2 = number of units of products A and B, respectively to be produced.
   Max $Z_x = 7x_1 + 5x_2$
   subject to (i) $3x_1 + x_2 \leq 48$ (ii) $2x_1 + x_2 \leq 40$
   and $x_1, x_2 \geq 0$
   Ans. $x_1 = 0, x_2 = 40$, Max Z_x = Rs. 200
   Dual
   y_1 and y_2 = worth of one unit of raw material and labour, respectively.

Min $Z_y = 48y_1 + 40y_2$

subject to (i) $3y_1 + 2y_2 \geq 7$ (ii) $y_1 + y_2 \geq 5$

and $y_1, y_2 \geq 0$

Ans. $y_1 = 0, y_2 = 5$ and Min Z_y = Rs. 200

8. *Primal*

x_1, x_2 and x_3 = units of the products X, Y and Z, respectively to be produced.

Max $Z_x = 40x_1 + 25x_2 + 50x_3$

subject to (i) $x_1 + x_2 + x_3 \leq 36$ (ii) $2x_1 + x_2 + 4x_3 \leq 60$

(iii) $2x_1 + 5x_2 + x_3 \leq 45$

and $x_1, x_2, x_3 \geq 0$.

Ans. $x_1 = 20, x_2 = 0, x_3 = 5$; Max Z_x = 1,050

Dual

y_1, y_2 and y_3 = worth (or shadow price) per unit of raw materials A, B and C, respectively.

Min $Z_x = 36y_1 + 60y_2 + 45y_3$

subject to (i) $y_1 + 2y_2 + 2y_3 \geq 40$ (ii) $y_1 + y_2 + 5y_3 \geq 25$

(iii) $y_1 + 4y_2 + y_3 \geq 50$

and $x_1, x_2, x_3 \geq 0$.

Ans. $y_1 = 0, y_2 = 0, y_3 = 10$; Min Z_y = 1,050

10. *Primal*

x_1, x_2 and x_3 = units of the products P, Q and R to be produced, respectively.

Max $Z = 3x_1 + 5x_2 + 4x_3$

subject to (i) $2x_1 + 3x_3 \leq 8$ (ii) $5x_1 + 2x_2 + 2x_3 \leq 10$

(iii) $5x_2 + 4x_3 \leq 15$

and $x_1, x_2, x_3 \geq 0$.

Dual

Min $Z = 8y_1 + 10y_2 + 15y_3$

subject to (i) $2y_1 + 5y_2 \geq 3$ (ii) $2y_2 + 5y_3 \geq 5$

(iii) $3y_1 + 2y_2 + 4y_3 \geq 4$

and $y_1, y_2, y_3 \geq 0$.

11. *Primal*

x_1 and x_2 = units of fresh food I and II, to be bought, respectively

Min $Z = 3x_1 + 2x_2$

subject to (i) $7x_1 + 2x_2 \geq 30$ (ii) $5x_1 + 4x_2 \geq 20$

(iii) $2x_1 + 8x_2 \geq 16$

and $x_1, x_2 \geq 0$.

Dual

y_1, y_2 and y_3 = worth per unit of vitamins A, B and C, respectively to the body.

Max $Z = 30y_1 + 20y_2 + 16y_3$

subject to (i) $7y_1 + 5y_2 + 2y_3 \leq 3$ (ii) $2y_1 + 4y_2 + 8y_3 \leq 2$

and $y_1, y_2, y_3 \geq 0$.

(b) The optimal solution to the dual problem is: $y_1 = 20/52$, $y_2 = 0$, $y_3 = 8/52$ and Max Z = Rs. 14.

12. *Primal*

x_1, x_2, x_3 and x_4 = quantities of coffee seeds, respectively which the company buys from each plantation.

Min = $3x_1 + 2x_2 + 1.2x_3 + 1.5x_4$

subject to (i) $0.4x_1 + 0.2x_2 + 0.6x_3 + 0.8x_4 \geq 0.3$

(ii) $0.6x_1 + 0.8x_2 + 0.4x_3 + 0.2x_4 \geq 0.7$

and $x_1, x_2, x_3, x_4 \geq 0$.

Dual

y_1 and y_2 = worth of coffee seeds A and B, respectively

Max $Z_y = 0.3y_1 + 0.7y_2$

subject to (i) $0.4y_1 + 0.6y_2 \leq 3$ (ii) $0.2y_1 + 0.8y_2 \leq 2$

(iii) $0.6y_1 + 0.4y_2 \leq 1.2$ (iv) $0.8y_1 + 0.2y_2 \leq 1.5$

and $y_1, y_2 \geq 0$

13. *Primal*

x_1, x_2 and x_3 = number of valves of the types A, B and C, respectively to be produced.

Max $Z_x = 15x_1 + 13.5x_2 + 10x_3$

subject to (i) $5x_1 + 3x_2 + 2x_3 \leq 1{,}200$

(ii) $7x_1 + 2x_2 + 4x_3 \leq 900$

(iii) $4x_1 + 10x_2 + 5x_3 \leq 1{,}300$

and $x_1, x_2, x_3 \geq 0$.

Ans. $x_1 = 22{,}400/217$, $x_2 = 5{,}500/62$, $x_3 = 0$ and Max Z_x = Rs. 58,125/31

Dual

y_1, y_2 and y_3 = work of Chambers 1, 2 and 3 production capacity, respectively.

Min $Z_y = 1{,}200y_1 + 900y_2 + 1{,}300y_3$

subject to (i) $5y_1 + 7y_2 + 4y_3 \geq 15$

(ii) $3y_1 + 2y_2 + 10y_3 \geq 13.5$

(iii) $2y_1 + 4y_2 + 5y_3 \geq 10$

and $y_1, y_2, y_3 \geq 0$.

Ans. $y_1 = 0, y_2 = -41/31, y_3 = -645/62$.

15. *Primal*

x_1, x_2 = number of units of product A and B respectively to be produced

Max $Z_x = 5x_1 + 3x_2$

subject to (i) $4x_1 + 2x_2 \leq 10$ (ii) $2x_1 + 2x_2 \leq 8$

and $x_1, x_2 \geq 0$

Ans. $x_1 = 1, x_2 = 3$ and Max Z_x = 14

Dual

y_1, y_2, y_3 = worth (or shadow price) per unit of resource operators in three shifts respectively.

Min $Z_y = 10y_1 + 8y_2$

subject to (i) $4y_1 + 2y_2 \geq 5$ (ii) $2y_1 + 2y_2 \geq 3$

and $y_1, y_2 \geq 0$.

Ans. $y_1 = 1, y_2 = 1/2$ and Min Z_y = 14.

16. *Primal*

x_1, x_2, x_3 = number of units of product P, Q and R to be produced, respectively.

Max $Z_x = 3x_1 + 5x_2 + 4x_3$

subject to (i) $2x_1 + 3x_3 \leq 8$ (ii) $5x_1 + 2x_2 + 2x_3 \leq 10$

(iii) $5x_2 + 4x_3 \leq 15$

and $x_1, x_2, x_3 \geq 0$.

Ans. $x_1 = 44/41, x_2 = 59/41, x_3 = 80/41$

Dual

y_1, y_2, y_3 = worth per unit of material m_1, m_2 and m_3 respectively.

Min $Z_y = 8y_1 + 10y_2 + 15y_3$

subject to (i) $2y_1 + 3y_2 \geq 3$ (ii) $2y_2 + 5y_3 \geq 5$

(iii) $3y_1 + 2y_2 + 4y_3 \geq 4$

and $y_1, y_2, y_3 \geq 0$.

17. *Primal*

x_1, x_2 = number of units of food A and B to be consumed, respectively.

Max $Z_x = 8.50x_1 + 4x_2$

subject to (i) $2x_1 + x_2 \geq 8$ (ii) $6x_1 + x_2 \geq 12$

(iii) $x_1 + 3x_2 \geq 9$

and $x_1, x_2 \geq 0$.

Ans. $x_1 = 1, x_2 = 6$ and Min $Z_x = 65/2$;

Dual

y_1, y_2, y_3 = worth per unit of proteins, carbohydrates and fats respectively.

Min $Z_y = 8y_1 + 12y_2 + 9y_3$

subject to (i) $2y_1 + 6y_2 + y_3 \leq 8.50$ (ii) $y_1 + y_2 + 3y_3 \leq 4$

and $y_1, y_2, y_3 \geq 0$

Ans. $y_1 = 31/8, y_2 = 1/8, y_3 = 0$ and Max $Z_y = 65/2$;

18. *Primal*

x_1, x_2, x_3 = number of units of articles X, Y and Z respectively.

Max $Z_x = 4x_1 + 3x_2 + 6x_3$

subject to (i) $x_1 + x_3 \geq 2$ (ii) $x_2 + x_3 \geq 5$

and $x_1, x_2, x_3 \geq 0$.

Ans. $x_1 = 0, x_2 = 3, x_3 = 2$ with Min $Z = 21$

Dual

y_1, y_2 = worth per unit of resources to be used, respectively

Min $Z_y = 2y_1 + 5y_2$

subject to (i) $y_1 \leq 4$ (ii) $y_2 \leq 3$

(iii) $y_1 + y_2 \leq 6$

and $y_1, y_2 \leq 0$

Ans. $y_1 = 3, y_2 = 3$ with Max $Z^* = 21$

19. *Primal*

x_1, x_2 = number of assortments of size I and II, respectively.

Max $Z_x = x_1 + x_2$

subject to (i) $20x_1 + 10x_2 \geq 120$ (ii) $50x_1 + 80x_2 \geq 740$

(iii) $10x_1 + 60x_2 \geq 240$

and $x_1, x_2 \geq 0$

Ans. $x_1 = 2, x_2 = 8$ and Min $Z_x = 10$

Dual

y_1, y_2, y_3 = worth per unit of biscuits of type A, B and C, respectively.

Min $Z_y = 120y_1 + 740y_2 + 240$

subject to (i) $20y_1 + 50y_2 + 10y_3 \leq 1$

(ii) $10y_1 + 80y_2 + 60y_3 \leq 1$

and $y_1, y_2, y_3 \geq 0$

20. *Primal*

Max $Z_x = 200x_1 + 300x_2$

subject to (i) $x_1 + 2x_2 \leq 32$ (ii) $0x_1 + x_2 \leq 8$

and $x_1, x_2 \geq 0$.

Dual

Min $Z_y = 32y_1 + 8y_2$

subject to (i) $y_1 + 0y_2 \geq 200$ (ii) $2y_1 + y_2 \geq 300$

and $y_1, y_2 \geq 0$

The optimum solution to the dual LP problem is: $y_1 = 200, y_2 = 0 s_1 = 0 s_2 = 100, Z^* =$ Rs. 600

Note that, y_1 = marginal increase in profit for an addition 1 hour of capacity. Since y_1 = Rs. 200, profit becomes Rs. 6400 + 200 = Rs. 6600. Also y_2 = marginal increase in profit given an addition hour of capacity in D_2. Since $y_2 = 0$, there would be no change in the profit.

In primal LP problem solution, $s_2 = 8$ indicates that there is an excess capacity of 8 hours in D_2. Thus, in dual solution we have $y_2 = 0$. Dual constraint (i) is written as: $y_1 + 0y_2 \geq 200$ or

$$\begin{bmatrix}\text{total value of hours in department } D_1 \text{ and } D_2 \\ \text{required to produce 1 unit of Product 1}\end{bmatrix} \geq \begin{bmatrix}\text{unit profit received from manufacturing} \\ \text{and selling 1 unit of Product 1}\end{bmatrix}$$

CHAPTER SUMMARY

In this chapter we discussed the relationship between an LP problem and its dual. The method of formulating a dual LP problem of the given LP problem was explained with several examples. The main focus, while solving a dual LP problem, was to find for each resource its best marginal value, also known as dual or shadow price. This value reflects the maximum additional price to be paid in order to obtain one additional unit of any resource in order to maximize profit (or minimize cost) under resource constraints.

CHAPTER CONCEPTS QUIZ

True or False

1. If the objective function of the primal LP problem is maximized, then the objective function of the dual is to be minimized.
2. A dual variable is not defined for each constraint in the primal LP problem.
3. The dual of dual is a primal problem.
4. The value in the $c_j - z_j$ row under columns of slack/surplus variables ignoring negative signs do not give the direct optimal values of the dual/primal basic variables.

5. The shadow price indicates how much increase in a right hand side coefficient will change the optimality range of an objective function coefficient.
6. The right hand side coefficient of some constraint in a primal LP problem appears in the corresponding dual as a coefficient in the objective function.
7. The change in the objective function value due to per unit decrease in the right hand side of a constraint is given by the negative of the shadow price for that constraint.
8. A variable which is in the solution of a linear programming problem is called basic variable.
9. The $c_j - z_j$ values in the non-basic slack variable columns is equal to the value of the basic variable.
10. The number of constraints in any LP problem usually equals the number of iterations required to solve the problem.

Fill in the Blanks

11. The optimal value of the objective function is same for the —— ———— and ———— problem.
12. The ———— variables provide the important economic interpretation of the final solution of the LP problem.
13. Duality is used to solve an LP problem by ———— method in which the initial solution is infeasible.
14. If either of the primal or dual LP problem has an ———— objective function, then the other problem has ———— solution.
15. If the *i*th primal constraint is of ———— type, then the *i*th dual variable will be ———— in sign and vice versa.
16. ———— property states that for a positive basic variable in the primal, the corresponding dual variable will be ———— to zero.
17. The maximum amount that should be paid for one additional unit of a resource is called its ———.
18. The ———— variables in the primal correspond to the dual ———— variables in the optimal solution.
19. Linear programming models are composed of three parts: ———, ————, and ———.
20. Associated with any linear programming problem is another linear problem called the ————, which can be used to evaluate the resources in the original problem. If the original linear program, called the ———— is a maximization problem, then this associated problem is a ———— problem.

Multiple Choice

21. The dual of the primal maximization LP problem having *m* constraints and *n* non-negative variables should
    (a) have *n* constraints and *m* non-negative variables
    (b) be a minimization LP problem
    (c) both (a) and (b)
    (d) none of the above
22. For any primal problem and its dual,
    (a) optimal value of objective functions is same
    (b) primal will have an optimal solution if and only if dual does too
    (c) both primal and dual cannot be infeasible
    (d) all of the above
23. The right-hand side constant of a constraint in a primal problem appears in the corresponding dual as
    (a) a coefficient in the objective function
    (b) a right-hand side constant of a constraint
    (c) an input-out coefficient
    (d) none of the above
24. Dual LP problem approach attempts to optimize resource allocation by ensuring that
    (a) marginal opportunity cost of a resource equals its marginal return
    (b) marginal opportunity cost of a resource is less than its marginal return
    (c) both (a) and (b)
    (d) none of the above
25. Shadow price indicates how much one unit change in the resource value will change the
    (a) optimality range of an objective function
    (b) optimal value of the objective function
    (c) value of the basic variable in the optimal solution
    (d) none of the above
26. Principle of complementary slackness states that
    (a) primal slack × dual main = 0
    (b) primal main × dual surplus = 0
    (c) both (a) and (b)
    (d) none of the above
27. If dual has an unbounded solution, primal has
    (a) no feasible solution
    (b) unbounded solution
    (c) feasible solution
    (d) none of the above
28. If at the optimality a primal constraint has a positive value of a slack variable, then
    (a) dual variable corresponding to that constraint has zero value
    (b) corresponding resource is not completely used up
    (c) corresponding resource have zero opportunity cost
    (d) both (b) and (c) but not (a)
29. The shadow price for a resources is the
    (a) price that is paid for the purchase of a resource
    (b) saving by eliminating one of the excess quantities of resource
    (c) increase in the objective function value by providing one additional unit of resource
    (d) none of the above
30. The value of dual variable
    (a) represents worth of each additional unit of resource
    (b) can be obtained by examining the z_j row of primal optimal simplex table
    (c) can be obtained by examining $c_j - z_j$ row of primal optimal simplex table
    (d) all of the above
31. If value of primal slack variable in the optimal simplex table is zero, then
    (a) LP problem has multiple solutions
    (b) dual variable value associated with that constraint is zero
    (c) resource measured by that constraint is not fully utilized
    (d) all of the above
32. The shadow price of a greater than or equal to (≥) constraint in a maximization LP problem can be
    (a) zero (b) positive
    (c) negative (d) all of the above
33. The imputed cost of used resources per unit of an activity is
    (a) less than or equal to the profit per unit of the activity
    (b) greater than or equal to the profit per unit of the activity
    (c) strictly equal to the profit per unit of the activity.
    (d) neither (a) nor (b)
34. Dual constraints for the maximization LP problem can be written as
    (a) $\Sigma a_{ij} y_i \geq c_j$ (b) $\Sigma a_{ji} y_i \geq c_j$
    (c) $\Sigma a_{ij} y_j \leq c_j$ (d) $\Sigma a_{ji} y_i \leq c_j$
35. If a primal LP problem has a finite solution then the dual LP problem should have
    (a) finite solution (b) infeasible solution
    (c) unbounded solution (d) none of the above

Answers to Quiz

1. T 2. F 3. T 4. F 5. F 6. T 7. T 8. T 9. T 10. T
11. Primal, dual 12. Dual 13. Simplex 14. Unbounded; infeasible 15. Equality; unrestricted
16. Complementary slackness; equal 17. Shadow price 18. Slack; basic
19. Decision variables; constraints; objective function 20. Dual; primal; minimization
21. (c) 22. (b) 23. (a) 24. (a) 25. (b) 26. (c) 27. (a) 28. (d) 29. (c) 30. (a) 31. (c)
32. (d) 33. (a) 34. (b) 35. (a)

CASE STUDY

Case 5.1: Bhawani Manufacturer

Bhawani Manufacturer has discontinued production of a certain unprofitable product line and this has created considerable excess production capacity. The management is considering to devote this excess capacity to produce one or more of three products 1, 2 and 3. The available excess capacity on the machines which might limit output, is summarized in the following table:

Machine type	*Available Excess Capacity (Machine Hours per Week)*
Milling machine	250
Lathe	150
Grinder	50

The number of machine-hours for each unit of the respective product is given below.

	Capacity Requirement (Machine-hours per Unit)		
Machine Type	*Product 1*	*Product 2*	*Product 3*
Milling machine	8	2	3
Lathe	4	3	0
Grinder	2	0	1

The per unit contribution would be Rs. 20, Rs. 6 and Rs. 8, respectively for products 1, 2 and 3.

Suggest to the management of the company a production plan to produce and sell its three products so as to maximize total revenue. Also, evaluate shadow price of each resource.

Case 5.2: Raman Traders

The procurement manager of Raman Traders that manufactures special gasoline additives must determine the proper amounts of each raw material to be purchased for the production of one of its products. Three raw materials are available. Each litre of the finished product must have at least a combustion point of 222°F. In addition, the gamma content (which cause hydrocarbon pollution) cannot exceed 6 per cent of volume. The zeta content (which is goods for cleaning the internal moving parts of engines) must be at least 12 per cent by volume. Each raw material contains three elements in varying amounts, as shown in the given table.

Raw material A costs Rs. 9.00 per litre, whereas raw materials B and C cost Rs. 6.00 and Rs. 7.50 per litre, respectively.

	Raw Material		
	A	*B*	*C*
Combustion point, °F	200	180	280
Gamma content, %	4	3	10
Zeta content, %	20	10	8

The procurement manager wishes to minimize the cost of raw materials per litre of product.

You as management trainee are expected to suggest method to find the optimal proportions of each raw material to be used in a litre of the finished product. Also write the dual of the given problem and give its economic interpretation.

APPENDIX: THEOREMS ON DUALITY

Theorem 5.1: The dual of the dual is the primal.

Proof: Consider the primal problem in canonical form:

$$\text{Minimize } Z_x = \mathbf{cx}$$
$$\text{subject to} \quad \mathbf{Ax} \geq \mathbf{b};\ \mathbf{x} \geq 0 \tag{1}$$

where $\mathbf{A}$ is an $m \times n$ matrix, $\mathbf{b}^T \in E^m$ and $\boldsymbol{c}, \boldsymbol{x}^T \in E^n$.

Applying the transformation rules, the dual of this problem is:

$$\text{Maximize } Z_y = \mathbf{b}^T\mathbf{y}$$
$$\text{subject to} \quad \mathbf{A}^T\mathbf{y} \leq \boldsymbol{c}^T;\ \mathbf{y} \geq 0 \tag{2}$$

This dual problem can also be written as:

$$\text{Minimize } Z_y^* = (-\mathbf{b})^T\mathbf{y} \tag{3}$$
$$\text{subject to} \quad (-\mathbf{A}^T)\mathbf{y} \geq (-\mathbf{c}^T);\ \mathbf{y} \geq 0, \text{ where } Z_y^* = -Z_y$$

If we consider LP problem (3) as primal, then its dual can be constructed by considering $\mathbf{x}$ as the dual variable. Thus, we have

$$\text{Maximize } Z_x^* = (-\mathbf{c}^T)^T\mathbf{x} = (-\mathbf{c})\mathbf{x}$$
$$\text{subject to} \quad (-\mathbf{A}^T)^T\mathbf{x} \leq (-\mathbf{b}^T)^T \text{ or } (-\mathbf{A})\mathbf{x} \leq (-\mathbf{b})$$
$$\text{and} \quad \mathbf{x} \geq 0 \tag{4}$$

But LP problem (4) is identical to the given primal LP problem (1). This completes the proof of the theorem.

Theorem 5.2: Let $\mathbf{x}^*$ be any feasible solution to the primal LP problem

$$\text{Maximize } Z_x = \mathbf{cx}$$
$$\text{subject to} \quad \mathbf{Ax} \leq \mathbf{b};\ \mathbf{x} \geq 0$$

and $\mathbf{y}^*$ be any feasible solution to the dual LP problem

$$\text{Maximize } Z_y = \mathbf{b}^T\mathbf{y}$$
$$\text{subject to} \quad \mathbf{A}^T\mathbf{y} \geq \mathbf{c}^T;\ \mathbf{y} \geq 0$$

of the above primal problem. Then prove that $\mathbf{cx}^* \leq \mathbf{b}^T\mathbf{y}^*$, i.e. $Z_x \leq Z_y$

Proof: Since $\mathbf{x}^*$ and $\mathbf{y}^*$ are the feasible solutions to the primal and dual LP problems, respectively, therefore from the constraints in primal and dual, we have

$$\mathbf{Ax}^* \leq \mathbf{b};\quad \mathbf{x}^* \geq 0 \tag{5}$$
$$\mathbf{A}^T\mathbf{y}^* \geq \mathbf{c}^T;\ \mathbf{y}^* \geq 0 \tag{6}$$

From inequality (6), we have $\mathbf{c}^T \leq \mathbf{A}^T\mathbf{y}^*$ or $\mathbf{c} \leq \mathbf{A}\ (^*\mathbf{y})^T$

Multiplying both sides by $\mathbf{x}^*$, we get

$$\mathbf{cx}^* \leq (\mathbf{Ay}^{*T})\mathbf{x}^* = \mathbf{y}^{*T}(\mathbf{Ax}^*) = \mathbf{y}^{*T}\mathbf{b} = \mathbf{b}^T\mathbf{y}^* \qquad [\text{since } \mathbf{y}^{*T}\mathbf{b} = \mathbf{b}^T\mathbf{y}^*]$$

This completes the proof of the theorem.

Theorem 5.3: If $\mathbf{x}^*$ is the feasible solution to the primal LP problem,

$$\text{Minimize } Z_x = \mathbf{cx}$$
$$\text{subject to} \quad \mathbf{Ax} \leq \mathbf{b};\ \mathbf{x} \geq 0$$

and y^* is the feasible solution to the dual problem

$$\text{Minimize } Z_y = \mathbf{b}^T\mathbf{y}$$
$$\text{subject to} \quad \mathbf{A}^T\mathbf{y} \geq \mathbf{c}^T;\ \mathbf{x} \geq 0$$

of the above primal problem, such that, $\mathbf{cx}^* = \mathbf{b}^T\mathbf{y}$, then $\mathbf{x}^*$ and $\mathbf{y}^*$ are the optimal solutions to their respective problems.

Proof: Let $\mathbf{x}_0^*$ be any other feasible solution to the primal. Then from Theorem 5.2 we have

$\mathbf{c}\,\mathbf{x}_0^* \le \mathbf{b}^T\mathbf{y}$ or $\mathbf{c}\,\mathbf{x}_0^* \le \mathbf{c}\mathbf{x}^*$ [since $\mathbf{c}\mathbf{x}^* = \mathbf{b}^T\mathbf{y}^*$]. Hence $\mathbf{x}^*$ is an optimal solution to the given maximization primal LP problem.

Similarly, for any other feasible solution $\mathbf{y}_0^*$ to the dual LP problem, we have $\mathbf{c}\mathbf{x}^* \le \mathbf{b}^T\mathbf{y}_0^*$ or $\mathbf{b}^T\mathbf{y}^* \le \mathbf{b}^T\mathbf{y}_0^*$ [since $\mathbf{c}\,\mathbf{x}^* = \mathbf{b}^T\mathbf{y}^*$]. Hence, $\mathbf{y}^*$ is also an optimal solution to the given minimization dual problem.

Theorem 5.4: If *i*th constraint in the primal is an equality, then the *i*th dual variable is unrestricted in sign.

Proof: Consider the primal LP problem in its standard form as

Minimize $Z_x = \mathbf{c}\mathbf{x}$

subject to $\mathbf{A}\mathbf{x} = \mathbf{b};\ \mathbf{x} \ge 0$

This problem can also be written in equivalent form as:

Maximize $Z_x = \mathbf{c}\mathbf{x}$

subject to $\mathbf{A}\mathbf{x} \le \mathbf{b}$ and $(-\mathbf{A})\,\mathbf{x} \le -\mathbf{b};\ \mathbf{x} \ge 0$

The dual of this problem can now be written as:

Minimize $Z_y = y_1\mathbf{b}^T + y_2(-\mathbf{b}^T) = (y_1 - y_2)\,\mathbf{b}^T$

subject to $\mathbf{A}^T y_1 + (-\mathbf{A}^T)\,y_2 \ge \mathbf{c};\ y_1, y_2 \ge 0$

Let $y = y_1 - y_2$. Then we can rewrite these equations as

Minimize $Z_y = \mathbf{y}\mathbf{b}^T$

subject to $\mathbf{A}^T\mathbf{y} \ge \mathbf{c};$ $\mathbf{y}$ unrestricted in sign.

Theorem 5.5 (*Unboundedness Theorem*): If either the primal or the dual LP problem has an unbounded objective function value, then the other problem has no feasible solution.

Proof: Let the given primal LP problem have an unbounded solution. Then for any value of the objective function, say $+\infty$, there exists a feasible solution say $\mathbf{x}$ yielding this solution, i.e. $\mathbf{c}\mathbf{x} \to \infty$.

From Theorem 5.2, for each feasible solution $\mathbf{y}^*$ of the dual, there exists a feasible solution $\mathbf{x}^*$ to the primal such that $\mathbf{c}\mathbf{x}^* \le \mathbf{b}^T\mathbf{y}^*$. That is, $\mathbf{b}^T\mathbf{y}^* \to +\infty$. As $\mathbf{b}$ is a constant number and $\mathbf{y}^*$ has to satisfy the constraint $\mathbf{A}^T\mathbf{y}^* \le \mathbf{c}^T$. Hence the dual objective function $Z_y = \mathbf{b}^T\mathbf{y}^*$ must be finite. This contradicts the result $\mathbf{b}^T\mathbf{y}^* \to \infty$. Hence the dual LP problem has no feasible solution.

A similar argument can be used to show that when the dual LP problem has an unbounded solution, the primal LP problem has no solution.

Theorem 5.6 (*Duality Theorem*): If either the primal or the dual problem has a finite optimal solution, then the other one also possess the same, and the optimal values of the objective functions of the two problems are equal, Max Z_x = Min Z_y.

Proof: Let the following LP problem represent a primal problem:

Maximize $Z_x = \mathbf{c}\mathbf{x}$

subject to $\mathbf{A}\mathbf{x} \le \mathbf{b}$ or $\mathbf{A}\mathbf{x} + \mathbf{I}\mathbf{s} = \mathbf{b};\ \mathbf{x} \ge 0$

where $\mathbf{I}$ is the identify matrix of order m and $\mathbf{s}$ is the slack variable.

Let $\mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b}$ be an optimal basic feasible solution to this primal problem and $\mathbf{c}_B$ the cost vector of the basic variables. Then according to the optimality condition, we have, $c_j - z_j \le 0$, for any vector $\mathbf{a}_j$ of $\mathbf{A}$ but not in $\mathbf{B}$. Therefore,

$$c_j - z_j = c_j - \mathbf{c}_B\mathbf{y}_j = c_j - \mathbf{c}_B(\mathbf{B}^{-1}\,\mathbf{a}_j) \le 0, \quad \text{for all } j$$

This is equivalent to $\mathbf{c} \le \mathbf{c}_B\mathbf{B}^{-1}\mathbf{A}$, when written in matrix notation.

Let $\mathbf{y}^* = \mathbf{c}_B\mathbf{B}^{-1}$. Then $\mathbf{c} \le \mathbf{c}_B\mathbf{B}^{-1}\mathbf{A}$ becomes $\mathbf{c} \le \mathbf{y}^*\mathbf{A}$ or $\mathbf{c}^T \le \mathbf{y}^*\,\mathbf{A}^T$. Hence $\mathbf{y}^*$ is a feasible solution to the dual of the given primal because it satisfies the dual constraints. The corresponding dual objective function is given by

$$Z_y = \mathbf{y}^*\mathbf{b}^T = \mathbf{c}_B\mathbf{B}^{-1}\,\mathbf{b}^T = \mathbf{c}_B\mathbf{x}_B = Z_x$$

Hence, it has been shown that $\mathbf{x}_B$ and $\mathbf{c}_B\mathbf{B}^{-1}$ are the feasible solution to the primal and dual LP problems, respectively for which Max Z_x = Min Z_y.

Theorem 5.7 (*Complementary Slackness Theorem*): If **x*** and **y*** be feasible solutions to the primal and dual LP problems, respectively, then a necessary and sufficient condition for **x*** and **y*** to be optimal solutions to their respective problems is,

$$y_i \cdot x_{n+i} = 0, \qquad i = 1, 2, \ldots, m$$

and $$x_j \cdot y_{m+j} = 0, \qquad j = 1, 2, \ldots, n$$

where x_{n+i} is the ith slack variable in the primal LP problem and y_{m+j} the jth surplus variable for the dual LP problem.

Proof: Let **x*** be the feasible solution to the primal LP problem,

$$\text{Maximize } Z_x = \sum_{j=1}^{n} c_j x_j$$

$$\text{subject to} \quad \sum_{j=1}^{n} a_{ij} x_j + x_{n+i} = b_i; \; x_j \geq 0 \tag{7}$$

and **y*** be the feasible solution to the dual of the above primal.

$$\text{Minimize } Z_y = \sum_{j=1}^{n} b_i y_i \tag{8}$$

$$\text{subject to} \quad \sum_{j=1}^{n} a_{ij} y_i - y_{m+j} = c_j; \; y_i \geq 0$$

Multiplying constraints in (7) by y_i, $i = 1, 2, \ldots, m$, and then adding we get

$$x_j \sum_{i=1}^{m} a_{ij} y_i + x_{n+i}\, y_i = \sum_{i=1}^{m} y_i\, b_i; \; j = 1, 2, \ldots, n \tag{9}$$

Subtracting (9) from the objective function of (7), we get

$$\left\{ c_j - \sum_{i=1}^{m} a_{ij} y_i \right\} x_j - x_{n+i}\, y_i = Z_x - \sum_{i=1}^{m} y_i\, b_i \tag{10}$$

Substituting the value of Z_y and y_{m+j} from (8) to (10) we get,

$$-\sum_{j=1}^{n} x_j\, y_{m+j} - \sum_{i=1}^{m} y_i x_{n+i} = Z_x - Z_y \tag{11}$$

If **x*** and **y*** are optimal solutions to the primal and dual problems, then we have $\mathbf{c}\,\mathbf{x}^* = \mathbf{b}^T\mathbf{y}^*$ (or $Z_x = Z_y$) Thus, from (11), we have

$$\sum_{j=1}^{n} x_j\, y_{m+j} = \sum_{i=1}^{m} y_i x_{n+i} = 0$$

Now it follows that, for each $x_j > 0, j$ (= 1, 2, . . ., m) implies $y_{m+j} = 0$ and $\sum_{i=1}^{m} a_{ij} y_i = c_j$, i.e. jth constraint in the dual is an equation. Also $x_j = 0$ implies $y_{m+i} = 0$ and $\sum_{i=1}^{m} a_{ij} y_i > c_j$.

Similarly, for each $y_i > 0$, i (= 1, 2, . . ., m) implies $x_{n+i} = 0$ and $\sum_{j=1}^{n} a_{ij} x_i = b_i$, i.e. ith constraint in the primal is an equation. Also $y_i = 0$ implies $x_{n+i} > 0$ and $\sum_{j=1}^{n} a_{ij} x_j < b_i$. This completes the proof of the theorem.

Chapter 6

Sensitivity Analysis in Linear Programming

"The most efficient way to produce anything is to bring together under one management as many as possible of the activities needed to turn out the product."

– Peter Drucker

PREVIEW

The purpose of sensitivity analysis is to evaluate the effect on the optimal solution of an LP problem due to variations in the input coefficients (also called parameters), one at a time.

LEARNING OBJECTIVES

After studying this chapter, you should be able to

- appreciate the significance of sensitivity analysis concept in managerial decision-making.
- perform sensitivity analysis on various parameters in an LP model without affecting the optimal solution.
- introduce a new variable and a constraint in the existing LP model with reformulation.

CHAPTER OUTLINE

6.1 Introduction

6.2 Sensitivity Analysis

- Conceptual Questions
- Self Practice Problems
- Hints and Answers

❑ Chapter Summary

❑ Chapter Concepts Quiz

❑ Case Study

6.1 INTRODUCTION

In an LP model, the coefficients (also known as parameters) such as: (i) profit (cost) contribution (c_j) per unit of a decision variable, x_j (ii) availability of a resources (b_i), and (iii) consumption of resource per unit of decision variables (a_{ij}), are assumed to be constant and known with certainty during a planning period. However, in real-world situations, these input parameters value may change due to dynamic nature of the business environment. Such changes in any of these parameters may raise doubt on the validity of the optimal solution of the given LP model. Thus, a decision-maker, in such situations, would like to know how changes in these parameters may affect the optimal solution and the range within which the optimal solution will remain unchanged.

Sensitivity analysis and parametric linear programming are the two techniques that are used to evaluate the effect on an optimal solution of any LP problem due to changes in its parameters.

Sensitivity analysis helps in evaluating the effect on optimal solution of any LP problem due to changes in its parameters, one at a time.

- *Sensitivity analysis determines the sensitivity range (both lower and upper limit) within which the LP model parameters can vary (one at a time) without affecting the optimality of the current optional solution.*
- *Parametric analysis is the study of measuring the effect on the optimal solution of the LP model due to changes at a time in more than one input parameters value outside the sensitivity range.*

However, while sensitivity analysis provides the sensitive ranges (both lower and upper limits) within which the LP model parameters can vary without changing the optimality of the current optimal solution; parametric linear programming provides information about changes in parameters value outside the sensitive range, as well as to changes in more than one parameter at a time.

6.2 SENSITIVITY ANALYSIS

While study of duality helps to identify only an increase (or decrease) in the value of objective function due to per unit variation in the amount of resources available, the sensitivity analysis reveals the magnitude of change in the optimal solution of an LP model due to discrete variations (changes) in its parameters. The possible change in the parameter values, can range from zero to a substantial change. Thus, aim of sensitivity analysis is to determine the range (or limit) within which the LP model parameters can change without affecting the current optimal solution. For this, instead of solving an LP problem again with new values of parameters, the current optimal solution is considered as an initial solution to determine the ranges, both lower and upper, within which a parameter may assume a value.

The sensitivity analysis is also referred to as *post-optimality analysis* because it does not begin until the optimal solution to the given LP model has been obtained. Different parametric changes in an LP problem discussed in this chapter are:

Parametric analysis helps in evaluating the effect on optimal solution of any LP problem due to changes in its parameter, simultaneously as a function of one parameter.

(i) Profit (or cost) per unit (c_j) associated with both basic and non-basic decision variables (i.e., coefficients in the objective function).
(ii) Availability of resources (i.e., right-hand side constants, b_i in constraints).
(iii) Consumption of resources per unit of decision variables x_j (i.e., coefficients of decision variables in the constraints, a_{ij}).
(iv) Addition of a new variable to the existing list of variables in LP problem.
(v) Addition of a new constraint to the existing list of constraints in the LP problem.

6.2.1 Change in Objective Function Coefficient (c_j)

The coefficient, c_j in the objective function of an LP model represents either the profit or the cost per unit of an activity (variable) x_j. The question that may now arise is: *What happens to the optimal solution and the objective function value when this coefficient is changed?* For example, let Rs 10 be the per unit profit coefficient of a particular variable in the objective function of an LP model. After obtaining an optimal solution, the decision-maker may realize that the true value of this coefficient might be any value between Rs 9 and Rs 11 per unit. The sensitivity analysis can help him to determine both lower and upper value (i.e., range) of this coefficient (or any other such coefficients) without affecting the current optimal solution. For this he may decide whether resources from other activities (variables) should be diverted to (diverted away from) a more profitable (or less profitable) activity.

Given an optimal basic feasible solution, suppose that the coefficient c_k of a variable x_k in the objective function is changed from c_k to $c_k + \Delta c_k$, where Δc_k represents the positive (or negative) amount of change in the value of c_k. In optimal simplex table, the feasibility of the solution remains unaffected due to changes in the

coefficients, c_j of basic variables in the objective function. However, any change in these coefficients (c_j's) only affect the optimality of the solution. Thus, such a change requires recomputing z_j values in $c_j - z_j$ row of the optimal simplex table because $c_k - z_k (c_k - \mathbf{c_B}\,\mathbf{B}^{-1}\,\mathbf{a_k})$ involves coefficients c_k. Obviously the effect of this change will be seen in the $c_j - z_j$ row of the optimal simplex table.

While recomputing $c_j - z_j$ values, the following two cases will arise

(i) The new $c_j - z_j$ values satisfy optimality condition and the current optimal feasible solution remains unchanged. However, optimal value of objective function may change.

(ii) The optimality condition is not satisfied. In such a case usual simplex method is used to obtain new optimal solution.

Case I: Change in the coefficient of a non-basic variable If $c_j - z_j \le 0$ for all non-basic variables in a maximization LP problem, then the current optimal feasible solution remains unchanged. Let c_k be the coefficient of a non-basic variable x_k in the objective function. Since c_k is the coefficient of non-basic variable x_k, therefore, it does not effect any of the c_j values listed in the '$\mathbf{c_B}$' column of optimal simplex table associated with basic variables. Since the calculation of $z_j = \mathbf{c_B}\,\mathbf{B}^{-1}\,\mathbf{a}_j$ values do not involve c_j, therefore changes in c_j do not change z_j values and hence $(c_j - z_j)$ values remain unchanged, except the $c_k - z_k$ value due to change in c_k.

If change in coefficient, c_k does not affect the feasibility of the optimal solution, then unit profit of x_k can be lowered to any level without causing the optimal solution to change. But any increase in c_k value, beyond a certain level, (i.e. upper limit) should make this variable eligible to be a basic variable in the new solution mix. Obviously, $c_k - z_k$ will no longer then be negative.

To retain optimality of the current optimal solution for a change Δc_k in c_k, we must have $(c_k + \Delta c_k) - z_k \le 0$ or $c_k + \Delta c_k \le z_k$. Hence, for an LP problem with an objective function of the maximization type, the value of c_k may be increased up to the value of z_k, and decreased to negative infinity ($-\infty$) without affecting the optimal solution.

Case II: Change in the coefficient of a basic variable In the maximization LP problem, the change in the coefficient, say c_k, of a basic variable x_k affects the $c_j - z_j$ values corresponding to all non-basic variables in the optimal simplex table. This is because the coefficient c_k is listed in the $\mathbf{c_B}$ column of the simplex table and affects the calculation of the z_j values.

Range of optimality is where the coefficient of any basic variable can change without causing change in the optimal solution mix

The sensitivity limits for coefficient, c_j of a basic variable are calculated as under:

$$\text{Lower limit} = \text{Original value, } c_k - \{\text{Lowest absolute value } \left|\frac{c_j - z_j}{a_{kj}} \le 0\right| \text{ or } -\infty$$

$$\text{Upper limit} = \text{Original value, } c_k + \{\text{Highest absolute value } \left|\frac{c_j - z_j}{a_{kj}} \ge 0\right| \text{ or } \infty$$

Remark While performing sensitivity analysis, the *artificial variable columns in the simplex table are ignored.* Thus, the improvement ratio, $(c_j - z_j)/a_{kj}$ using coefficients corresponding to artificial variables should not be considered.

Case III: Change in the coefficient of non-basic variables In the minimization LP problem, the value of cost coefficient, c_j can be increased to any arbitrary level but its values cannot be decreased more than per unit improvement value, without making it eligible to become a basic variable. The sensitivity limits can be calculated as:

Lower limit = Original value c_k − Absolute value of $c_k - z_k$
Upper limit = Infinity ($+\infty$)

Alternative method If we add Δc_k to the objective function coefficient c_k of the basic variable x_k, then its coefficient c_{Bk} will become $c^*_{Bk} = c_{Bk} + \Delta c_{Bk}$. The new $c_j - z^*_j$ can be calculated as follows:

$$c_j - z_j^* = c_j - \sum_{i=1}^{m} c_{Bi} y_{ij} = c_j - \left\{ \sum_{i \ne k}^{m} c_{Bi} y_{ij} + (c_{Bk} + \Delta c_{Bk})\, y_{kj} \right\}$$

$$= c_j - \left\{ \sum_{i=k}^{m} c_{Bi} y_{ij} + \Delta c_{Bk} y_{kj} \right\} = c_j - \{z_j + \Delta c_{Bk} y_{kj}\} = (c_j - z_j) - \Delta c_{Bk} y_{kj}$$

For a current basic feasible solution to remain optimal for a maximization LP problem, we must have $c_j - z_j^* \le 0$. That is,

$$c_j - z_j^* = (c_j - z_j) - \Delta c_{Bk}\, y_{kj} \le 0 \quad \text{or} \quad c_j - z_j \le \Delta c_{Bk}\, y_{kj}$$

or

$$(c_j - z_j)/y_{kj} \le \Delta c_{Bk} \quad \text{when } y_{kj} > 0$$

$$(c_j - z_j)/y_{kj} \ge \Delta c_{Bk} \quad \text{when } y_{kj} < 0$$

Hence, the value of Δc_{Bk} that satisfies the optimality criterion can be determined by solving the following system of linear inequalities:

$$\text{Min}\left\{\frac{c_j - z_j}{y_{kj} < 0}\right\} \ge \Delta c_{Bk} \ge \text{Max}\left\{\frac{c_j - z_j}{y_{kj} > 0}\right\} \tag{1}$$

Here it may be noted that y_{kj}s are entries in the non-basic variable columns (i.e. variables) of the optimal simplex table. The new value of the objective function becomes:

$$Z^* = \sum_{i \ne k}^{m} c_{Bi} x_{Bi} + (c_{Bk} + \Delta c_{Bk})\, x_{Bk} = \sum_{i=1}^{m} c_{Bi} x_{Bi} + \Delta c_{Bk}\, x_{Bk} = Z + \Delta c_{Bk}\, x_{Bk}$$

Hence if Δc_{Bk} satisfies inequality (1), then the optimal solution will remain unchanged but the value of Z will improve by an amount $\Delta c_{Bk}\, x_{Bk}$.

Example 6.1 Use simplex method to solve the following LP problem:

Maximize $Z = 3x_1 + 5x_2$

subject to the constraints

(i) $3x_1 + 2x_2 \le 18$, (ii) $x_1 \le 4$, (iii) $x_2 \le 6$

and $x_1, x_2 \ge 0$.

Discuss the change in c_j on the optimality of the optimal basic feasible solution.

Solution The standard form of the given LP problem is stated as follows:

Maximize $Z = 3x_1 + 5x_2 + 0s_1 + 0s_2 + 0s_3$

subject to the constraints

(i) $3x_1 + 2x_2 + s_1 = 18$, (ii) $x_1 + s_2 = 4$ (iii) $x_2 + s_3 = 6$

and $x_1, x_2, s_1, s_2, s_3 \ge 0$,

The optimal solution: $x_1 = 2, x_2 = 6$ and Max Z = 36 of the given LP problem is shown in Table 6.1. The cost coefficients (c_j) associated with basic variables x_1, x_2, and s_2 in the objective function are $\mathbf{c_B} = (c_1, c_2, c_4) = (3, 5, 0)$. The changes in c_j can be classified as under:

(a) ***Changes in the coefficients c_j (i.e. c_3 and c_5) of non-basic variables s_1 and s_3:*** Let us add Δc_3 and Δc_5 to the objective function coefficients c_3 and c_5. Then the new objective function coefficients will become $c_3' = c_3 + \Delta c_3 = 0 + \Delta c_3 = \Delta c_3$ and $c_5' = c_5 + \Delta c_5 = 0 + \Delta c_5 = \Delta c_5$. Thus, the new values of $c_3 - z_3$ and $c_5 - z_5$ will become $\Delta c_3 - 1$ and $\Delta c_5 - 3$, respectively. In order to maintain optimality, we must have

$$\Delta c_3 - 1 \le 0 \text{ and } \Delta c_5 - 3 \le 0 \quad \text{or} \quad \Delta c_3 \le 1 \text{ and } \Delta c_5 \le 3$$

(b) ***Change in the coefficients c_j (i.e. c_1, c_2 and c_4) of basic variables x_1, x_2 and s_2:*** The value of additional increment Δc_k, $k = 1, 2, 4$ in the coefficients c_1, c_2 and c_4 which satisfy the optimality condition can be determined by solving the following system of linear inequalities [refer to Eqn. (1)].

$$\text{Min}\left\{\frac{c_j - z_j}{y_{kj} < 0}\right\} \ge \Delta c_{Bk} \ge \text{Max}\left\{\frac{c_j - z_j}{y_{kj} > 0}\right\}$$

		$c_j \rightarrow$	3	5	0	0	0
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	s_1	s_2	s_3
3	x_1	2	1	0	1/3	0	–2/3
0	s_2	0	0	0	–2/3	1	4/3
5	x_2	6	0	1	0	0	1
$Z = 36$		$c_j - z_j$	0	0	–1	0	–3

Table 6.1 Optimal Solution

For $k = 1$ (i.e. basic variable x_1 in row 1), we have

$$\text{Min}\left\{\frac{-3}{-2/3}\right\} \geq \Delta c_1 \geq \text{Max}\left\{\frac{-1}{1/3}\right\}; \quad j = 3, 5 \quad \text{or} \quad 9/2 \geq \Delta c_1 \geq -3$$

It may be noted that $y_{kj} = (y_{13} = 1/3, y_{15} = -2/3)$ are only for those columns corresponding to which variables are not in the optimal basis (i.e. non-basic variables). The current optimal solution will not change as long as

$$\left(3 + \frac{9}{2}\right) \geq c_1 \geq (3 - 3) \quad \text{or} \quad \frac{15}{2} \geq c_1 \geq 0$$

Hence, the current optimal solution will not change as long as: $0 \leq c_1 \leq 15/2$

For $k = 3$ (i.e. basic variable x_2 in row 3), we have

$$\text{Min}\left\{\frac{-1}{0}\right\} \geq \Delta c_2 \geq \left\{\frac{-3}{1}\right\}; \quad \text{or} \quad \infty \geq \Delta c_2 \geq -3, \quad \text{for} \quad j = 3, 5$$

Hence, the current optimal solution will not change as long as: $(5 + \infty) \geq c_2 \geq (5 - 3)$ or $\infty \geq c_2 \geq 2$.

Example 6.2 A company wants to produce three products: A, B and C. The per unit profit on these products is Rs 4, Rs 6 and Rs 2, respectively. These products require two types of resources, manpower and raw material. The LP model formulated for determining the optimal product mix is as follows:

Maximize $Z = 4x_1 + 6x_2 + 2x_3$

subject to the constraints

(i) $x_1 + x_2 + x_3 \leq 3$ (Manpower required), (ii) $x_1 + 4x_2 + 7x_3 \leq 9$ (Raw material available)

and $x_1, x_2, x_3 \geq 0$

where x_1, x_2 and x_3 = number of units of products A, B and C, respectively, to be produced.

(a) Find the optimal product mix and the corresponding profit of the company.

(b) Find the range of the profit contribution of product C (i.e. coefficient c_3 of variable x_3) in the objective function, such that the current optimal product mix remains unchanged.

(c) What shall be the new optimal product mix when per unit profit from product C is increased from Rs 2 to Rs 10?

(d) Find the range of the profit contribution of product A (i.e. coefficient c_1 of variable x_1) in the objective function such that the current optimal product mix remains unchanged.

Solution (a) Introducing the slack variables s_1 and s_2 in the constraints to convert LP problem into its standard form as follows:

Maximize $Z = 4x_1 + 6x_2 + 2x_3 + 0s_1 + 0s_2$

subject to the constraints

(i) $x_1 + x_2 + x_3 + s_1 = 3$, (ii) $x_1 + 4x_2 + 7x_3 + s_2 = 9$

and $x_1, x_2, x_3, s_1, s_2 \geq 0$

The optimal solution obtained by applying the simplex method is shown in Table 6.2.

Table 6.2 Optimal Solution

	$c_j \rightarrow$	4	6	2	0	0
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2
4	x_1	1	1	0	– 1	4/3	– 1/3
6	x_2	2	0	1	2	– 1/3	1/3
$Z = 16$		z_j	4	6	8	10/3	2/3
		$c_j - z_j$	0	0	– 6	– 10/3	– 2/3

The optimal solution is: $x_1 = 1$, $x_2 = 2$ and Max Z = Rs 16.

(b) *Effect of change in the coefficient c_3 of non-basic variable x_3 for product C* In Table 6.2, the variable x_3 is non-basic and its coefficient $c_3 = 2$ is not listed in the $\mathbf{c_B}$ column of the table. This means a further decrease in its profit contribution c_3 (= Rs 2) per unit will have no effect on the current optimal product mix. But, if c_3 is increased beyond a certain value, the product may become profitable to be produced. Hence, there is only an upper limit on c_3 for which the current optimal product mix will be effected.

Table 6.3 Table with a Δc_3 change in c_3

	$c_j \rightarrow$	4	6	$2 + \Delta c_3$	0	0
Basic Variables c_B	*Basic Variables* B	***Basic Variables*** *Value* $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2
4	x_1	1	1	0	– 1	4/3	– 1/3
6	x_2	2	0	1	2	– 1/3	1/3
$Z = 16$		z_j	4	6	8	10/3	2/3
		$c_j - z_j$	0	0	$\Delta c_3 - 6$	– 10/3	– 2/3

Now a change Δc_3 in c_3 will cause a change in z_3 and $c_3 - z_3$ value in the x_3-column. The change in c_3 results in the modified simplex Table 6.3.

For an optimal solution shown in Table 6.3 to remain unchanged, we must have: $\Delta c_3 - 6 \leq 0$ or $\Delta c_3 \leq 6$.

Recalling that $c_3 = 2 + \Delta c_3$, or $\Delta c_3 = c_3 - 2$, after substituting this amount in the above inequality, we get $c_3 - 2 \leq 6$ or $c_3 \leq 8$. This implies that as long as the profit contribution per unit of product C is less than Rs 8 (i.e. change should not be more than Rs 8) it is not profitable to produce it and therefore the current optimal solution will remain unchanged.

(c) *If the value of c_3 is increased from Rs 2 to Rs 10, the new value of $c_3 - z_3$ will be*

$$c_3 - z_3 = c_3 - \mathbf{c_B a_3} = 10 - [4 \quad 6] \begin{bmatrix} -1 \\ 2 \end{bmatrix} = 10 - (-4 + 12) = 2\ (> 0)$$

Thus, if coefficient of variable x_3 is increased from Rs 2 to Rs 10, the value of $c_3 - z_3$ will become positive as shown in Table 6.4. The variable x_3 (corresponding to product C) becomes eligible to enter into the basis. Hence, the solution cannot remain optimal any more.

Table 6.4

	$c_j \rightarrow$	4	6	10	0	0	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2	*Min Ratio* x_B/x_3
4	x_1	1	1	0	– 1	4/3	– 1/3	–
6	x_2	2	0	1	(2)	– 1/3	1/3	2/2 = 2 →
$Z = 16$		z_j	4	6	8	10/3	2/3	
		$c_j - z_j$	0	0	2 ↑	– 10/3	– 2/3	

Applying the following row operations to enter variable x_3 into the new solution mix

R_2 (new) $\rightarrow R_2$ (old)/2 (*key element*) ; R_1 (new) $\rightarrow R_1$ (old) + R_2 (new)

The new optimal solution is shown in Table 6.5.

	$c_j \rightarrow$	4	6	10	0	0
Basic Variables Coefficient c_B	*Basic Variables* **B**	*Basic Variables Value* b $(= x_B)$	x_1	x_2	x_3	s_1	s_2
4	x_1	2	1	1/2	0	7/6	– 1/6
10	x_3	1	0	1/2	1	– 1/6	1/6
$Z = 18$		z_j	4	7	10	3	1
		$c_j - z_j$	0	– 1	0	– 3	– 1

Table 6.5

Since all $c_j - z_j \leq 0$ in Table 6.5, the new optimal solution is: $x_1 = 2$, $x_2 = 0$ and $x_3 = 1$ and Max Z = Rs 18.

(d) *Effect of change in the coefficient c_1 of basic variable x_1 for product A* In the optimal simplex Table 6.2, variable x_1 is the basic variable. This means that any decrease in its coefficient (profit contribution per unit) c_1 (= Rs 4) will make product A less profitable to produce and therefore the current optimal product mix will be affected. Also, an increase in the value of c_1, beyond a certain limit, will make the product A much more profitable and may force the decision-maker to decide to only produce product A. Thus, in either case, the current optimal product mix will be affected and hence we need to know both the lower and upper limit on the value of c_1, within which the current optimal solution will remain unchanged.

Referring again to Table 6.2, the range of change in the value of c_1 (and/or also in c_2) that does not affect the current optimal product mix can be determined once again by calculating $c_j - z_j$, values that correspond to non-basic variables x_3, s_1 and s_2 respectively. For this reproduce Table 6.2 once again with unknown value of c_1 as shown in Table 6.6.

	$c_j \rightarrow$	$4 + \Delta c_1$	6	2	0	0
Basic Variables Coefficient c_B	*Basic Variables* **B**	*Basic Variables Value* b $(= x_B)$	x_1	x_2	x_3	s_1	s_2
$4 + \Delta c_1$	x_1	1	1	0	– 1	4/3	– 1/3
6	x_2	2	0	1	2	– 1/3	1/3
$Z = 12 + (4 + \Delta c_1)$		z_j	$4 + \Delta c_1$	6	$8 - \Delta c_1$	$10/3 + 4\Delta c_1/3$	$2/3 - \Delta c_1/3$
		$c_j - z_j$	0	0	$\Delta c_1 - 6$	$-10/3 - 4\Delta c_1/3$	$\Delta c_1/3 - 2/3$

Table 6.6
A Change Δc_1 in Coefficient c_1

For the solution shown in Table 6.6 to remain optimal we must have all $c_j - z_j \leq 0$. That is

$$\Delta c_1 - 6 \leq 0 \quad \text{gives} \quad \Delta c_1 \leq 6$$

$$-\frac{10}{3} - \frac{4\Delta c_1}{3} \leq 0 \quad \text{gives} \quad \Delta c_1 \geq -5/2$$

$$\frac{\Delta c_1}{3} - \frac{2}{3} \leq 0 \quad \text{gives} \quad \Delta c_1 \leq 2$$

Thus, the range of values within which c_1 may change without affecting the current optimal solution is:

$$-\frac{5}{2} \leq \Delta c_1 \leq 2 \text{ or } 4 - \frac{5}{2} \leq c_1 \leq 4 + 2, \text{ i.e. } \frac{3}{2} \leq c_1 \leq 6$$

From these calculations, it may be concluded that in the case of competitive pressure or in the absence of any competition, the decision-maker knows to what extent the prices can be adjusted without changing the optimal product mix.

Example 6.3 Given the following LP problem

Maximize $Z = -x_1 + 2x_2 - x_3$,

subject to the constraints

(i) $3x_1 + x_2 - x_3 \le 10$, (ii) $-x_1 + 4x_2 + x_3 \ge 6$, (iii) $x_2 + x_3 \le 4$

and $x_1, x_2, x_3 \ge 0$.

Determine the effect of discrete changes in c_j ($j = 1, 2, ..., 6$) on the optimal basic feasible solution shown in Table 6.7.

Table 6.7 Optimal Solution

		$c_j \rightarrow$	–1	2	–1	0	0	0	–M
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* b $(=x_B)$	x_1	x_2	x_3	s_1	s_2	s_3	A_1
0	s_1	6	3	0	–2	1	0	–1	0
2	x_2	4	0	1	1	0	0	1	0
0	s_2	10	1	0	3	0	1	4	–1
$Z = 8$		z_j	0	2	2	0	0	2	0
		$c_j - z_j$	–1	0	–3	0	0	–2	–M

Solution In Table 6.7, variable x_1 is a non-basic variable with coefficient $c_1 = -1$. If value of c_1 further decreases, then x_1 will not become eligible to enter in the basis and hence will not affect the optimality of the solution. Thus there is no lower limit on the value of c_1. However, if the value of c_1 increases and exceeds a certain value, it may become eligible to enter into the basis. Thus, there should be an upper limit on the value of c_1.

Solution shown in Table 6.7 will remain optimal provided largest value of c_1 should be

$$c_1 - z_1 = c_1 - \boldsymbol{c_B a_1} = c_1 - (0, 2, 0)\begin{bmatrix}3\\0\\1\end{bmatrix} \le 0 \quad \text{or} \quad c_1 \le 0.$$

Since variable x_2 in Table 6.7 is also a basic variable, the solution will remain optimal so long its largest value is,

$$\bar{c}_1 = c_1 - z_1 = -1 - (0, c_2, 0)\begin{bmatrix}3\\0\\1\end{bmatrix} \le 0 \quad \text{or} \quad \bar{c}_1 = -1 (\le 0)$$

$$\bar{c}_3 = c_3 - z_3 = -1 - (0, c_2, 0)\begin{bmatrix}-2\\1\\3\end{bmatrix} \le 0 \text{ or } -1 - c_2 \le 0, \text{ i.e., } c_2 \ge -1$$

$$\bar{c}_6 = c_6 - z_6 = 0 - (0, c_2, 0)\begin{bmatrix}-1\\1\\4\end{bmatrix} \le 0 \text{ or } 0 - c_2 \le 0, \text{ i.e., } c_2 \ge 0$$

$$\bar{c}_7 = c_7 - z_7 = -M - (0, c_2, 0)\begin{bmatrix}0\\0\\-1\end{bmatrix} \le 0 \text{ or } -M \le 0, \text{ i.e., } M \ge 0$$

The variable x_3 in Table 6.7 is a non-basic variable, and for the current optimal solution to remain optimal, the largest value of c_3 should be

$$c_3 - z_3 = c_3 - \boldsymbol{c_B a_B} = c_3 - (0, 2, 0)\begin{bmatrix}-2\\1\\3\end{bmatrix} \le 0 \text{ or } c_3 - 2 \le 0 \text{ or } c_3 \le 2.$$

The variable s_1 in Table 6.7 is a basic variable and its limiting value is obtained from largest value $\bar{c}_1, \bar{c}_3, \bar{c}_6$ and $\bar{c}_7$ of c_1, c_3, c_6 and c_7, respectively as follows:

$$\bar{c}_1 = -1 - (c_4, 2, 0)\begin{bmatrix}3\\0\\1\end{bmatrix} \le 0 \text{ or } -1 - 3c_4 \le 0 \text{ or } c_4 \ge -\frac{1}{3}$$

$$\bar{c}_3 = -1 - (c_4, 2, 0)\begin{bmatrix}-2\\1\\3\end{bmatrix} \le 0 \text{ or } -1 + 2c_4 - 2 \le 0 \text{ or } c_4 \le \frac{3}{2}$$

$$\bar{c}_6 = 0 - (c_4, 2, 0)\begin{bmatrix}-2\\1\\3\end{bmatrix} \le 0 \text{ or } 0 + c_4 - 2 \le 0 \text{ or } c_4 \le 2$$

$$\bar{c}_7 = -M - (c_4, 2, 0)\begin{bmatrix}0\\0\\-1\end{bmatrix} \le 0 \text{ or } M \ge 0,$$

Hence, limits for c_4 are: $-1/3 \le c_4 \le 3/2$

The variable s_2 in Table 6.7 is a basic variable and its limiting value is obtained from largest value $\bar{c}_1, \bar{c}_3, \bar{c}_6$ and $\bar{c}_7$ of c_1, c_3, c_6 and c_7 respectively as follows:

$$\bar{c}_1 = -1 - (0, 2, c_5)\begin{bmatrix}3\\0\\1\end{bmatrix} \le 0 \quad \text{or} \quad -1 - c_5 \le 0 \text{ or } c_5 \ge -1,$$

$$\bar{c}_3 = -1 - (0, 2, c_5)\begin{bmatrix}-2\\1\\3\end{bmatrix} \le 0 \quad \text{or} \quad -1 - (2 + 3c_5) \le 0 \text{ or } c_5 \ge -1,$$

$$\bar{c}_6 = 0 - (0, 2, c_5)\begin{bmatrix}-1\\1\\4\end{bmatrix} \le 0 \quad \text{or} \quad 0 - (2 + 4c_5) \le 0 \text{ or } c_5 \ge -\frac{1}{2},$$

$$\bar{c}_7 = -M - (0, 2, c_5)\begin{bmatrix}0\\0\\-1\end{bmatrix} \le 0 \quad \text{or} \quad -M + c_5 \le 0 \text{ or } c_5 \le M.$$

Hence, limits for c_5 are: $-1/2 \le c_5 \le M$

The variable s_3 in Table 6.7 is non-basic variable and to find its limiting value $\bar{c}_6 \le 0$, we have

$$c_6 - z_6 = c_6 - \boldsymbol{c_B a_6} = c_6 - (0, 2, 0)\begin{bmatrix}-1\\1\\4\end{bmatrix} \le 0 \text{ or } c_6 - 2 \le 0 \text{ or } c_6 \le 2.$$

6.2.2 Change in the Availability of Resources (b_i)

Case I : When slack variable is not a basic variable Since at optimal solution of any LP problem, $c_j - z_j$ values (ignoring negative sign) corresponding to the slack variable columns represent shadow prices (or dual prices) of the available resources. The shadow price provides information about the change in the value of objective function due to per unit increase or decrease in the RHS (resource value) of a constraint.

The optimal simplex Table 6.2 of example 6.2 is reproduced as Table 6.8. Recall that slack variable s_1 represents the availability of manpower resource and s_2 represents the availability of raw material. The

$c_4 - z_4 = -10/3$ corresponding to s_1-column represents shadow price for an additional hour of manpower time. This information helps to determine how many hours we can actually afford to increase (or decrease) in the manpower time so as to increase the profit. That is, we need to determine the range within which the shadow price, $c_4 - z_4 = -10/3$ remain same.

In Table 6.8, slack variable is the basic variable. The procedure for finding the range for 'resource values' within which the current optimal solution remains unchanged is summarized below.

(a) Treat the slack variable corresponding to *resource value* as an entering variable in the solution. For this, calculate exchange ratio (minimum ratio) for every row.

$$\text{Minimum ratio} = \frac{\text{Basic variable value, } \mathbf{x_B}}{\text{Coefficients in a slack variable column}}$$

(b) Find both the lower and upper sensitivity limits.

Lower limit = Original value – Least (smallest) positive ratio or $-\infty$

Upper limit = Original value + Smallest absolute negative ratio or ∞

Shadow price is the value of one additional unit of a scarce resource.

		$c_j \rightarrow$	4	6	2	0	0
Basic Variable Coefficient c_B	*Basic Variables* B	*Basic Variables Value* b $(= x_B)$	x_1	x_2	x_3	s_1	s_2
4	x_1	1	1	0	– 1	4/3	– 1/3
6	x_2	2	0	1	2	– 1/3	1/3
$Z = 16$		z_j	4	6	8	10/3	2/3
		$c_j - z_j$	0	0	– 6	– 10/3	– 2/3

Table 6.8

To illustrate the method of finding the range of variation in the availability of resources, we repeat s_1 column and the *Basic Variable Values* column from Table 6.8 and calculate ratios as shown below:

Basis Variables B	*Solution Values* b $(= x_B)$	*Coefficients in* s_1*-Column*	*Minimum Ratio* $(= x_B/s_1)$	*Coefficients in* s_2*-Column*	*Minimum Ratio* $(= x_B/s_2)$
x_1	1	4/3	1/(4/3) = 3/4	– 1/3	1/(– 1/3) = – 3
x_2	2	– 1/3	2/(– 1/3) = – 6	1/3	2/(1/3) = 6

The smallest positive ratio (= 3/4) in x_1-row indicates the number of hours that can be decreased from the manpower resource while the smallest absolute negative ratio (= –6) indicates the number of hours that can be increased (added), without changing the current optimal product mix. Thus, the shadow price for manpower resource (= Rs 10/3) and raw material resource (= Rs 2/3) will remain unchanged over the range as follows.

Solution Mix	*Lower Limit*	*Upper Limit*
Manpower (x_1)	3 – (3/4) = 9/4	3 – (– 6) = 9
Raw material (x_2)	9 – 6 = 3	9 – (– 3) = 12

Case II: When a slack variable is a basic variable When a slack variable is the basic variable, the procedure for finding the range of variation for the corresponding *resource value* (RHS in a constraint) is as follows:

Lower limit = Original value – Solution value of slack variable

Upper limit = Infinity (∞)

Case III: Changes in right-hand side when constraints are of the mixed type

- *When surplus variable is not in the basis* (*Basic variable column,* **B**)

  Lower limits = Original value – Smallest absolute value of negative minimum ratios or $-\infty$
  Upper limit = Original value + Smallest positive minimum ratio or ∞

- *When surplus variable is in the basis* (*Basic variable column,* **B**)

  Lower limit = Minus infinity ($-\infty$)
  Upper limit = Original value + Solution value of surplus variable.

Alternative methods

1. Any change in the right-hand side of the constraints does not affect the optimality condition. Since in the determination of solution values, ($\mathbf{x_B} = \mathbf{B}^{-1}\,\mathbf{b}$) value of resource (i.e. **b**) is involved, therefore any change in b_i values affect the value of basic variables and value of objective function $Z(= \boldsymbol{c_B x_B})$. Thus, if resource b_k is changed to $b_k + \Delta b_k$, then the new values of resources becomes

   The $c_j - z_j$ values in the slack variable columns with negative sign are the shadow prices.

$$\mathbf{b}^* = (b_1, b_2, \ldots, b_k + \Delta b_k, \ldots, b_m)$$
$$= (b_1, b_2, \ldots, b_m) + (0, 0, \ldots, \Delta b_k, \ldots, 0) = \mathbf{b} + \Delta b_k$$

   The range of values within which Δb_k can vary without affecting the optimality of the current solution is determined as follows:

$$\mathbf{x}_B^* = \mathbf{B}^{-1}(\mathbf{b} + \Delta\mathbf{b}) = \mathbf{B}^{-1}\mathbf{b} + \mathbf{B}^{-1}(\Delta\mathbf{b})$$
$$= \mathbf{B}^{-1}\mathbf{b} + \mathbf{B}^{-1}(0, 0, \ldots, \Delta b_k, \ldots, 0)$$
$$= \mathbf{x}_B + (\boldsymbol{\beta}_1, \boldsymbol{\beta}_2, \ldots, \boldsymbol{\beta}_k, \ldots, \boldsymbol{\beta}_m)(0, 0, \ldots, \Delta b_k, \ldots, 0) = \mathbf{x}_B + \boldsymbol{\beta}_k(\Delta b_k)$$
$$x_{Bi}^* = x_{Bi} + \beta_{ik}(\Delta b_k); \quad \text{for } i\text{th basic variable,}$$

   where β_{ik} is the (i, k) element of $\mathbf{B}^{-1}$ and is the kth column vector of $\mathbf{B}^{-1}$. In order to maintain the feasibility of the solution at each iteration, the solution values, $\mathbf{x_B}$ must be non-negative. That is, we must have:

$$\mathbf{x}_B^* = x_{Bi} + \beta_{ik}(\Delta b_k) \geq 0; \quad i = 1, 2, \ldots, m$$

   Hence, the range of variation in b_k can be obtained by solving the following system of inequalities

$$\text{Min}\left\{\frac{-x_{Bi}}{\beta_{ik} < 0}\right\} \geq \Delta b_k \geq \text{Max}\left\{\frac{-x_{Bi}}{\beta_{ik} > 0}\right\}$$

2. The range of variation in the availability of resources (b_i), can also be obtained by using condition of feasibility of the current optimal solution, i.e. $\mathbf{x_B} = \mathbf{B}^{-1}\,\mathbf{b} \geq 0$,

   where $\mathbf{B}^{-1}$ = matrix of coefficients corresponding to slack variables in the optimal simplex table
   Δb_k = amount of change in the resource k
   $\mathbf{x_B}$ = basic variables appearing in **B**-column of simplex table

Remarks 1. If one or more entries in the $\mathbf{x_B}$-column of the simplex table are negative, the dual simplex method can be used to get an optimal solution to the new problem by maintaining feasibility.

2. A resource whose shadow price is higher in comparison to others, should be increased first to ensure the best marginal increase in the objective function value.

Example 6.4 Solve the following LP problem

Maximize $Z = 4x_1 + 6x_2 + 2x_3$

subject to the constraints

(i) $x_1 + x_2 + x_3 \leq 3$, (ii) $x_1 + 4x_2 + 7x_3 \leq 9$

and $x_1, x_2, x_3 \geq 0$

(a) Discuss the effect of discrete change in the availability of resources from $[3, 9]^T$ to $[9, 6]^T$.

(b) Which resource should be increased (or decreased) in order to get the best marginal increase in the value of the objective function? [*AMIE, 2004*]

Solution The given LP problem in its standard form can be stated as follows:

Maximize $Z = 4x_1 + 6x_2 + 2x_3 + 0.s_1 + 0.s_2$

subject to the constraints

(i) $x_1 + x_2 + x_3 + s_1 = 3$, (ii) $x_1 + 4x_2 + 7x_3 + s_2 = 9$

and $x_1, x_2, x_3, s_1, s_2 \geq 0$

Applying the simplex method, the optimal solution: $x_1 = 1, x_2 = 2$ and Max $Z = 16$ so obtained is shown in Table 6.9.

(a) If new values of the right-hand side constants in the constraints are $[\,9,\ 6\,]^T$, then the new values of the basic variables ($\mathbf{x_B} = \mathbf{B}^{-1}\,\mathbf{b}$) shown in Table 6.9 will become:

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4/3 & -1/3 \\ -1/3 & 1/3 \end{bmatrix} \begin{bmatrix} 9 \\ 6 \end{bmatrix} = \begin{bmatrix} 10 \\ -1 \end{bmatrix} \quad \text{or} \quad x_1 = 10, \text{ and } x_2 = -1$$

Table 6.9 Optimal Solution

		$c_j \rightarrow$	4	6	2	0	0
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2
4	x_1	1	1	0	−1	4/3	−1/3
6	x_2	2	0	1	2	−1/3	1/3
$Z = 16$		z_j	4	6	8	10/3	2/3
		$c_j - z_j$	0	0	−6	−10/3	−2/3

Since the value of x_2 is negative, the optimal solution shown in Table 6.9 is infeasible. Apply dual simplex to remove this infeasibility. For this, reproducing Table 6.9 with new values of x_1 and x_2 as shown in Table 6.10.

Table 6.10

		$c_j \rightarrow$	4	6	2	0	0
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2
4	x_1	10	1	0	−1	4/3	−1/3
6	x_2	−1	0	1	2	(−1/3)	1/3 →
$Z = 34$		z_j	4	6	8	10/3	2/3
		$c_j - z_j$	0	0	−6	−10/3 ↑	−2/3

Since $x_2 = -1$, therefore x_2-row becomes the key row and so x_2 is the outgoing variable. Calculating the minimum ratio to identify key column as follows.

$$\text{Min}\left\{\frac{c_j - z_j}{y_{rj}},\ y_{rj} < 0\right\} = \left\{\frac{-10/3}{-1/3}\right\} = 10 \text{ (column } s_1)$$

Hence column 's_1' is the key column and variable s_1 will enter into the basis. Revise the simplex table using the following row operations:

$$R_2 \text{ (new)} \rightarrow R_2 \text{ (old)} \times -3 \qquad R_1 \text{ (new)} \rightarrow R_1 \text{ (old)} - (4/3)\ R_2 \text{ (new)}$$

The new solution is shown in Table 6.11.

Since all $c_j - z_j \leq 0$ and all $b_i > 0$, the solution: $x_1 = 6, x_2 = 0, x_3 = 0$ and Max $Z = 24$ given in Table 6.11 is optimal.

	$c_j \rightarrow$	4	6	2	0	0
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2
4	x_1	6	1	4	7	0	1
0	s_1	3	0	−3	−6	1	−1
$Z = 24$		z_j	4	16	28	0	4
		$c_j - z_j$	0	−10	−26	0	−4

Table 6.11 Optimal Solution

(b) In Table 6.10, $c_4 - z_4 = 10/3$ and $c_5 - z_5 = 2/3$ (ignoring negative sign) corresponding to s_1 and s_2 columns. These values represent shadow prices of resources 1 and 2 respectively. Thus, any increase in the amount of resources 1 and 2 will increase the value of objective function Z by Rs 10/3 and Rs 2/3, respectively. In order to know how much these resources may be increased without disturbing current optimal solution. Let Δb_1 be an increase in first resource (RHS of first constraint) so that

$$\mathbf{x_B} = \mathbf{B^{-1}}\ \mathbf{b} \geq 0$$

or
$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4/3 & -1/3 \\ -1/3 & 1/3 \end{bmatrix} \begin{bmatrix} 3+\Delta b_1 \\ 9 \end{bmatrix} = \begin{bmatrix} 1+(4/3)\Delta b_1 \\ 2-(1/3)\Delta b_1 \end{bmatrix} \geq 0$$

i.e. $1 + (4/3)\Delta b_1 \geq 0$ or $\Delta b_1 \geq -3/4$ and $2 - (1/3)\Delta b_1 \geq 0$ or $\Delta b_1 \leq 3$

From these inequalities, we have: $-3/4 \leq \Delta b_1 \leq 3$ or $3 - (3/4) \leq b_1 \leq 3 + 6$ or $9/4 \leq b_1 \leq 9$

Hence, value of first resource can be increased from 3 units to a maximum limit of 9 units. Similarly, second resource can be increased from 9 units to a maximum limit of 20 units (approx.).

Example 6.5 Solve the following LP problem

Maximize $Z = 5x_1 + 12x_2 + 4x_3$,

subject to the constraints

(i) $x_1 + 2x_2 + x_3 \leq 5$, (ii) $2x_1 - x_2 + 3x_3 = 2$

and $x_1, x_2, x_3 \geq 0$.

(a) Discuss the effect of changing the requirement vector from $[5, 2]^T$ to $[7, 2]^T$ on the optimum solution.

(b) Discuss the effect of changing the requirement vector from $[5, 2]^T$ to $[3, 9]^T$ on the optimum solution.

(c) Which resource should be increased and how much to achieve the best marginal increase in the value of the objective function? *[Dayalbagh Edu. Inst., M Tech., 2006]*

Solution The given LP problem in its standard form can be stated as follows:

Maximize $Z = 5x_1 + 12x_2 + 4x_3 + 0s_1 - MA_1$

subject to the constraints

(i) $x_1 + 2x_2 + x_3 + s_1 = 5$, (ii) $2x_1 - x_2 + 3x_3 + A_1 = 2$,

and $x_1, x_2, x_3, s_1, A_1 \geq 0$.

An initial solution: $s_1 = 5, A_1 = 2$ and Max $Z = -2M$ is obtained as shown in Table 6.12, by putting decesion variables $x_1 = x_2 = x_3 = 0$

	$c_j \rightarrow$	5	12	4	0	–M	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	s_1	A_1	*Min Ratio* x_B/x_3
0	s_1	5	1	2	1	1	0	5/1
–M	A_1	2	2	–1	(3)	0	1	2/3
$Z = -2M$		z_j	–2M	M	–3M	0	–M	
		$c_j - z_j$	5 + 2 M	12 – M	4 + 3 M ↑	0	0	

Table 6.12 Initial Solution

Applying Big-M simplex method, the optimal solution: $x_1 = 9/5, x_2 = 8/5, x_3 = 0$ and max $Z = 141/5$ is shown in Table 6.13.

Table 6.13 Optimal Solution

		$c_j \rightarrow$	5	12	4	0	–M
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (=x_B)$	x_1	x_2	x_3	s_1	A_1
12	x_2	8/5	0	1	– 1/5	2/5	– 2/5
5	x_1	9/5	1	0	7/5	1/5	2/5
$Z = 141/5$		z_j	5	12	23/5	29/5	– 2/5
		$c_j - z_j$	0	0	– 3/5	– 29/5	$-M + 2/5$

(a) If new values of right hand side constants in the constraints are changed from $[5, 2]^T$ to $[7, 2]^T$, then the new values of the basic variables ($\boldsymbol{x_B} = \boldsymbol{B}^{-1}\boldsymbol{b}$) shown in Table 6.13 will become

$$\begin{bmatrix} x_2 \\ x_1 \end{bmatrix} = \boldsymbol{B}^{-1}\boldsymbol{b} = \begin{bmatrix} 2/5 & -1/5 \\ 1/5 & 2/5 \end{bmatrix} \begin{bmatrix} 7 \\ 2 \end{bmatrix} = \begin{bmatrix} 14/5 - 2/5 \\ 7/5 + 4/5 \end{bmatrix} = \begin{bmatrix} 12/5 \\ 11/5 \end{bmatrix}$$

Since both x_1 and x_2 are non-negative, the current solution remains feasible and optimal with new values: $x_1 = 11/5, x_2 = 12/5, x_3 = 0$ and Max Z = 199/5.

(b) New values of the current basic variables are

$$\begin{bmatrix} x_2 \\ x_1 \end{bmatrix} = \boldsymbol{B}^{-1}\boldsymbol{b} = \begin{bmatrix} 2/5 & -1/5 \\ 1/5 & 1/5 \end{bmatrix} \begin{bmatrix} 3 \\ 9 \end{bmatrix} = \begin{bmatrix} 6/5 - 9/5 \\ 3/5 + 18/5 \end{bmatrix} = \begin{bmatrix} -3/5 \\ 21/5 \end{bmatrix}$$

Since value of x_2 becomes negative, the current optimal solution becomes infeasible. Applying dual simplex method to remove infeasibility. Rewriting Table 6.13 with new values as shown in Table 6.14.

Table 6.14 Optimal (but infeasible) solution

		$c_j \rightarrow$	5	12	4	0	–M
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (=x_B)$	x_1	x_2	x_3	s_1	A_1
12	x_2	– 3/5	0	1	(– 1/5)	2/5	– 1/5
5	x_1	21/5	1	0	7/5	1/5	2/5
$Z = 69/5$		z_j	5	12	23/5	29/5	– 2/5
		$c_j - z_j$	0	0	– 3/5 ↑	– 29/5	$-M + 2/5$

Since $x_2 = -3/5$ (< 0), the x_2-row is the key row and x_2 is the outgoing variable. For identifying the key column, calculate the following ratio

$$\text{Min}\left\{\frac{c_j - z_j}{y_{r_j}},\quad y_{r_j} < 0\right\} = \left\{\frac{-3/5}{-1/5}\right\} = 3\ (x_3 - \text{column})$$

Revise the solution shown in Table 6.14 using following row operations:

$$R_1\ (\text{new}) \rightarrow R_1\ (\text{old}) \times -5 \qquad R_2\ (\text{new}) \rightarrow R_2\ (\text{old}) - (7/5)\, R_1\ (\text{new})$$

The new solution is shown in Table 6.15.

Table 6.15 Optimal Solution

		$c_j \rightarrow$	5	12	4	0	0
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (=x_B)$	x_1	x_2	x_3	s_1	A_1
4	x_3	3	0	–5	1	–2	1
5	x_1	0	1	7	0	3	–1
$z = 36$		z_j	5	15	4	7	–1
		$c_j - z_j$	0	–3	0	–7	$-M + 1$

Since all $c_j - z_j \le 0$ and $b_i \ge 0$, the current solution: $x_1 = 0, x_2 = 0, x_3 = 3$, Max $Z = 12$ shown in Table 6.15 is optimal.

(c) To find the value of any resource that should be increased (or decreased), write the dual objective function as: $Z_y = 5y_1 + 2y_2$, where, $y_1 = 29/5$ an $y_2 = 3/5$ are the optimal dual variables (Table 6.13). Since value of y_1 is higher than the value of y_2, the first resource should be increased as each additional unit of the first resource increases the objective function by 29/5. This requirement will be met so long as the primal problem remains feasible. Let Δb_1 be the increase in the first resource, so that

$$\begin{bmatrix} x_2 \\ x_1 \end{bmatrix} = \boldsymbol{B^{-1}}\, \boldsymbol{b} = \begin{bmatrix} 2/5 & -1/5 \\ 1/5 & 2/5 \end{bmatrix} \begin{bmatrix} 5 + \Delta b_1 \\ 2 \end{bmatrix} = \begin{bmatrix} 10/5 + 2\Delta b_1/5 - 2/5 \\ 5/5 + \Delta b_1/5 + 4/5 \end{bmatrix} = \begin{bmatrix} \{8 + 2\Delta b_1\}/5 \\ \{9 + \Delta b_1\}/5 \end{bmatrix} \ge 0$$

i.e. $\quad 8 + 2\Delta b_1 \ge 0$ or $\Delta b_1 \ge -4$ and $9 + \Delta b_1 \ge 0$ or $\Delta b_1 \ge -9$

Since x_1 and x_2 remain feasible (≥ 0) for all values of $\Delta b_1 \ge 0$, the first resource can be increased indefinitely while maintaining the condition that each additional unit will increase the objective function by 29/5.

The second resource should be decreased as each additional unit of the second resource decreases the objective function by 3/5. Let Δb_2 be the decrease in the second resource, so that

$$\begin{bmatrix} x_2 \\ x_1 \end{bmatrix} = \boldsymbol{B^{-1}}\, \boldsymbol{b} = \begin{bmatrix} 2/5 & -1/5 \\ 1/5 & 2/5 \end{bmatrix} \begin{bmatrix} 5 \\ 2 - \Delta b_2 \end{bmatrix} = \begin{bmatrix} 10/5 - 2/5 + \Delta b_2/5 \\ 5/5 + 4/5 - 2\Delta b_2/5 \end{bmatrix} \begin{bmatrix} \{8 + \Delta b_2\}/5 \\ \{9 - 2\Delta b_2\}/5 \end{bmatrix} \ge 0$$

Thus, x_1 remains positive only so long as $9 - 2\Delta b_2 \ge 0$ or $\Delta b_2 \le 9/2$. If $\Delta b_2 > 9/2$, x_1 becomes negative and must leave the solution.

Example 6.6 A factory manufactures three products *A*, *B* and *C* for which the data is given in the table below. Find the optimal product mix if the profit/unit is Rs. 32, Rs. 30 and Rs. 40 for product *A*, *B* and *C* respectively.

	Product			*Available Resources*
	A	*B*	*C*	
Material required (kg/unit)	5	4	3	2,500 kg
Machine hours required/unit	2	3	1	1,275 hours
Labour hours require/unit	3	2	4	2,100 hours

(a) Find the optimal solution if machine hours available become 1,350 instead of 1,275.
(b) Find the optimal solution if labour hours become 2,000 instead of 2,100.
(c) Find the optimal solution if 10 units of product *A* are to be produced.

Solution Let us define the following decision variables:

x_1, x_2 and x_3 = Number of units of product *A*, *B* and *C* to be produced, respectively.

Then the LP model based on problem data is written as:

Maximize $Z = 32x_1 + 30x_2 + 40x_3$

subject to the constraints

(i) $5x_1 + 4x_2 + 3x_3 \le 2{,}500$, (ii) $2x_1 + 3x_2 + x_3 \le 1{,}275$, (iii) $3x_1 + 2x_2 + 4x_3 \le 2{,}100$

and $\quad x_1, x_2, x_3 \ge 0$.

Initial basic feasible solution: $x_1 = 0, x_2 = 0, x_3 = 0, s_1 = 2{,}500, s_2 = 1{,}275, s_3 = 2{,}100$ and Max $Z = 0$ is shown in Table 6.16

		$c_j \rightarrow$	*32*	*30*	*40*	*0*	*0*	*0*	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3	*Min Ratio* x_B / x_3
0	s_1	2500	5	4	3	1	0	0	2500/3
0	s_2	1275	2	3	1	0	1	0	1275/1
0	s_3	2100	3	2	(4)	0	0	1	2100/4 →
$Z = 0$		z_j	0	0	0	0	0	0	
		$c_j - z_j$	32	30	40 ↑	0	0	0	

Table 6.16 Initial Solution

The initial solution shown in Table 6.16 is updated to obtain the optimal solution shown in Table 6.17 using following row operations:

R_3 (new) $\rightarrow R_3$ (old)/4 (*key element*); R_2 (new) $\rightarrow R_2$ (old) $- R_3$ (new);
R_1 (new) $\rightarrow R_1$ (old) $- 3\, R_3$ (new)

Table 6.17
Optimal Solution

		$c_j \rightarrow$	32	30	40	0	0	0
Basic Variables Coefficients c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3
0	s_1	125	3/2	0	0	1	–1	–1/2
30	x_2	300	1/2	1	0	0	2/5	–1/10
40	x_3	375	1/2	0	1	0	–1/5	3/10
$Z = 24,000$		z_j	35	30	40	0	4	9
		$c_j - z_j$	– 3	0	0	0	– 4	– 9

The optimal solution shown in Table 6.17 is: $x_1 = 0, x_2 = 300$ units, $x_3 = 375$ units and Max Z = Rs. 24,000. Since $s_2 = 0$ and $s_3 = 0$, this implies that while producing product B and C all the machine hours and labour hours available are consumed. However, $s_1 = 125$ kg implies this only 125 kg of material remains unutilized. From Table 6.17, the following conclusions may be drawn:

(a) s_2-column shows that if one machine hour is increased, 1 kg of material will remain unconsumed and product B will increase by 2/5 unit and C will reduce by –1/5 unit. Thus, if machine hours available are increased by 75(= 1,350 – 1,275), the product B (= variable x_2) will increase by (2/5) × 75 = 30 units and product C (= variable x_3) will decrease by (1/5) × 75 = 15 units. Thus the optimal solution will be: $x_2 = 330$, $x_3 = 360$. For every additional machine hour used, there is a gain of Rs. 4 and hence profit will increase by Rs. 75 × 4 = Rs. 300 to attain a value of Z = Rs. 24,300 (= 0 × 125 + 30 × 330 + 40 × 360).

(b) s_3-column shows that if one labour hour is decreased, it would change the value of x_2 by – 1 × (–1/10) = (1/10) units and of x_3 by – 1 × (3/10) = – (3/10) units. Therefore, a reduction of 100 (= 2,100 – 2,000) labour hours will increase value of variable x_2 by (1/10) × 100 = 10 units and decrease value of variable, x_3 by (3/10) × 100 = 30 units and the new optimal solution would be: $x_1 = 0, x_2 = 310$ units, $x_3 = 345$ units and Max Z = Rs. 23,100.

(c) x_1-column shows that if one unit of product A is produced, it would reduce the production of B and C each by 1/2 unit each. Therefore, if $x_1 = 10, x_2 = 300 - (1/2) \times 10 = 295, x_3 = 375 - (1/2) \times 10 = 370$. The profit contribution of Rs. 3 by product A (coefficient of variable x_1) will also reduces and hence it will reduced to Rs. 30 and become Z = Rs. 23,970 (= 10 × 32 + 30 × 295 + 40 × 370).

6.2.3 Changes in the Input-Out Coefficients (a_{ij}'s)

Suppose that the elements of coefficient matrix **A** are changed. Then two cases arise

(i) Change in a coefficient, when variable is a basic variable, and
(ii) Change in a coefficient, when variable is a non-basic variable.

Case I: When a non-basic variable column, $\mathbf{a}_k \notin \mathbf{B}$ is changed to $\mathbf{a}_k^*$, the only effect of such change will be on the optimality condition. Thus the solution will remain optimal, if

$$c_k - z_k^* = c_k - \mathbf{c_B}\,\mathbf{B}^{-1}\mathbf{a}_k^* \leq 0$$

However, the range for the discrete change Δa_{ij} in the coefficients of non-basic variable, x_j in constraint, i can be determined by solving following linear inequalities:

$$\text{Max}\left\{\frac{c_j - z_j}{\mathbf{c_B}\,\boldsymbol{\beta}_i > 0}\right\} \leq \Delta a_{ij} \leq \text{Min}\left\{\frac{c_j - z_j}{\mathbf{c_B}\,\boldsymbol{\beta}_i < 0}\right\}$$

Here $\boldsymbol{\beta}_i$ is the ith column unit matrix **B**. If $\mathbf{c_B}\,\boldsymbol{\beta}_i = 0$, then Δa_{ij} is unrestricted in sign.

Alternative method The change in the values of coefficients (a_{ij}'s) associated with non-basic variables can be analysed by forming a corresponding dual constraint from the original set of constraints:

$$\sum_{i=1}^{m} a_{ji}\, y_i \geq c_j;\ \text{for } x_j \text{ non-basic variable.}$$

The value of dual variables y_i's can be obtained from the optimal simplex table. The reason behind this dual constraint formulation is that if shadow price of all resources needed to produce one unit of an activity become equal to its per unit contribution towards value of objective function.

Case II : Suppose a basic variable column $\mathbf{a}_k \in \mathbf{B}$ is changed to $\mathbf{a}_k^*$. Then conditions to maintain both feasibility and optimality of the current optimal solution are:

(a) $$\underset{k \neq p}{\text{Max}} \left\{ \frac{-x_{Bk}}{x_{Bk}\,\beta_{pi} - x_{Bp}\,\beta_{ki} > 0} \right\} \leq \Delta a_{ij} \leq \underset{k \neq p}{\text{Min}} \left\{ \frac{-x_{Bk}}{x_{Bk}\,\beta_{pi} - x_{Bp}\,\beta_{ki} < 0} \right\}$$

(b) $$\text{Max} \left\{ \frac{c_j - z_j}{(c_j - z_j)\beta_{pi} - y_{pj}\,\mathbf{c_B}\,\boldsymbol{\beta}_i > 0} \right\} \leq \Delta a_{ij} \leq \text{Min} \left\{ \frac{c_j - z_j}{(c_j - z_j)\beta_{pi} - y_{pj}\,\mathbf{c_B}\,\boldsymbol{\beta}_i < 0} \right\}$$

Example 6.7 Solve the following LP problem

Maximize $Z = -x_1 + 3x_2 - 2x_3$

subject to the constraints

(i) $3x_1 - x_2 + 2x_3 \leq 7$, (ii) $-2x_1 + 4x_2 \leq 12$, (iii) $-4x_1 + 3x_2 + 8x_3 \leq 10$

and $x_1, x_2, x_3 \geq 0$.

Discuss the effect of the following changes in the optimal solution.

(a) Determine the range for discrete changes in the coefficients a_{13} and a_{23} consistent with the optimal solution of the given LP problem.

(b) 'x_1'-column in the LP problem is changed from $[\,3, -2, -4\,]^T$ to $[\,3, 2, -4\,]^T$.

(c) 'x_3'-column in the LP problem is changed from $[\,2, 0, 8\,]^T$ to $[\,3, 1, 6\,]^T$. [*AMIE, 2008*]

Solution The given LP problem in its standard form can be stated as follows:

Maximize $Z = -x_1 + 3x_2 - 2x_3 + 0.s_1 + 0.s_2 + 0.s_3$

subject to the constraints

(i) $3x_1 - x_2 + 2x_3 + s_1 = 7$, (ii) $-2x_1 + 4x_2 + s_2 = 12$, (iii) $-4x_1 + 3x_2 + 8x_3 + s_3 = 10$

and $x_1, x_2, x_3, s_1, s_2, s_3 \geq 0$

Applying the simplex (Big-*M*) method, the optimal solution so obtained is shown in Table 6.18.

		$c_j \rightarrow$	*– 1*	*3*	*– 2*	*0*	*0*	*0*
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3
– 1	x_1	4	1	0	4/5	2/5	1/10	0
3	x_2	5	0	1	2/4	1/5	3/10	0
0	s_3	11	0	0	10	1	– 1/2	1
$Z = 11$	z_j	–1	3	2/5	1/5	4/5	0	
		$c_j - z_j$	0	0	– 12/5	– 1/5	– 4/5	0

Table 6.18 Optimal Solution

The optimal basic feasible solution shown in Table 6.18 is: $x_1 = 4$, $x_2 = 5$, $x_3 = 0$ and Max $Z = 11$.

In table 6.18, the inverse of basis matrix, **B** is

$$\mathbf{B}^{-1} = \begin{bmatrix} 2/5 & 1/10 & 0 \\ 1/5 & 3/10 & 0 \\ 1 & -1/2 & 1 \end{bmatrix} = [\boldsymbol{\beta}_1, \boldsymbol{\beta}_2, \boldsymbol{\beta}_3]$$

Thus, we have $\mathbf{c_B}\,\boldsymbol{\beta}_1 = -1(2/5) + 3(1/5) + 0(1) = 1/5$

$\mathbf{c_B}\,\boldsymbol{\beta}_2 = -1(1/10) + 3(3/10) + 0(-1/2) = 8/10$

$\mathbf{c_B}\,\boldsymbol{\beta}_3 = -1(0) + 3(0) + 0(1) = 0$

Since variables x_1, x_2 and s_3 are in the basis, therefore any discrete change in coefficients, belonging to any of these column vectors may affect both feasibility and optimality of the original optimal basic feasible solution, whereas any discrete change in the non-basic variables (i.e. x_3, s_1 and s_2) column vectors may affect only the optimality condition.

(a) Ranges for discrete change in coefficients a_{13} and a_{23} in the x_3-column vector of Table 6.18 are computed as:

$$\text{Max}\left\{\frac{c_3 - z_3}{\mathbf{c_B}\,\boldsymbol{\beta}_1}\right\} = \text{Max}\left\{\frac{-12/5}{1/5}\right\} \le \Delta a_{13} \quad \text{or} \quad \Delta a_{13} \ge -12$$

and

$$\text{Max}\left\{\frac{c_3 - z_3}{\mathbf{c_B}\,\boldsymbol{\beta}_2}\right\} = \text{Max}\left\{\frac{-12/5}{8/10}\right\} \le \Delta a_{23} \quad \text{or} \quad \Delta a_{23} \ge -3$$

(b) To measure the change Δa_{21} in the coefficient a_{21} (= 2) of variable x_1 in the second constraint of the original set of constraints, we need to check both the feasibility as well as optimality conditions. This needs to be done because variable x_1 is the basic variable, as shown in Table 6.18.

(i) *Feasibility Condition:* For $i = 2$ (second constraint), $p = 1$ (first column), and $k = 2, 3$ (columns of basis matrix), we have

For $k = 2$ $\quad x_{B2}\,\beta_{12} - x_{B1}\,\beta_{22} = 5(1/10) - 4(3/10) = -7/10$

For $k = 3$ $\quad x_{B3}\,\beta_{12} - x_{B1}\,\beta_{32} = 11(1/10) - 4(-1/2) = 31/10$

Hence, the range to maintain feasibility of the existing optimal solution is

$$\frac{-5}{31/10} \le \Delta a_{21} \le \frac{-5}{-7/10}$$

$$2 - (50/31) \le a_{21} \le 2 + (50/7) \quad \text{or} \quad -12/31 \le a_{21} \le 64/7$$

(ii) *Optimality Condition*

$$(c_3 - z_3)\,\beta_{12} - y_{13}\,\boldsymbol{c}_B\,\beta_2 = -\frac{12}{5}\left(\frac{1}{10}\right) - \frac{4}{5}\left(\frac{8}{10}\right) = -\frac{44}{50}$$

$$(c_4 - z_4)\,\beta_{12} - y_{14}\,\boldsymbol{c}_B\,\beta_2 = -\frac{1}{5}\left(\frac{1}{10}\right) - \frac{2}{5}\left(\frac{8}{10}\right) = -\frac{17}{50}$$

$$(c_5 - z_5)\,\beta_{12} - y_{15}\,\boldsymbol{c}_B\,\beta_2 = -\frac{4}{5}\left(\frac{1}{10}\right) - \frac{1}{10}\left(\frac{8}{10}\right) = -\frac{16}{100}$$

Hence, the range to maintain optimality of the existing optimal solution is

$$-\infty \le \Delta a_{21} \le \text{Min}\left\{\frac{-12/5}{-44/50};\ \frac{-1/5}{-17/50};\ \frac{-4/5}{-16/100}\right\}$$

$$-\infty \le \Delta a_{21} \le 10/17 \quad \text{or} \quad -\infty \le a_{21} \le 44/17$$

(c) Suppose column vector $\mathbf{a}_3$ (x_3-column in Table 6.18) of original LP model is changed from $[2, 0, 8]^T$ to $[3, 1, 6]^T$. Then new value of $c_3 - z_3^*$ for this column is:

$$\mathbf{B}^{-1}\,\mathbf{a}_3^* = \begin{bmatrix} 2/5 & 1/10 & 0 \\ 1/5 & 3/10 & 0 \\ 1 & -1/2 & 1 \end{bmatrix}\begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 13/10 \\ 9/10 \\ 17/2 \end{bmatrix}$$

$$c_3 - z_3^* = c_3 - \boldsymbol{c}_B \mathbf{B}^{-1} \mathbf{a}_3^* = -2 - [-1,\ 3,\ 0] \begin{bmatrix} 13/10 \\ 9/10 \\ 17/2 \end{bmatrix} = -\frac{34}{10}$$

Since all entries in the x_3-column are non-zero we must replace column x_3 with new entries in Table 6.18 and proceed to get new optimal solution.

Example 6.8 Find the effect of the following changes on the optimal solution (Table 6.19) of the following LP problem.

Maximize $Z = 45x_1 + 100x_2 + 30x_3 + 50x_4$

subject to the constraint

(i) $7x_1 + 10x_2 + 4x_3 + 9x_4 \geq 1200$, (ii) $3x_1 + 40x_2 + x_3 + x_4 \leq 800$,

and $x_2, x_2, x_3, x_4 \geq 0$.

	$c_j \rightarrow$	*45*	*100*	*30*	*50*	*0*	*0*
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	x_4	s_1	s_2
30	x_3	800/3	5/3	0	1	7/3	4/5	–1/15
100	x_2	40/3	1/30	1	0	–1/30	–1/150	2/75
$Z = 28{,}000/3$		z_j	1600/30	100	30	–11/3	110/15	50/75
		$c_j - z_j$	–25/3	0	0	–55/3	–22/3	–2/3

Table 6.19 Optimal Solution

(a) 'x_1'-column in the problem changes from $[7, 3]^T$ to $[7, 5]^T$.

(b) 'x_1'-column changes from $[7, 3]^T$ to $[5, 8]^T$.

Solution (a) The variable x_1 is a non-basic variable in the optimal solution shown in Table 6.19. The upper limit $\bar{c}_1$ the coefficient c_1 of x_1 is calculated as follows:

$$\bar{c}_1 = c_1 - \boldsymbol{c}_B \bar{\boldsymbol{a}}_1 = c_1 - \boldsymbol{c}_B \boldsymbol{B}^{-1} \boldsymbol{a}_1 = c_1 - \hat{y}_1 \boldsymbol{a}_1, \quad \text{where} \quad c_1 = 45,\ \boldsymbol{a}_1 = \begin{bmatrix} 7 \\ 5 \end{bmatrix},$$

and

$$\hat{y} = \boldsymbol{c}_B \boldsymbol{B}^{-1} = (30, 100) \begin{bmatrix} 4/15 & -1/15 \\ -1/150 & 2/7 \end{bmatrix} = \left[\frac{22}{3}, \frac{2}{3}\right].$$

Thus

$$\bar{c}_1 = 45 - \left[\frac{22}{3}, \frac{2}{3}\right] \begin{bmatrix} 7 \\ 5 \end{bmatrix} = 45 - \left(\frac{154}{3} + \frac{10}{3}\right) = 45 - \frac{164}{3} = -\frac{29}{3}.$$

Since value of $\bar{c}_1$ is negative, the current optimal solution remains optimal for the new problem also.

(b) $\bar{c}_1 = c_1 - \boldsymbol{c}_B \bar{\boldsymbol{a}}_1 = c_1 - \boldsymbol{c}_B \boldsymbol{B}^{-1} \boldsymbol{a}_1 = c_1 - \hat{y}_1 \boldsymbol{a}_1 = 45 - \left[\frac{22}{3}, \frac{2}{3}\right] \begin{bmatrix} 5 \\ 8 \end{bmatrix} = 45 - \left(\frac{110}{3} + \frac{16}{3}\right) = 3.$

Since value of $\bar{c}_1$ is positive, the current optimal solution can be improved. Also

$$\bar{\boldsymbol{a}}_1 = \boldsymbol{B}^{-1} \boldsymbol{a}_1 = \begin{bmatrix} 4/15 & -1/5 \\ -1/150 & 2/75 \end{bmatrix} \begin{bmatrix} 5 \\ 8 \end{bmatrix} = \begin{bmatrix} 4/5 \\ 27/150 \end{bmatrix}.$$

	$c_j \rightarrow$	*45*	*100*	*30*	*50*	*0*	*0*	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	x_4	s_1	s_2	*Min Ratio* x_B / x_1
30	x_3	800/3	4/5	0	1	7/3	4/15	– 1/15	1000/3
100	x_2	40/3	(27/150)	1	0	– 1/30	– 1/150	2/75	2000/27 →
$Z = 28{,}000/3$		z_j	126/3	100	30	11/3	110/15	50/75	
		$c_j - z_j$	3 ↑	0	0	– 55/3	– 22/3	– 2 / 3	

Table 6.20 Optimal Solution

Introducing non-basic variable x_1 into the basis to replace basic variable x_2 using suitable row operations: In Table 6.21, all $c_j - z_j \le 0$, the solution is optimal with: $x_1 = 2000/27, x_2 = 0, x_3 = 5600/27$ and Max $Z = 86{,}000/9$.

Table 6.21 Optimal Solution

	$c_j \rightarrow$	*45*	*100*	*30*	*0*	*0*	*0*
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	x_4	s_1	s_2
30	x_3	5600/27	0	–40/9	1	67/27	8/27	–5/27
45	x_1	2000/27	1	50/9	0	–5/7	–1/27	4/27
$Z = 86,000/9$		z_j	45	350	30	595/9	65/9	10/9
		$c_j - z_j$	0	–50/3	0	–145/9	–65/9	–10/9

6.2.4 Addition of a New Variable (Column)

Let an extra variable x_{n+1} with coefficient c_{n+1} be added in the system of original constraint $\mathbf{Ax} = \mathbf{B}$, $\mathbf{x} \ge 0$. This in turn creates an extra column $\boldsymbol{a_{n+1}}$ in the matrix $\mathbf{A}$ of coeffcients. To see the impact of this addition on the current optimal solution, we compute

$$\mathbf{y_{n+1}} = \mathbf{B}^{-1}\ \mathbf{a_{n+1}}$$

and

$$c_{n+1} - z_{n+1} = c_{n+1} - \mathbf{c_B}\,\mathbf{y_{n+1}}$$

Two cases of the maximization LP model may arise:

(a) If $c_{n+1} - z_{n+1} \le 0$, then $\mathbf{x_B} = 0$, and hence current solution remains optimal.

(b) If $c_{n+1} - z_{n+1} > 0$, then the current optimal solution can be improved upon by the introduction of a new column $\mathbf{a_{n+1}}$ into the basis to find the new optimal solution.

Example 6.9 Discuss the effect on optimality by adding a new variable to the following LP problem with column coefficients $(3, 3, 3)^T$ and coefficient 5 in the objective function.

Minimize $Z = 3x_1 + 8x_2$

subject to the constraints

(i) $x_1 + x_2 = 200$, (ii) $x_1 \le 80$, (iii) $x_2 \ge 60$

and $x_1, x_2 \ge 0$

Solution The given LP problem in its standard form can be expressed as:

Minimize $Z = 3x_1 + 8x_2 + 0s_1 + 0s_2 + MA_1 + MA_2$

subject to the constraints

(i) $x_1 + x_2 + A_1 = 200$, (ii) $x_1 + s_1 = 80$, (iii) $x_2 - s_2 + A_2 = 60$

and $x_1, x_2, s_1, s_2, A_1, A_2 \ge 0$

Applying the simplex method to obtain optimal solution: $x_1 = 80$, $x_2 = 120$ and Min $Z = 1{,}200$ as shown in Table 6.22.

Table 6.22 Primal-Dual Relationship

	$c_j \rightarrow$	*3*	*8*	*0*	*0*	*M*
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	s_1	s_2	A_1
0	s_2	60	0	0	– 1	1	1
3	x_1	80	1	0	1	0	0
8	x_2	120	0	1	– 1	0	1
$Z = 1{,}200$		z_j	3	8	– 5	0	8
		$c_j - z_j$	0	0	5	0	$M - 8$

Given that $c_7 = 5$ and the column, $\mathbf{a}_7 = (3, 3, 3)^T$, the changes in the optimal solution can be evaluated as follows:

$$c_7 - z_7 = c_7 - \mathbf{c_B}\,\mathbf{B}^{-1}\,\mathbf{a}_7 = 5 - (0, 3, 8)\begin{bmatrix} -1 & 1 & 1 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix}\begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} = -4$$

$$\mathbf{a}_7^* = \mathbf{B}^{-1}\,\mathbf{a}_7 = \begin{bmatrix} -1 & 1 & 1 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix}\begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 0 \end{bmatrix}$$

Since $c_7 - z_7 = -4$, which is a negative number, the existing optimal solution can be improved. For this, adding entries corresponding to variable x_7 in Table 6.22, we get values as shown in Table 6.23.

	$c_j \rightarrow$	3	8	0	0	M	5	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	s_1	s_2	A_1	x_7	*Min. Ratio* x_B/x_7
0	s_2	60	0	0	−1	1	1	(3)	60/3 →
3	x_1	80	1	0	1	0	0	3	80/3
8	x_2	120	0	1	−1	0	1	0	—
$Z = 1{,}200$		z_j	3	8	−5	0	8	9	
		$c_j - z_j$	0	0	5	0	$M - 8$	−4 ↑	

Table 6.23 New Column with Variable x_7 Added

The variable x_7 in Table 6.23 must enter into the solution and s_2 should leave it. The new solution is shown in Table 6.24.

	$c_j \rightarrow$	3	8	0	0	M	5
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	s_1	s_2	A_1	x_7
5	x_7	20	0	0	−1/3	1/3	1/3	1
3	x_1	20	1	0	2	−1	−1	0
8	x_2	120	0	1	−1	0	1	0
$Z = 1{,}120$		z_j	3	8	−11/3	−4/3	−4/3	5
		$c_j - z_j$	0	0	11/13	4/3	$M + 4/3$	0

Table 6.24

Since all $c_j - z_j \geq 0$ in Table 6.24, the solution is optimal, with $x_1 = 20$, $x_2 = 120$, $x_7 = 20$ and Min $Z = 1{,}120$.

6.2.5 Addition of a New Constraint (Row)

After solving an LP model, the decision-maker may recall that a particular resource constraint was overlooked during the model formulation or perhaps he may desire to know the effect of adding a new resource to enhance the objective function value. The addition of a constraint in the existing constraints will cause a simultaneous change in the objective function coefficients (c_j), as well as coefficients a_{ij} of a corresponding non-basic variable. Consequently, it will affect the optimality of the LP problem. This means that the new variable should enter into the basis only if it improves the value of the objective function.

Suppose that a new constraint

$$\mathbf{a}_{m+1,1}\,x_1 + \mathbf{a}_{m+1,2}\,x_2 + \cdots + \mathbf{a}_{m+1,n}\,\mathbf{x}_n \leq \mathbf{b}_{m+1}$$

is added to the system of original constraints $\mathbf{Ax} = \mathbf{b}$, $\mathbf{x} \geq 0$, where b_{m+1} is positive, zero or negative. Then the following two cases may arise.

1. The optimal solution ($\mathbf{x_B}$) of the original problem satisfies the new constraint. If this is the case, then the solution remains feasible as well as optimal. This is because the new constraint either reduces or leaves unchanged the feasible region of the given LP problem.

2. The optimal solution ($\mathbf{x_B}$) of the original problem does not satisfy the new constraint. In this case the optimal solution to the modified LP problem should be re-obtained. Let $\mathbf{B}$ be the basis matrix for the original problem and $\mathbf{B}_1$ be the basis matrix for the new problem with $m + 1$ constraints. That is, matrix $\mathbf{B}_1$ of order $(m + 1)$ is given by

$$\mathbf{B}_1 = \begin{bmatrix} \mathbf{B} & 0 \\ \boldsymbol{\alpha} & 1 \end{bmatrix}$$

where the second column of $\mathbf{B}_1$ corresponds to slack, surplus or artificial variables added to the new constraint and $\boldsymbol{\alpha} = (a_{m+1,1}, a_{m+1,2}, \ldots, a_{m+1,n})$ is a row vector containing the coefficients in the new constraint and corresponds to variables in the optimal basis. In order to prove that the new solution

$$\mathbf{x}_\mathbf{B}^* = \begin{bmatrix} \mathbf{x_B} \\ \mathbf{s} \end{bmatrix} \quad \mathbf{s} = \text{slack variable}$$

is a basic feasible solution to the new LP problem, we shall compute the inverse of $\mathbf{B}_1$ using partitioned methods, as given below:

$$\mathbf{B}_1^{-1} = \begin{bmatrix} \mathbf{B}^{-1} & 0 \\ -\boldsymbol{\alpha}\mathbf{B}^{-1} & 1 \end{bmatrix}$$

Since each column vector in the new LP problem is given by $\mathbf{a}_j^* = (\mathbf{a}_j, \mathbf{a}_{m+1,j})$, the new columns y_j^* are given by

$$\mathbf{y}_j^* = \mathbf{B}^{-1}\mathbf{a}_j^* = \begin{bmatrix} \mathbf{B}^{-1} & 0 \\ -\boldsymbol{\alpha}\mathbf{B}^{-1} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{a}_j \\ \mathbf{a}_{m+1,j} \end{bmatrix}$$

$$= \begin{bmatrix} \mathbf{B}^{-1}\mathbf{a}_j \\ -\boldsymbol{\alpha}\mathbf{B}^{-1}\mathbf{a}_j + \mathbf{a}_{m+1,j} \end{bmatrix} = \begin{bmatrix} \mathbf{y}_j \\ \mathbf{a}_{m+1,j} - \boldsymbol{\alpha}\mathbf{y}_j \end{bmatrix}$$

The entries in the $c_j - z_j^*$ row of the simplex table for any non-basic variable x_j in the new problem are computed as follows:

$$c_j - z_j^* = c_j - z_j^*\mathbf{y}_j^* = c_j - [\mathbf{c_B}, \mathbf{c}_{Bm+1}] \begin{bmatrix} \mathbf{y}_j \\ a_{m+1,j} - \boldsymbol{\alpha}\mathbf{y}_j \end{bmatrix}$$

$$= c_j - (\mathbf{c_B}\mathbf{y}_j + \mathbf{c}_{Bm+1}\, a_{m+1} - \mathbf{c}_{Bm+1}\, \boldsymbol{\alpha}\mathbf{y}_j)$$

where c_{Bm+1} is the coefficient associated with the new variable introduced in the basis of the new LP problem.

If the new variable introduced in the basis of the new LP problem is slack or surplus variable, then $c_{Bm+1} = 0$. Hence, we have

$$c_j - z_j^* = c_j - \mathbf{c_B}\mathbf{y}_j = c_j - z_j$$

That is, the entries in the $c_j - z_j$ row are the same for the initial as well as the new LP problem. The value of objective function is given by:

$$Z^* = [\mathbf{c_B},\ 0] \begin{bmatrix} \mathbf{x_B} \\ \mathbf{s} \end{bmatrix} = \mathbf{c_B}\,\mathbf{x_B} = Z$$

This shows that the optimal simplex table of the original problem remains unchanged even after adding the new constraint. However, the slack or surplus variable appears with a negative value because the optimal solution of the original problem does not satisfy the new constraints. Thus, the dual simplex method may be used to get an optimal solution.

Remarks If the new constraint added is an equation and an artificial variable appears in the basis of the new problem, then the following two cases may arise:

1. If an artificial variable appears in the basis at negative value, a zero cost may be assigned to it. Apply the dual simplex method in order to obtain an optimal solution.
2. If the artificial variable appears in the basis at a positive value, then $-M$ cost may be assigned to it. Apply the usual simplex method to obtain an optimal solution.

Example 6.10 Consider the following LP problem

Maximize $Z = 3x_1 + 5x_2$

subject to the constraints

(i) $3x_1 + 2x_2 \le 18$, (ii) $x_1 + 2x_2 \le 4$, (iii) $x_2 \le 6$

and $x_1, x_2 \ge 0$

Obtain an optimal solution of the given LP problem.

(a) Suppose variable x_6 is added to the given LP problem. Then obtain an optimal solution to the resulting LP problem. It is given that the coefficients of x_6 in the constraint of the problem are 1, 1 and 1, and that its coefficient in the objective function is 2.

(b) Discuss the effect on the optimal basic feasible solution by adding a new constraint $2x_1 + x_2 \le 8$ to the given set of constraints. [*AMIE, 2008*]

Solution The optimal solution: $x_1 = 2, x_2 = 6$ with Max $Z = 36$ to the given LP problem is shown in Table 6.25.

		$c_j \rightarrow$	3	5	0	0	0
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	s_1	s_2	s_3
3	x_1	2	1	0	1/3	0	−2/3
0	s_2	0	0	0	−2/3	1	4/3
5	x_2	6	0	1	0	0	1
$Z = 36$		$c_j - z_j$	0	0	−1	0	−3

Table 6.25 Optimal Solution

(a) After adding the new variable, say x_6 in the given problem, the new LP problem becomes:

Maximize $Z = 3x_1 + 5x_2 + 2x_6$

subject to the constraints

(i) $3x_1 + 2x_2 + x_6 \le 18$, (ii) $x_1 + x_6 \le 4$, (iii) $x_2 + x_6 \le 6$

and $x_1, x_2, x_6 \ge 0$

Given that the column vector associated with variable, x_6 is $\mathbf{a}_6 = (1, 1, 1)$, with the help of Table 6.25, we compute

$$\mathbf{y}_6 = \mathbf{B}^{-1}\mathbf{a}_6 = \begin{bmatrix} 1/3 & 0 & -2/3 \\ -2/3 & 1 & 4/3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1/3 \\ 5/3 \\ 1 \end{bmatrix}$$

Since in the current optimal solution, $\mathbf{c_B} = (3, 0, 5)$, we have

$$c_6 - z_6 = c_6 - \mathbf{c_B}\mathbf{y}_6 = 2 - (3, 0, 5)\begin{bmatrix} -1/3 \\ 5/3 \\ 1 \end{bmatrix} = -2\ (\le 0)$$

As $c_6 - z_6 \le 0$, the optimality of the current solution remains unaffected with the addition of x_6.

(b) The optimal basic feasible solution given in Table 6.25 does not satisfy the additional constraint $2x_1 + x_2 \le 8$. Thus, a new optimal solution is obtained by adding slack variable s_4 to this constraint. Then, it is written together with the entries in Table 6.25, as shown in Table 6.26.

		$c_j \rightarrow$	3	5	0	0	0	0
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	s_1	s_2	s_3	s_4
3	x_1	2	1	0	1/3	0	−2/3	0
0	s_2	0	0	0	−2/3	1	4/3	0
5	x_2	6	0	1	0	0	1	0
0	s_4	8	2	1	0	0	0	1
$Z = 36$		$c_j - z_j$	0	0	−1	0	−3	0

Table 6.26

In Table 6.26 it may be noted that the basis matrix **B** has been disturbed due to row 4. Thus, the coefficients in row 4 must become zero. This can be done by using the following row operations.

$$R_4 \text{ (new)} \rightarrow R_4 \text{ (old)} - 2R_1 - R_3$$

We get a new table as shown in Table 6.27.

Table 6.27

		$c_j \rightarrow$	3	5	0	0	0	0
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b\ (= x_B)$	x_1	x_2	s_1	s_2	s_3	s_4
3	x_1	2	1	0	1/3	0	− 2/3	0
0	s_2	0	0	0	− 2/3	1	4/3	0
5	x_2	6	0	1	0	0	1	0
0	s_4	− 2	0	0	(− 2/3)	0	1/3	1 →
$Z = 36$		$c_j - z_j$	0	0	− 1 ↑	0	− 3	0

Since the solution given in Table 6.27 is optimal but not feasible, apply the dual simplex method to get an optimal basic feasible solution. Introduce s_1 into the basis and remove s_4 from the basis. The new solution is shown in Table 6.28.

Table 6.28
Optimal Solution

		$c_j \rightarrow$	3	5	0	0	0	0
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b\ (= x_B)$	x_1	x_2	s_1	s_2	s_3	s_4
3	x_1	1	1	0	0	0	− 1/2	1/2
0	s_2	2	0	0	0	1	1	− 1
5	x_2	6	0	1	0	0	1	0
0	s_1	3	0	0	1	0	− 1/2	− 3/2
$Z = 33$		$c_j - z_j$	0	0	0	0	− 7/2	− 3/2

Since all $c_j - z_j \leq 0$, the solution shown in Table 6.28 is the optimal basic feasible solution: $x_1 = 1$, $x_2 = 6$ and Max $Z = 33$. It may be noted here that the additional constraint has decreased the optimal value of the objective function from 36 to 33.

Example 6.11 Solve the following LP problem

Maximize $Z = 3x_1 + 4x_2 + x_3 + 7x_4$

subject to the constraints

(i) $8x_1 + 3x_2 + 4x_3 + x_4 \leq 7$, (ii) $2x_1 + 6x_2 + x_3 + 5x_4 \leq 3$

(iii) $x_1 + 4x_2 + 5x_3 + 2x_4 \leq 8$

and $x_1, x_2, x_3, x_4 \geq 0$

(a) Discuss the effect of discrete changes in (i) RHS of constraints, (ii) Coefficients of variables in the objective function, (iii) Structural coefficients in the constraints on the optimal basic feasible solution of the LP problem.

(b) Discuss the effect on the optimal solution of the LP problem of adding an additional constraint: $2x_1 + 3x_2 + x_3 + 5x_4 \leq 4$.

Solution The optimal solution of the LP problem is shown in Table 6.29.

	$c_j \rightarrow$	3	4	1	7	0	0	0
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (=x_B)$	x_1	x_2	x_3	x_4	s_1	s_2	s_3
3	x_1	16/19	1	9/38	1/2	0	5/38	–1/38	0
7	x_4	5/19	0	21/19	0	1	–1/19	4/19	0
0	s_3	126/19	0	59/38	9/2	0	–1/38	–15/38	1
$Z = 83/9$		z_j	3	321/38	3/2	7	1/38	53/38	0
		$c_j - z_j$	0	–169/38	–1/2	0	–1/38	–53/38	0

Table 6.29 Optimal Solution

The optimal solution is : $x_1 = 16/19$, $x_2 = 0$, $x_3 = 0$ and $x_4 = 5/19$ and Max $Z = 83/9$.

(a) (i) The range of variation in RHS values $b_k (k = 1, 2, 3)$ can be obtained by solving the following system of inequalities:

$$\text{Max}\left\{\frac{-x_{Bi}}{\beta_{ik>0}}\right\} \le \Delta b_k \le \text{Min}\left\{\frac{-x_{Bi}}{\beta_{ik<0}}\right\}$$

For $k = 1$, i.e. $b_1 = 7$ we have

$$\text{Max}\left\{\frac{-16/19}{5/28}\right\} \le \Delta b_1 \le \text{Min}\left\{\frac{-5/29}{-1/19}, \frac{-126/19}{-1/38}\right\}$$

$$-32/5 \le \Delta b_1 \le 5$$

$$7 - (32/5) \le b_1 \le 7 + 5, \text{ or } 3/5 \le b_1 \le 12$$

For $k = 2$, i.e. $b_2 = 3$, we have

$$\text{Max}\left\{\frac{-5/19}{4/19}\right\} \le \Delta b_2 \le \text{Min}\left\{\frac{-16/19}{-1/38}, \frac{-126/19}{-15/38}\right\}$$

$$-5/4 \le \Delta b_2 \le 85/5$$

$$3 - (5/4) \le b_2 \le 3 + (85/5) \text{ or } 7/4 \le b_2 \le 20$$

(ii) The range of variation in the coefficients $c_j\ (j = 1, 2, \ldots, 7)$ of variables in the objective function without disturbing the current optimal solution is obtained by solving the following system of inequalities:

$$\text{Max}\left\{\frac{c_j - z_j}{y_{kj} > 0}\right\} \le \Delta c_k \le \text{Min}\left\{\frac{c_j - z_j}{y_{kj} < 0}\right\}$$

The range of variation in the coefficients of basic variables x_1, x_4 and s_3 is given by

For $k = 1$, i.e. $c_1 = 3$

$$\text{Max}\left\{\frac{-169/38}{9/38}, \frac{-1/2}{1/2}, \frac{-1/38}{5/38}\right\} \le \Delta c_1 \le \text{Min}\left\{\frac{-53/38}{-1/38}\right\}$$

$$-1/5 \le \Delta c_1 \le 53$$

$$3 - (1/5) \le c_1 \le 3 + 53, \quad \text{or} \quad 14/5 \le c_1 \le 56$$

For $k = 4$, i.e. $c_4 = 7$

$$\text{Max}\left\{\frac{-169/38}{21/19}, \frac{-53/38}{4/19}\right\} \le \Delta c_4 \le \text{Min}\left\{\frac{-1/38}{-1/19}\right\}$$

$$-169/42 \le \Delta c_4 \le 1/2$$

or $$7 - (169/42) \le c_4 \le 7 + (1/2), \text{ or } 125/42 \le c_4 \le 15/2$$

For $k = 7$, i.e. $c_7 = 0$

$$\text{Max}\left\{\frac{-169/38}{59/38}\right\} \le \Delta c_7 \le \text{Min}\left\{\frac{-1/38}{-1/38}, \frac{-53/38}{-15/38}\right\}$$

$$-1/9 \le \Delta c_7 \le 1$$

The range of variation in the coefficients of non-basic variables x_2, x_3, s_1 and s_2, must satisfy the upper limit $c_k + \Delta c_k \le z_k$ or $\Delta c_k \le z_k - c_k$ thus:

For $k = 2$, $\Delta c_2 \le 169/28$; For $k = 3$, $\Delta c_3 \le 1/2$

For $k = 5$, $\Delta c_5 \le 1/38$; For $k = 6$ $\Delta c_6 \le 53/38$

(iii) Effect of change in coefficients a_{ij} of non-basic variables x_2 , x_3, s_1 and s_2 on the optimal solution is determined by solving the following system of inequalities

$$\text{Max}\left\{\frac{c_j - z_j}{c_B \beta_i > o}\right\} \le \Delta c_{ij} \le \text{Min}\left\{\frac{c_j - z_j}{c_B \beta_i < 0}\right\}$$

Since $$c_B \beta_i = (3\ 7\ 0)\begin{bmatrix} 5/28 & -1/38 & 0 \\ -1/19 & 4/19 & 0 \\ -1/38 & -15/38 & 1 \end{bmatrix}, \text{ therefore}$$

$$c_B \beta_1 = 3\ (3/28) + 7\ (-1/19) + 0\ (-1/38) = 1/38$$
$$c_B \beta_2 = 3\ (-1/38) + 7(4/19) + 0(-15/38) = 53/38$$
$$c_B \beta_3 = 3\ (0) + 7\ (0) + 0\ (1) = 0$$

For row $i = 1, 2, 3$ and column $j = 2$, we have

$$(-169/38)/(1/38) \le \Delta a_{12} \text{ or } \Delta a_{12} \ge -169$$
$$(-169/38)/(53/38) \le \Delta a_{22} \text{ or } \Delta a_{22} \ge -169/53$$
$$(-169/38)/(0) \le \Delta a_{32} \text{ or } -\infty \le \Delta a_{32} \le \infty$$

For row $i = 1, 2, 3$ and column $j = 3$

$$(-1/2)/(1/38) \le \Delta a_{13} \text{ or } \Delta a_{13} \ge -19$$
$$(-1/2)/(53/38) \le \Delta a_{23} \text{ or } \Delta a_{23} \ge -19/53$$
$$(-1/2)/(0) \le \Delta a_{23} \text{ or } -\infty \le \Delta a_{33} \le \infty$$

For row $i = 1, 2, 3$ and column $j = 5$

$$(-1/38)/(1/38) \le \Delta a_{15} \text{ or } \Delta a_{15} \ge -1$$
$$(-1/38)/(53/38) \le \Delta a_{25} \text{ or } \Delta a_{25} \ge -1/53$$
$$(-1/38)/0 \le \Delta a_{35} \text{ or } -\infty \le \Delta a_{35} \le \infty$$

For row $i = 1, 2, 3$ and column $j = 6$

$$(-53/38)/(1/38) \le \Delta a_{16} \text{ or } a_{16} \ge -53$$
$$(-53/38)/(53/38) \le \Delta a_{26} \text{ or } \Delta a_{26} \ge -1$$
$$(-53/38)/0 \le \Delta a_{36} \text{ or } -\infty \ge \Delta a_{36} \le \infty$$

The effect of change in coefficient a_{ij} of basic variables x_1, x_4 and s_3 on the optimal solution is determined by solving the followings system of inequalities. Any change in the coefficients of basic variables may affect both feasibility and optimality of the solution. Considering a change in the element belonging to basic variable $x_4 = \beta_2$, i.e. a_{24}.

Feasible condition: To maintain the feasibility of the solution we need to solve the following inequalities:

$$\underset{k \ne p}{\text{Max}}\left\{\frac{-x_{Bk}}{x_{Bk}\ \beta_{pi} - x_{Bp}\ \beta_{ki} > 0}\right\} \le \Delta\ a_{ij} \le \text{Min}\left\{\frac{-x_{Bk}}{x_{Bk}\ \beta_{pi} - x_{Bp}\ \beta_{ki} < 0}\right\}$$

where β_{ij} are the entries under column s_1, s_2 and s_3 in the optimal simplex table.

Since
$$x_{B1}\ \beta_{24} - x_{B2}\ \beta_{21} = x_{B1}\ \beta_{22} - x_{B2}\ \beta_{21}$$
$$= (16/19)\ (4/19) - (15/19)\ (-1/19) = 7/38$$
$$x_{B3}\ \beta_{24} - x_{B2}\ \beta_{22} = x_{B1}\beta_{22} - x_{B2}\ \beta_{21}$$
$$= (126/19)\ (4/19) - (15/19)\ (-15/38) = 3/2$$

Thus the range of variation in the element a_{24} maintaining feasibility is given by

$$\text{Max}\left\{\frac{-16/19}{7/38}, \frac{-126/19}{3/2}\right\} \le \Delta a_{24} \quad \text{or} \quad -84/19 \le \Delta a_{24}$$

Optimality condition: Further, consider the discrete changes in a_{ij} belonging to column vector. $x_4 = \beta_2$ in the basis matrix $\boldsymbol{B}$.

$$\beta_{22}\,(c_2 - z_2) - y_{22} C_B\,\beta_2 = \frac{8}{38}\left(-\frac{169}{38}\right) - \frac{21}{19}\left(\frac{53}{38}\right) = -2.47$$

$$\beta_{22}\,(c_3 - z_3) - y_{23}\,C_B\,\beta_2 = \frac{8}{38}\left(-\frac{1}{2}\right) - 0\left(\frac{53}{38}\right) = -0.105$$

$$\beta_{22}\,(c_5 - z_5) - y_{25}\,c_B\,\beta_2 = \frac{8}{38}\left(-\frac{1}{38}\right) - \left(-\frac{1}{19}\right)\left(\frac{53}{38}\right) = 0.078$$

$$\beta_{22}\,(c_6 - z_6) - y_{26}\,c_B\,\beta_2 = \frac{8}{38}\left(-\frac{53}{38}\right) - \frac{4}{19}\left(\frac{53}{38}\right) = 0.293$$

Thus, the range of variation in the element a_{24}, maintaining optimality, is given by:

$$\text{Max}\left\{\frac{-1/38}{0.078}, -\frac{53/38}{0.293}\right\} \le \Delta a_{24} \le \text{Min}\left\{\frac{-169/38}{-2.47}, \frac{-1/2}{-0.105}\right\}$$

$$\text{Max}\,\{-0.337, -4.760\} \le \Delta\, a_{24} \le \text{Min}\,\{1.80, 4.901\}$$

$$-0.337 \le \Delta a_{24} \le 1.80$$

Since both the feasibility and the optimality conditions need to be maintained for the element a_{24}, we have: $-0.337 \le \Delta\, a_{24} \le 1.80$.

Similarly, the ranges for discrete change in elements a_{14}, a_{23}, . . . etc., can also be determined.

The optimal basic feasible solution in Table 6.29 does not satisfy the additional constraint. $2x_1 + 3x_2 + x_3 + 5x_4 \le 2$. Thus a new optimal solution is obtained by adding slack variable s_4 to this constraint. It is written together with entries in Table 6.29 as shown in Table 6.30.

		$c_j \rightarrow$	3	4	1	7	0	0	0	0
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	x_4	s_1	s_2	s_3	s_4
3	x_1	16/19	1	9/38	1/2	0	5/38	–1/38	0	0
7	x_4	5/19	0	21/19	0	1	–1/38	–15/38	0	0
0	s_3	126/19	0	59/38	9/2	0	–1/38	–15/38	1	0
0	s_4	2	2	3	1	5	0	0	0	1
Z = 83/9		z_j	3	321/38	3/2	7	1/38	53/38	7	0
		$c_j - z_j$	0	−169/38	−1/2	0	−1/38	−53/38	−7	0

Table 6.30 Addition of New Constraint

In Table 6.30 the basis matrix B has been disturbed due to Row 4. Thus the coefficients in Row 4 under column x_1 and x_4 should become zero. This can be done by applying following row operations.

$$R_4(\text{new}) = R_4\ (\text{old}) - 2R_1 - 5R_2$$

The new solution so obtained is shown in Table 6.31.

Table 6.31 Infeasible Solution

		$c_j \rightarrow$	3	4	1	7	0	0	0	0	
Basic Variables Coefficient $\mathbf{c_B}$	*Basic Variables* $\boldsymbol{B}$	*Basic Variables Value* $\mathbf{b}\ (= \mathbf{x_B})$	x_1	x_2	x_3	x_4	s_1	s_2	s_3	s_4	
3	x_1	16/19	1	9/38	1/2	0	5/38	–1/38	0	0	
7	x_4	5/19	0	21/19	0	1	–1/19	4/19	0	0	
0	s_3	126/19	0	59/38	9/2	0	–1/38	–15/38	1	0	
0	s_4	–1	0	–3	0	0	0	(–1)	0	1	→
$Z = 83/19$		$c_j - z_j$	0	–159/38	–1/2	0	–1/38	–53/38 ↑	0	0	

The solution shown in Table 6.31 is not feasible because $s_4 = -1$. Thus the dual simplex method is applied to the obtained optimal basic feasible solution. As per rule of dual simplex method variable s_4 should obviously leave the basis and s_2 should enter into the basis. The new solution is shown in Table 6.32.

Table 6.32 Optimal and Feasible Solution

		$c_j \rightarrow$	3	4	1	7	0	0	0	0
Basic Variables Coefficient $\mathbf{c}_B$	*Basic Variables* $\boldsymbol{B}$	*Basic Variables Value* $\mathbf{b}\ (= \mathbf{x}_B)$	x_1	x_2	x_3	x_4	s_1	s_2	s_3	s_4
3	x_1	33/38	1	6/19	1/2	0	5/38	0	0	–1/38
7	x_4	1/19	0	9/19	0	1	– 1/19	0	0	4/19
0	s_3	267/38	0	52/19	1/2	0	– 1/38	0	1	–15/38
0	s_4	1	0	3	0	0	0	1	0	–1
$Z = 113/38$		$c_j - z_j$	0	– 5/19	– 1/12	0	– 1/38	0	0	– 53/38

Since all $c_j - z_j \leq 0$ in Table 6.32 and all $x_{Bi} \geq 0$, therefore the current solution is optimal. The new solution value are: $x_1 = 33/38$, $x_2 = 0$, $x_3 = 0$ $x_4 = 1/19$ and Max $Z = 113/38$.

CONCEPTUAL QUESTIONS

1. Write a short note on sensitivity analysis.
2. Discuss the role of sensitivity analysis in linear programming. Under what circumstances is it needed, and under what conditions do you think it is not necessary?
3. (a) Explain how a change in a input-output coefficient can affect a problem's optimal solution?
   (b) How can a change in resource availability affect a solution?
4. What do you understand by the term 'sensitivity analysis'? Discuss the effect of (i) variation of c_j, (ii) variation of b_i and (iii) addition of a new constraint.
5. Discuss the changes in the coefficients a_{ij} for the given LP problem: Max $Z = \mathbf{cx}$, subject to $\mathbf{Ax} = \mathbf{b}$, $\mathbf{x} \geq 0$.
6. Given that the problem: Max $Z = \mathbf{cx}$ such that $\mathbf{Ax} = \mathbf{b}$, $\mathbf{x} \geq 0$ has an optimal solution, can one obtain a linear programming problem which has an unbounded solution changing $\mathbf{b}$ alone?
7. Consider the LP problem: Max $Z = \mathbf{cx}$; subject to $\mathbf{Ax} = \mathbf{b}$ and $\mathbf{x} \geq 0$, where $\mathbf{c}$, $\mathbf{x}^T \in E^n$, $\mathbf{b}^T \in E^m$ and $\mathbf{A}$ is $m \times n$ coefficients matrix. Determine how much can components of the cost vector $\mathbf{c}$ be changed without affecting the optimal solution of the LP problem.
8. Find the limits of variation of element a_{ik} so that the optimal feasible solution of $\mathbf{Ax} = \mathbf{b}$, $\mathbf{x} \geq 0$, Max $Z = \mathbf{c}x$ remains the optimal feasible solution when (i) $\mathbf{a}_k \in \mathbf{B}$, (ii) $\mathbf{a}_k \notin \mathbf{B}$.

SELF PRACTICE PROBLEMS

1. In a LP problem, Max Z = $\boldsymbol{cx}$, subject to $\boldsymbol{Ax} = \boldsymbol{b}$, $\boldsymbol{x} \geq 0$, obtain the variation in coefficients c_j which are permitted without changing the optimal solution. Find this for the following LP problem

   Max $Z = 3x_1 + 5x_2$
   subject to (i) $x_1 + x_2 \leq 1$; (ii) $2x_1 + x_2 \leq 1$
   and $x_1, x_2 \geq 0$. [*Meerut Univ., BSc, 2004*]

2. Discuss the effect of changing the requirement vector from [6 4 24] to [6 2 12] on the optimal solution of the following LP problem:

Max $Z = 3x_1 + 6x_2 + x_3$
subject to (i) $x_1 + x_2 + x_3 \geq 6$; (ii) $x_1 + 5x_2 - x_3 \geq 4$;
(iii) $x_1 + 5x_2 + x_3 \leq 24$
and $x_1, x_2, x_3 \geq 0$.

3. Discuss the effect of discrete changes in the parameter b_i (i = 1, 2, 3) for the LP problem.

Max $Z = 3x_1 + 4x_2 + x_3 + 7x_4$
subject to (i) $8x_1 + 3x_2 + 4x_3 + x_4 \leq 7$;
(ii) $2x_1 + 6x_2 + x_3 + 5x_4 \leq 3$;
(iii) $x_1 + 4x_2 + 5x_3 + 2x_4 \leq 8$
and $x_1, x_2, x_3, x_4 \geq 0$

4. Given the LP problem

Max $Z = -x_1 + 2x_2 - x_3$
subject to (i) $3x_1 + x_2 - x_3 \leq 10$; (ii) $-x_1 + 4x_2 + x_3 \geq 6$;
(iii) $x_2 + x_3 \leq 4$
and $x_1, x_2, x_3 \geq 0$.

Determine the range for discrete changes in the resource values $b_2 = 10$ and $b_3 = 6$ of the LP model so as to maintain optimality of the current solution. [*Meerut Univ., MSc* (*Maths*), *2007*]

5. Given the LP problem

Max $Z = 3x_1 + 5x_2$
subject to (i) $3x_1 + 2x_2 \leq 18$; (ii) $x_1 \leq 4$; (iii) $x_2 \leq 6$
and $x_1, x_2 \geq 0$.

Discuss the effect on the optimality of the solution when the objective function is changed to $3x_1 + x_2$.

6. In an LP problem, Max $Z = \mathbf{cx}$, subject to $\mathbf{Ax} = \mathbf{b}$; $\mathbf{x} \geq 0$, obtain the variation in coefficients c_j which are permitted without changing the optimal solution. Conduct sensitivity analysis on c_j's for the following LP problem.

Max $Z = 3x_1 + 5x_2$
subject to (i) $x_1 + x_2 \leq 1$; (ii) $2x_1 + x_2 \leq 1$
and $x_1, x_2 \geq 0$.

7. Given the LP problem

Max $Z = x_1 + 2x_2 + x_3$
subject to (i) $3x_1 + x_2 - x_3 \leq 10$; (ii) $-x_1 + x_2 + x_3 \leq 6$;
(iii) $x_2 + x_3 \leq 4$
and $x_1, x_2, x_3 \geq 0$.

(a) Determine optimal solution to the problem.
(b) Determine the effect of discrete changes in those components of the cost vector which corresponds to the basic variable.

8. Find the optimal solution to the LP problem

Max $Z = 15x_1 + 45x_2$
subject to (i) $x_1 + 16x_2 \leq 250$, (ii) $5x_1 + 2x_2 \leq 162$,
(iii) $x_2 \leq 50$
and $x_1, x_2 \geq 0$.

If Max $Z = \Sigma\, c_j x_j$, j = 1, 2 and c_2 is kept fixed at 45, determine how much can c_1 be changed without affecting the optimal solution of the problem.

9. Given the LP problem

Max $Z = -x_1 + 2x_2 - x_3$
subject to (i) $3x_1 + x_2 - x_3 \leq 10$, (ii) $-x_1 + 4x_2 + x_3 \geq 6$
(iii) $x_2 + x_3 \leq 4$
and $x_1, x_2, x_3 \geq 0$.

Determine the range for discrete changes in the resource values $b_1 = 10$ and $b_2 = 6$ of the LP model so as to maintain optimality of the current solution.

10. Consider the following LP problem

Max $Z = 4x_1 + 6x_2$
subject to (i) $x_1 + 2x_2 \leq 8$, (ii) $6x_1 + 4x_2 \leq 24$
and $x_1, x_2 \geq 0$.

(a) What is the optimal solution?
(b) If the first constraint is altered as: $x_1 + 3x_2 \leq 8$, does the optimal solution change?

11. The following LP model applies for Shastri & Sons Wood Furniture Company which makes tables (T), chairs (C) and book shelves (B), along with other items.

Max $Z = 200x_T + 150x_C + 150x_B$
subject to
$10x_T + 3x_C + 10x_B \leq 100$ (Wood)
$5x_T + 5x_C + 5x_B \leq 60$ (Labour)
and $x_T, x_C, x_B \geq 0$

The optimal simplex table is shown below:

		$c_j \rightarrow$	*200*	*150*	*150*	*0*	*0*
c_B	*Basic Variables* ***B***	*B.V. Values* $\mathbf{b}$ (= $\mathbf{x}_B$)	x_T	x_C	x_B	s_w	s_L
200	T	64/7	1	0	1	1/7	– 6/70
150	C	20/7	0	1	0	– 1/7	2/7
Z = 15,800/7		z_j	200	150	200	50/7	80/7
		$c_j - z_j$	0	0	– 50	– 50/7	– 80/7

where s_W and s_L are the slack variables for unused word and labour, respectively.

(a) Determine the sensitivity limits for the available wood and labour within which the present product mix will remain optimal.
(b) Find the new optimal solution when the available wood is 90 board feet and labour is 100 hours.

12. Refer to the data in Practice Problem 11.
(a) Determine the sensitivity limits for unit profits within which the current optimal solution will remain unchanged.
(b) What is the total profit when each table yields a profit of Rs. 180 and each chair yields a profit of Rs. 100.

13. A company makes two products A and B. The production of both products requires processing time in two departments I and II. The hourly capacity of I and II, unit profits for products: A and B and the processing time requirements in I and II are given in the following table:

Department	*Product*		*Capacity*
	A	*B*	(*Hours*)
I	1	2	32
II	0	1	8
Unit profit (Rs.)	200	300	

Now the company is considering the addition of a new product C to its line. Product C requires one hour each of departments I and II. What must be product C's unit profit in order to profitably add it to the firm's product line?

14. A company produces three products A, B and C. Each product requires two raw materials: steel and aluminium. The following LP model describes the company's product mix problem.

Max $Z = 30x_A + 10x_B + 50x_C$
subject to
$6x_A + 3x_B + 5x_C \leq 450$ (Steel)
$3x_A + 4x_B + 5x_C \leq 300$ (Aluminium)
and $x_A, x_B, x_C \geq 0$
The optimal production plan is given in the following table:

		$c_j \rightarrow$	30	10	50	0	0
c_B	Basic Variables B	B.V. Values b (= x_B)	x_A	x_B	x_C	s_S	s_A
0	s_S	150	3	– 1	0	1	– 1
50	x_C	60	3/5	4/5	1	0	1/5
Z = 3,000		$c_j - z_j$	0	– 30	0	0	– 10

where s_s and s_A are the slack variables for unused steel and aluminium quantity, respectively.

(a) Suppose an additional 300 tonnes of steel may be procured at a cost of Rs. 100 per tonne. Should the company procure the additional steel?

(b) Unit profit of product A is Rs. 30. How much should this price be increased so that A is produced by the company?

15. A pig farmer is attempting to analyse his feeding operation. The minimum daily requirement of the three nutritional elements for the pigs and the number of units of each of these nutritional elements in two feeds is given in the following table:

Required Nutritional Element	Units of Nutritional Elements (in kg)		Minimum Requirement
	Food 1	Food 2	
A	20	30	200
B	40	25	350
C	30	45	430
Cost per kg (Rs.)	5	3	

(a) Formulate and solve this problem as an LP model.

(b) Assume that the farmer can purchase a third feed at a cost of Rs. 2 per kg, which will provide 35 units of nutrient A, 30 units of nutrient B and 50 units of nutrient C. Would this change the optimal mix of feeds? If yes, how?

16. A company sells two different products A and B. The selling price and incremental cost information is as follows:

	Product A	Product B
Selling price (Rs.)	60	40
Incremental cost (Rs.)	30	10
Incremental profit (Rs.)	30	30

The two products are produced in a common production process and are sold in two different markets. The production process has a capacity of 30,000 labour hours. It takes three hours to produce a unit of A and one hour to produce a unit of B. The market has been surveyed and company officials feel that the maximum number of units of A that can be sold is 8,000; and that of B is 12,000 units.

(a) Find the optimal product mix.

(b) Suppose maximum number of units of A and B that can be sold is actually 9,000 units and 13,000 units, respectively instead of as given in the problem, what effect does this have on the solution? What is the effect on the profit? What is the shadow price for the constraint on the sales limit for both these products.

(c) Suppose there are 31,000 labour hours available instead of 30,000 as in the base case, what effect does this have on the solution? What is the effect on the profit? What is the shadow price for the constraint on the number of labour hours.

17. An organization can produce a particular component for passenger cars, jeeps and trucks. The production of the component requires utilization of sheet metal working and painting facilities, the details of which are given below.

Resource	Consumption (in hrs) to Produce a Unit for			Availability (hrs)
	Passenger Car	Jeep	Truck	
Sheet metal working	0.25	1	0.5	12
Painting	0.5	1	2	30

The profits that can be earned by the three categories of components, i.e. for passenger cars, jeeps and trucks are Rs. 600, Rs. 1,400, and Rs. 1,300, respectively.

(a) Find the optimal product mix for that organization.

(b) What additional profit would be earned by increasing the availability of (i) Sheet metal working shop by an hour only, and (ii) Painting shop by an hour only.

(c) What would be the effect on the profit earned if at least one component for jeep had to be produced?

18. A firm uses three machines in the manufacture of three products. Each unit of product I requires 3 hours on machine 1, 2 hours on machine 2 and 1 hour on machine 3. Each unit of product II requires 4 hours on machine 1, 1 hour on machine 2 and 3 hours on machine 3. Each unit of product III requires 2 hours on machine 1, 2 hours on machine 2, and 2 hours on machine 3. The contribution margin of the three products is Rs. 30, Rs. 40 and Rs. 35 per unit, respectively. Available for scheduling are 90 hours of machine 1 time, 54 hours of machine 2 time, and 93 hours of machine 3 time.

(a) What is the optimal production schedule for the firm?

(b) What is the marginal value of an additional hour of time on machine?

(c) What is the opportunity cost associated with product 1? What interpretation should be given to this opportunity cost?

(d) Suppose that the contribution margin for product I is increased from Rs. 30 to Rs. 43, would this change the optimal production plan? Give reasons.

HINTS AND ANSWERS

1. $-\infty \le c_1 \le 3 + (1/3)$; $5 - (1/2) \le c_2 \le 5 + \infty$ or $9/2 \le c_2 \le \infty$
2. The current basic feasible solution consisting of s_1, x_1 and x_3 remains feasible and optimal at the new values: $x_1 = 7, x_3 = 5$ and $x_2 = 0$ with Min $Z = 26$.
3. $5 \ge b_1 \ge -32/5$; $84/5 \ge b_2 \ge -5/4$; $\Delta b_3 \ge -126/19$
4. $b_2 < 10$; $-5/2 \le b_3 \ge 6$
5. Initial optimal solution: $x_1 = 2, x_2 = 6$ and Max $Z = 36$. Where $c_3 - z_3 = -1$ and $c_5 - z_5 = -3$. After having $c_1 = 3$ and $c_2 = 1$, calculate new values of $c_3 - z_3 = 1$ and $c_5 - z_5 = 2/3$. Thus, introduce s_3 in the basis to get the new solution: $x_1 = 2, x_2 = 6$ and Max $Z = 12$.
7. (a) $x_1 = 0, x_2 = 4, x_3 = 0$ and Max $Z = 8$ (b) $-(1/2) \le c_2$
8. $(15 - 195/16) \le c_1 \le (15 + 195/16)$
9. $b_2 < 10$; $-5/2 \le b_3 \ge 6$
11. (a) $36 \le x_w \le 120$; $50 \le x_L \le 500/3$
    (b) $x_T = 54/7$; $40/7$; $x_C = 30/7$; $100/7$; $Z = 15{,}300/7$, $23{,}000/7$.
12. $15 \le x_T \le 50$; $6 \le x_C \le 20$; $-\infty \le x_B \le 20$
16. (a) $x_A = 6{,}000$, $x_B = 12{,}000$, Max Z = Rs. 5,40,000
    (b) As $x_A \le 8{,}000$ is not binding because $x_A = 6{,}000$ in the solution. Thus, an increase in the limit on A will have no effect on the

solution or profit. Shadow price is zero. For increase in the limit of B, the new solution is, $x_A = 5{,}670$ and $x_B = 13{,}000$ and Max Z = Rs. 5,60,000. The shadow price is (5,60,000 – 5,40,000) = Rs. 20,000.

(c) New solution is $x_A = 6{,}330$ and $x_B = 12{,}000$ with Max Z = Rs. 5,50,000. The shadow price is (5,50,000 – 5,40,000) = Rs. 10,000.

CHAPTER SUMMARY

Sensitivity analysis provides the sensitive ranges (both lower and upper limits) within which LP model parameters can vary without changing the optimally of the current solution. Formulae to conduct sensitivity analysis followed by numerical examples have been solved to illustrate how changes in the parameters specially, (i) the change in the objective function coefficients; right hand-side value of a constraint and (ii) input-output coefficients in any constraint of a LP model, affect its optimal solution.

CHAPTER CONCEPTS QUIZ

True or False

1. The solution to the dual yields shadow prices.
2. The absolute values of the numbers in the $c_j - z_j$ row under the slack variables represent the solutions to the dual problem.
3. The transpose of the primal constraints coefficients become the dual constraint coefficients.
4. The range over which shadow prices remain valid is called right-hand-side ranging.
5. The shadow price is the value of additional profit margin from an activity to be conducted.
6. The solution in optima as long as all $c_j - z_j \geq 0$.
7. Non-basic variables are those that have a value of zero.
8. Testing basic variables does not require reworking the final simplex table during sensitivity analysis.
9. In a simplex table, if all of the substitution rates in the key column are negative, then it indicates an infeasible solution.
10. Making changes in the resources value result in changes in the feasible region and often the optima solution.

Fill in the Blanks

11. The _____ for a constrain is the value of an additional unit of the resource.
12. The range of significance are values over which a _____ coefficient can vary without causing a change in the optimal solution mix.
13. The range of optimality are values over which a _____ coefficient can change without causing a change in the optimal solution mix.
14. Right-hand-side ranging method is used to find the range over which _____ remain valid.
15. In the optimal simplex table the absolute value of cj – zj numbers corresponds to slack variable column represent _____ of the available resources.
16. The addition of a constraint in the existing constraints will cause a _____ change in the objective function coefficients.
17. The addition of new constraint _____ the feasible region of the given LP problem.
18. Sensitivity analysis provides the range within which a parameter may change without offering _____.
19. When an additional variable is added in the LP problem, the _____ solution can be improved if $c_j - z_j \leq 0$.
20. A non-basic variable should be brought into new solution mix provided its _____ is $c_j < c_j + (z_j - c_j)$.

Multiple Choice

21. Sensitivity analysis
    (a) is also called post-optimality analysis as it is carried out after the optimal solution is obtained
    (b) allows the decision-maker to get more meaningful information about changes in the LP model parameters
    (c) provides the range within which a parameter may change without affecting optimality
    (d) all of the above
22. When an additional variable is added in the LP model, the existing optimal solution can further be improved if
    (a) $c_j - z_j \geq 0$ (b) $c_j - z_j \leq 0$
    (c) both (a) and (b) (d) none of the above
23. Addition of an additional constraint in the existing constraints will cause a
    (a) change in objective function coefficients (c_j)
    (b) change in coefficients a_{ij}
    (c) both (a) and (b)
    (d) one of the above
24. If the additional constraint added is an equation and an artificial variable appears in the basis of the new problem, the new optimal solution is obtained by
    (a) assigning zero cost coefficient to the artificial variable if it appears in the basis at negative value
    (b) assigning $-M$ cost coefficient to the artificial variable if it appears in the basis at positive value
    (c) either (a) and (b)
    (d) none of the above
25. To ensure the best marginal increase in the objective function value, a resource value may be increased whose shadow price is comparatively
    (a) larger (b) smaller
    (c) neither (a) nor (b) (d) both (a) and (b)
26. A non-basic variable should be brought into the new solution mix provided its contribution rate (c_j) is
    (a) $c_j^* = c_j + (z_j - c_j)$ (b) $c_j^* > c_j + (z_j - c_j)$
    (c) $c_j^* < c_j + (z_j - c_j)$ (d) none of the above
27. While performing sensitivity analysis, the upper bound infinity on the value of the right-hand side of a constraint means that
    (a) there is slack in the constraint
    (b) the constraint is redundant
    (c) the shadow price for that constraint is zero
    (d) none of the above
28. The entering variable in the sensitivity analysis of objective function coefficients is always a
    (a) decision variable (b) non-basic variable
    (c) basic variable (d) slack variable

29. To maintain optimality of current optimal solution for a change Δc_k in the coefficient c_k of non-basic variable x_k, we must have
    (a) $\Delta c_k = c_k - z_k$ (b) $\Delta c_k = z_k$
    (c) $c_k + \Delta c_k = z_k$ (d) $\Delta c_k \geq z_k$
30. In sensitivity analysis of the coefficient of the non-basic variable in a cost minimization LP problem, the upper sensitivity limit is
    (a) original value + Lowest positive value of improvement ratio
    (b) original value – Lowest absolute value of improvement ratio
    (c) positive infinity
    (d) negative infinity
31. If the RHS value of a constraint is changed to another value within specified limits, then the new optimal solution would have
    (a) same optimal solution
    (b) a lower objective function value
    (c) a higher objective function value
    (d) same set of variables in the solution mix
32. A change in the objective function for a non-basic variable can affect
    (a) $c_j - z_j$ values of all non-basic variables
    (b) $c_j - z_j$ values of all basic variables
    (c) Only the $c_j - z_j$ value of that variable
    (d) none of the above
33. The 100 per cent rule for RHS values says that if the sum of the percentage change is less than or equal to 100%, then
    (a) dual prices do not change
    (b) different dual prices may exist
    (c) both (a) and (b) may occur
    (d) none of the above
34. The 100 per cent rule for objective function coefficients says that if the sum of the percentage change is less than or equal to 100%, then
    (a) optimal solution does not change
    (b) a different optimal solution may exist
    (c) both (a) and (b) may occur
    (d) none of the above
35. The right-hand side range is often referred to as the range of
    (a) improvement (b) feasibility
    (c) infeasibility (d) optimality

Answers to Quiz

1. T	2. T	3. T	4. T	5. F	6. F	7. T	8. F	9. F	10. T
11. shadow price		12. non-basic variable		13. basic variable		14. shadow price		15. shadow price	
16. simultaneous		17. leaves unchanged		18. optimality		19. existing optimal		20. contribution rate	
21. (d)	22. (a)	23. (c)	24. (c)	25. (a)	26. (c)	27. (d)	28. (d)	29. (c)	30. (c) 31. (d)
32. (c)	33. (a)	34. (a)	35. (b)						

CASE STUDY

Case 6.1: Utensil Manufacturer

A stainless steel utensil manufacturer makes three types of items. The restrictions, profits and requirements are tabulated below:

Utensil Type	*I*	*II*	*III*
Raw material requirement (kg per unit)	6	3	5
Welding and finishing time (hours per unit)	3	4	5
Profit per unit (Rs.)	3	1	4

If stainless steel (raw material) availability is 25 kg and welding and finishing time available is 20 hours per day, then the optimum product mix problem is expressed as:

$$\text{Max } Z = 3x_1 + x_2 + 4x_3$$

subject to

$$6x_1 + 3x_2 + 5x_3 \leq 25 \text{ (raw material restriction)}$$
$$3x_1 + 4x_2 + 5x_3 \leq 20 \text{ (time restriction)}$$

and $$x_1, x_2, x_3 \geq 0$$

where x_j ($j = 1, 2, 3$) is the number of units of the jth type of the item to be produced.

Questions for Discussion

(a) The second type of utensil would change the current optimal basis if its profit per unit is: ≥ 1; ≥ 1.02; ≥ 1.06; ≥ 2; ≥ 3.

(b) The simplex multiplier associated with the machine time restriction of 20 hours is (– 3/5). Thus, the multiplier remains unchanged for the upper limit on the machine time availability of 25; 27.5; 35; 42.5; 32.5 hours.

(c) The increase in the objective function for each unit availability of machine time higher than the upper limit indicated in part (b) is ... (show calculations).

(d) The profit of the third type of utensil is Rs. 4 per unit. The lower unit on its profitability such that the current basis is still optimal is 4; 3; 2.5; 2; < 2.